Dictionary of TQFT

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1 Basic Mathematic

1.1 Homogeneous Space

Core Idea: A homogeneous space is a highly symmetric space where every point is equal under the action of a symmetry group. **Mathematical Definition:** Let G be a Lie group acting smoothly on a manifold M. Then M is called a *G-homogeneous space* if the action is transitive:

$$\forall\, p, q \in M,\ \exists\, g \in G \text{ such that } g \cdot p = q.$$

Construction – Quotient Space G/H:

A fundamental result in Lie theory states that:

> Every homogeneous space M can be written as a quotient G/H, where $H \subseteq G$ is a closed subgroup (called the *stabilizer* or *isotropy subgroup* of a point).

Examples

1. The 2-Sphere $S^2 = SU(2)/U(1)$

- $G = SU(2)$: The group of 2×2 unitary matrices with determinant 1. Topologically, it is the 3-sphere S^3.
- Fix a point (e.g., the "north pole" p). The subgroup of $SU(2)$ that leaves p invariant consists of matrices of the form
$$\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix},$$
which is a $U(1)$ subgroup.
- Therefore, we have $S^2 \simeq SU(2)/U(1)$.

2. Real Projective Space $\mathbb{RP}^n = S^n/\mathbb{Z}_2$

- Identify antipodal points $x \sim -x$ on the sphere S^n. This is the action of the group $\mathbb{Z}_2$.
- So we define $\mathbb{RP}^n = S^n/\mathbb{Z}_2$.

Important Relation: $\mathbb{RP}^3 \simeq SO(3)$

- Physically, a 3D rotation is parameterized by an axis (a point on S^2) and an angle $\theta \in [0, \pi]$.
- A rotation by π around an axis is the same as a rotation by π around the opposite axis $\Rightarrow$ identify antipodal points.
- This space is a 3-ball with antipodal boundary points identified, which is homeomorphic to $\mathbb{RP}^3$.

3. Complex Projective Space $\mathbb{CP}^n$

Construction:

- Remove the origin from $\mathbb{C}^{n+1}$, then identify vectors that differ by a complex scalar multiple (i.e., rays).
- So $\mathbb{CP}^n$ is the set of complex lines through the origin in $\mathbb{C}^{n+1}$.

Special Case: $\mathbb{CP}^1 \simeq S^2$ (Riemann Sphere)

- A ray in $\mathbb{C}^2$ can be described by a complex coordinate z.
- Adding the point at infinity compactifies it into the Riemann sphere S^2.

Connection to Generalized Symmetry in Gauge Theory

- In a pure $SU(2)$ gauge theory, global symmetries act in a standard way.
- In an $SO(3) = SU(2)/\mathbb{Z}_2$ gauge theory, something new appears: a *1-form symmetry* acting on line operators (e.g., Wilson loops).
- This 1-form symmetry group is exactly the $\mathbb{Z}_2$ that was quotiented out.
- Understanding this subtle difference is key to understanding *generalized global symmetries* in non-Abelian gauge theories.

1.2 Hopf fibration

It describes a non-trivial fiber bundle structure where the 3-sphere is "woven" from circles (the fibers) over the base space of a 2-sphere.

In the standard language of fiber bundles, the Hopf fibration is expressed as:

$$S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$$

- **Total Space (E):** The 3-sphere S^3
- **Base Space (M):** The 2-sphere S^2
- **Fiber (F):** The circle S^1
- **Projection Map (π):** A continuous surjective map from S^3 to S^2

The key property is **local triviality**: For a small open neighborhood $U \subset S^2$, the preimage $\pi^{-1}(U)$ is homeomorphic to the direct product $U \times S^1$. However, globally, S^3 is *not* homeomorphic to $S^2 \times S^1$, which makes the bundle **non-trivial**.

1.3 Codimension

Let M be a smooth manifold of dimension n, and let $N \subset M$ be a submanifold of dimension k. Then the **codimension** of N in M is defined as

$$\operatorname{codim}(N, M) = \dim M - \dim N = n - k.$$

In other words, codimension measures how many dimensions are "missing" in the submanifold N compared to the ambient manifold M.

Examples:

- A curve (1-dimensional) inside $\mathbb{R}^3$ has codimension $3 - 1 = 2$.
- A hypersurface in $\mathbb{R}^n$ (i.e., a subset of dimension $n - 1$) has codimension 1.

1.4 Poincaré Duality

Let M be a closed, connected, oriented n-dimensional manifold. Then for each integer $0 \leq k \leq n$, there is a natural isomorphism between the cohomology and homology groups:

$$H^k(M;\mathbb{Z}) \cong H_{n-k}(M;\mathbb{Z}).$$

This isomorphism is called the **Poincaré duality**.

More generally, for any coefficient ring R, if M is a closed oriented n-manifold, then

$$H^k(M;R) \cong H_{n-k}(M;R),$$

where the isomorphism depends on the orientation of M and is realized via the cap product with the fundamental class $[M] \in H_n(M;R)$:

$$\mathrm{PD} : H^k(M;R) \xrightarrow{\cong} H_{n-k}(M;R), \quad \alpha \mapsto \alpha \frown [M].$$

1.5 Morse Theory

Definition

Let M be a smooth manifold and let $f : M \to \mathbb{R}$ be a smooth real-valued function, called a *Morse function*. The critical points of f are the points $p \in M$ such that

$$df(p) = 0.$$

A critical point p is said to be *non-degenerate* if the Hessian matrix

$$H_{ij}(p) = \frac{\partial^2 f}{\partial x^i \partial x^j}\Big|_p$$

has no zero eigenvalues. The number of negative eigenvalues of $H_{ij}(p)$ is called the *Morse index* of p.

Topology Change via Critical Points

Let $f : M \to \mathbb{R}$ be a Morse function. Suppose $a < b \in \mathbb{R}$, and assume that the interval $[a, b]$ contains exactly one critical value c, corresponding to a unique critical point p of index λ. Then the sublevel set

$$M_b = \{x \in M \mid f(x) \leq b\}$$

has the homotopy type of

$$M_a \cup e^\lambda,$$

where e^λ is a λ-dimensional cell (handle) attached to M_a.

Morse Homology and Topological Invariants

Let $\mathrm{Crit}_\lambda(f)$ denote the set of critical points of index λ. Then we can define a chain complex $(C_*(f), \partial)$, where:

$$C_\lambda(f) := \bigoplus_{p \in \mathrm{Crit}_\lambda(f)} \mathbb{Z} \cdot p,$$

and $\partial : C_\lambda \to C_{\lambda-1}$ counts (with orientation signs) the number of gradient flow lines between critical points of index difference 1.

The homology of the Morse complex $(C_*(f), \partial)$ is isomorphic to the singular homology of the manifold:

$$H_*(C_*(f), \partial) \cong H_*(M;\mathbb{Z}).$$

Physical Interpretation and Applications

Morse theory plays a significant role in several areas of theoretical physics:

- **Supersymmetric Quantum Mechanics (SUSY QM):** As shown by Witten (1982), the deformed Laplacian operator
$$H_t = d_t^\dagger d_t + d_t d_t^\dagger, \quad \text{with} \quad d_t = e^{-tf} d e^{tf},$$
has its ground states localized at critical points of f. The associated quantum system's cohomology is isomorphic to de Rham cohomology, bridging quantum mechanics and topology.
- **Path Integral Formulation:** In the semi-classical approximation, dominant contributions come from critical points of the action functional, mirroring Morse theory's focus on critical points and their indices.
- **String Theory and Moduli Spaces:** Morse functions are used to probe the topology of complicated moduli spaces by analyzing their critical sets.
- **Condensed Matter and Topological Phase Transitions:** Changes in the topological phase can be associated with appearance/disappearance of critical points of effective energy functionals.

In field theory, especially in supersymmetric theories, the scalar potential $V(\phi)$ plays the role of a Morse function on the space of field configurations $\{\phi\}$.

- The *critical points* of V correspond to vacuum configurations satisfying
$$\frac{\partial V}{\partial \phi_i} = 0.$$
- The *non-degenerate vacua* correspond to isolated minima of V, while *flat directions* (moduli) correspond to degenerate critical manifolds.

Thus, the study of the vacuum moduli space in supersymmetric theories is, mathematically, a study of the Morse structure induced by $V(\phi)$ on the configuration space.

Relation to Topological Field Theory

In Witten's topological quantum field theory (TQFT), the path integral localizes around the critical points of an action functional $S[\phi]$, which acts as a Morse function on the infinite-dimensional configuration space of fields. The supersymmetry operator Q plays the role of the differential in the Morse complex, and the cohomology of Q corresponds to the topological invariants of the underlying manifold.

Therefore, the supersymmetric vacuum equations $F = D = 0$ can be viewed as the finite-dimensional analogue of the localization equations in TQFT, and the corresponding moduli space encodes the topological information of the theory.

1.6 Mayer–Vietoris Sequence

Basic Concept

The *Mayer–Vietoris sequence* is an important tool in algebraic topology, mainly used to compute the homology groups of a topological space. Its key idea is:

> If a topological space X can be decomposed into two simpler open subsets U and V such that $X = U \cup V$, then the homology of X can be determined from the homology of U, V, and their intersection $U \cap V$.

This is especially useful for handling complicated spaces by breaking them into simpler parts and then "gluing" the homology information together.

—

The Sequence

Let X be a topological space, $U, V \subset X$ open sets with $X = U \cup V$, and $H_n(\cdot)$ denote the singular homology. Then there exists a long exact sequence:

$$\cdots \to H_n(U \cap V) \xrightarrow{(i_*, j_*)} H_n(U) \oplus H_n(V) \xrightarrow{k_* - l_*} H_n(X) \xrightarrow{\partial} H_{n-1}(U \cap V) \to \cdots$$

where

- $i_* : H_n(U \cap V) \to H_n(U)$ and $j_* : H_n(U \cap V) \to H_n(V)$ are induced by inclusion,
- $k_* : H_n(U) \to H_n(X)$ and $l_* : H_n(V) \to H_n(X)$ are inclusion maps,
- ∂ is the boundary map.

This sequence is *exact*, meaning that the image of each map equals the kernel of the next map.

Examples

1. Circle S^1:

- Take U and V as two open arcs covering S^1, then $U \cup V = S^1$.
- U and V are contractible: $H_*(U) = H_*(V) = 0$ for $* > 0$.
- $U \cap V$ consists of two disjoint intervals: $H_0(U \cap V) = \mathbb{Z} \oplus \mathbb{Z}$.
- Applying the sequence, we find $H_1(S^1) \cong \mathbb{Z}$ and $H_0(S^1) \cong \mathbb{Z}$.

2. Two Spheres Glued Together:

- Decompose the space into two overlapping balls (or disks).
- The homology of the total space can be computed from the homology of each ball and their intersection using the Mayer–Vietoris sequence.

1.7 Riemann Surfaces

Definition

A *Riemann surface* X is a one-dimensional complex manifold, i.e., a topological space such that:

- Each point $p \in X$ has a neighborhood U homeomorphic to an open subset of $\mathbb{C}$.
- The transition maps between local charts are *holomorphic*.

Properties

- Compact Riemann surfaces are *compact orientable 2-dimensional manifolds*.
- Classified by *genus* g, which counts the number of "holes" in the surface:
  - $g = 0$: Riemann sphere S^2
  - $g = 1$: torus T^2
  - $g > 1$: higher genus surfaces
- Admit holomorphic functions, meromorphic functions, and holomorphic 1-forms.

Examples

1. Riemann sphere: $\mathbb{C} \cup \{\infty\}$
2. Torus: $T^2 = \mathbb{C}/\Lambda$, where Λ is a 2-dimensional lattice
3. Higher genus surface: quotient of the unit disk by a Fuchsian group $\mathbb{D}/\Gamma$

1.8 Algebraic Curves

Definition

An *algebraic curve* C over $\mathbb{C}$ is a one-dimensional variety defined as the zero locus of a polynomial in the complex projective plane $\mathbb{CP}^2$:

$$C = \{[x : y : z] \in \mathbb{CP}^2 \mid F(x, y, z) = 0\},$$

where F is homogeneous and C is *smooth*, i.e.,

$$\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \text{ do not simultaneously vanish at any point on } C.$$

Compactification

- Affine curves in $\mathbb{C}^2$ can be compactified to $\mathbb{CP}^2$. - Compactification ensures the resulting curve is a *compact Riemann surface.*

Genus Formula

For a smooth plane curve of degree d:

$$g = \frac{(d-1)(d-2)}{2}.$$

—

Divisors and Linear Systems

Divisor

A *divisor* D on a Riemann surface X is a formal finite sum:

$$D = \sum_i n_i P_i, \quad n_i \in \mathbb{Z},\ P_i \in X.$$

Meromorphic Functions and Linear System

For a divisor D, define

$$L(D) = \{f \in \mathbb{C}(X) \mid (f) + D \geq 0\} \cup \{0\},$$

and

$$\ell(D) := \dim_{\mathbb{C}} L(D).$$

Canonical Divisor

The *canonical divisor* K is associated with holomorphic 1-forms on X. Its degree satisfies

$$\deg(K) = 2g - 2.$$

—

1.9 The Riemann–Roch Theorem

Statement

Let X be a compact Riemann surface of genus g, D a divisor on X, and K the canonical divisor. Then

$$\boxed{\ell(D) - \ell(K - D) = \deg(D) - g + 1}.$$

- $\ell(D)$ counts the dimension of meromorphic functions with poles constrained by D.
- $\ell(K - D)$ is a "correction term" measuring constraints.
- Intuitively:

$$\text{freedom of functions} - \text{constraints} = \deg(D) - g + 1$$

Examples

1. Zero divisor $D = 0$: $\ell(0) = 1$ (only constants)
2. Large degree $\deg(D) > 2g - 2$: $\ell(D) = \deg(D) - g + 1$
3. Riemann sphere $g = 0$: $\ell(D) = \deg(D) + 1$
4. Torus $g = 1$, divisor of degree $d > 0$: $\ell(D) = d$

2 Anomaly

The symmetry of a classical theory is mainly divided into two types. One kind is the local symmetry, which depends on spacetime coordinates and is also called gauge symmetry. The other kind is the global symmetry, which does not depend on spacetime coordinates.

Quantum anomalies are divided into three categories:

2.1 Gauge anomaly:

This is the anomaly associated with gauge symmetry. In other words, at the quantum level, gauge symmetry is broken, and we cannot obtain a consistent theory. To build a consistent Quantum Field Theory, **all gauge anomalies must cancel**. For the triangle diagram of a gauge current J^μ:

$$\sum_{\text{Fermions}} \langle J^\mu J^\nu J^\rho \rangle \neq 0 \quad \Rightarrow \quad \text{The theory is inconsistent and this must vanish.}$$

The simplest way to avoid gauge anomalies is to use **vector-like** theories:

- Fermions are **Dirac fermions**.
- Gauge fields couple *identically* to left-handed and right-handed Weyl components.

Such theories are automatically anomaly-free.

In **chiral gauge theories**, left- and right-handed fermions transform in *different* representations of the gauge group.

- These theories are *necessarily* coupled to massless fermions (mass terms are forbidden by the gauge symmetry).
- Anomaly cancellation is *not automatic* and imposes strict constraints.

The anomaly arises from triangle diagrams. Cancellation requires the sum of contributions from all fermions to vanish. For the three-current correlation function $\langle J^\mu J^\nu J^\rho \rangle$, the Ward identities require:

$$p_{1\mu} \langle J^\mu J^\nu J^\rho \rangle = 0, \quad p_{2\nu} \langle J^\mu J^\nu J^\rho \rangle = 0, \quad q_\rho \langle J^\mu J^\nu J^\rho \rangle = 0,$$

where p_1, p_2, and q are the incoming momenta, satisfying $p_1 + p_2 + q = 0$.

2.1.1 Abelian Gauge Group [U(1)]

For a set of Weyl fermions with charges Q_a, the anomalies cancel if:

$$\sum_a Q_a^3 = 0 \quad \text{and} \quad \sum_a Q_a = 0$$

The second condition ensures cancellation of the mixed gauge-gravitational anomaly.

2.1.2 Non-Abelian Gauge Group [e.g., SU(N)]

For Weyl fermions in representations $\mathbf{R}_i$, the anomaly cancels if:

$$\sum_i d^{abc}(\mathbf{R}_i) = 0, \quad \text{where} \quad d^{abc}(\mathbf{R}) = \operatorname{tr} T^a \{T^b, T^c\}$$

and T^a are the generators in representation $\mathbf{R}$.

2.1.3 Special Case: The SU(2) Anomaly

- $SU(2)$ has no *perturbative* gauge anomaly (all representations are real/pseudo-real).
- However, it has a **non-perturbative (global) anomaly**.
- An $SU(2)$ gauge theory is inconsistent with an **odd number of Weyl fermions** in doublet representations.
- Weyl fermions must come in pairs (i.e., as Dirac fermions).

2.2 ABJ-type anomaly(global anomaly)(perturbative):

This is the anomaly associated with global symmetry. In other words, at the quantum level, global symmetry is broken.

For the triangle diagram of an axial current J_A^μ:

$$\sum_{\text{Fermions}} \langle J^\mu J^\nu J_A^\rho \rangle \neq 0 \quad \Rightarrow \quad \text{Interesting physics, This is the ABJ anomaly.}$$

$\boxed{\text{Method 1}}$: There is a step-by-step derivation of the 4D axial-vector-vector (AVV) triangle diagram, which leads to the Adler–Bell–Jackiw (ABJ) anomaly. The conventions are:

- Metric signature $(+, -, -, -)$
- $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$
- $\operatorname{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^5) = -4i\varepsilon^{\mu\nu\rho\sigma}$, with $\varepsilon^{0123} = +1$

We aim to compute the AVV triangle amplitude

$$T^{\mu\nu\rho}(p,q) = \int \frac{d^4k}{(2\pi)^4} \mathrm{Tr}\Big[\gamma^\mu \frac{1}{\not{k}-m}\gamma^\nu \frac{1}{\not{k}+\not{p}-m}\gamma^\rho\gamma^5 \frac{1}{\not{k}-\not{q}-m}\Big], \tag{2.2.1}$$

and extract the divergence of the axial current $(p+q)_\rho T^{\mu\nu\rho}$, which leads to the anomaly

$$\partial_\rho J_5^\rho = \frac{e^2}{16\pi^2}\varepsilon^{\mu\nu\alpha\beta}F_{\mu\nu}F_{\alpha\beta}. \tag{2.2.2}$$

Consider the interaction Lagrangian $\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu$. The AVV triangle (axial vertex at the third point) in momentum space is:

$$T^{\mu\nu\rho}(p,q) = \int \frac{d^4k}{(2\pi)^4} \frac{\mathrm{Tr}\Big[\gamma^\mu(\not{k}+m)\gamma^\nu(\not{k}+\not{p}+m)\gamma^\rho\gamma^5(\not{k}-\not{q}+m)\Big]}{(k^2-m^2)((k+p)^2-m^2)((k-q)^2-m^2)}. \tag{2.2.3}$$

We contract with $(p+q)_\rho$:

$$(p+q)_\rho T^{\mu\nu\rho} = \int \frac{d^4k}{(2\pi)^4} \mathrm{Tr}\Big[\gamma^\mu \frac{\not{k}+m}{k^2-m^2}\gamma^\nu \frac{\not{k}+\not{p}+m}{(k+p)^2-m^2}(p+q)_\rho\gamma^\rho\gamma^5 \frac{\not{k}-\not{q}+m}{(k-q)^2-m^2}\Big]. \tag{2.2.4}$$

Using $(p+q)_\rho\gamma^\rho = \not{p}+\not{q}$, we rewrite as a difference of propagators:

$$(p+q)_\rho\gamma^\rho\gamma^5 = \gamma^5\left[(\not{k}+\not{p}-m)-(\not{k}-\not{q}-m)\right]. \tag{2.2.5}$$

Substituting back, we obtain two terms:

$$I_1 = \int \frac{d^4k}{(2\pi)^4} \mathrm{Tr}\Big[\gamma^\mu \frac{\not{k}+m}{k^2-m^2}\gamma^\nu \frac{\not{k}+\not{p}+m}{(k+p)^2-m^2}\gamma^5\Big], \tag{2.2.6}$$

$$I_2 = \int \frac{d^4k}{(2\pi)^4} \mathrm{Tr}\Big[\gamma^\mu\gamma^5 \frac{\not{k}-\not{q}+m}{(k-q)^2-m^2}\gamma^\nu \frac{\not{k}+m}{k^2-m^2}\Big], \tag{2.2.7}$$

so that

$$(p+q)_\rho T^{\mu\nu\rho} = I_1 - I_2. \tag{2.2.8}$$

Naively, shifting $k \to k-q$ in I_2 would give $I_2 = I_1$, implying $(p+q)_\rho T^{\mu\nu\rho} = 0$. However, in divergent integrals, such shifts are not allowed unless the regularization preserves the shift invariance. Standard regularizations (Pauli–Villars) cannot preserve both axial and vector Ward identities simultaneously. The surface term arising from this shift is the anomaly.

Introduce a heavy regulator fermion with mass M and opposite sign contribution:

$$T^{\mu\nu\rho}_{\text{reg}} = T^{\mu\nu\rho}(m) - T^{\mu\nu\rho}(M). \tag{2.2.9}$$

For $(p+q)_\rho T^{\mu\nu\rho}_{\text{reg}}$, at high momentum k, only the leading terms in k survive. The trace in the numerator is

$$\mathrm{Tr}(\gamma^\mu\not{k}\gamma^\nu\not{k}\gamma^5\not{k}) \sim \mathrm{Tr}(\gamma^\mu\gamma^\alpha\gamma^\nu\gamma^\beta\gamma^5\gamma^\gamma)k_\alpha k_\beta k_\gamma. \tag{2.2.10}$$

After symmetrization, only terms contributing to ε-tensor survive. Using

$$\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^5) = -4i\varepsilon^{\mu\nu\rho\sigma}, \tag{2.2.11}$$

The contracted amplitude becomes

$$(p+q)_\rho T^{\mu\nu\rho} = -\frac{i}{2\pi^2}\varepsilon^{\mu\nu\alpha\beta}p_\alpha q_\beta. \tag{2.2.12}$$

Relating to the field strength $F_{\mu\nu}(p) = i(p_\mu A_\nu - p_\nu A_\mu)$, we obtain the coordinate space form

$$\boxed{\partial_\mu J_5^\mu(x) = \frac{e^2}{16\pi^2}\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}(x)F_{\rho\sigma}(x)} \neq 0 \tag{2.2.13}$$

This result is nonzero: axial current is not conserved due to the anomaly.

$\boxed{\text{Method 2}}$: There is the detailed derivation of the axial (chiral) anomaly using the path integral measure approach introduced by Fujikawa.

Consider a Dirac fermion in an external gauge field A_μ. The Euclidean action is

$$S_E[\bar{\psi}, \psi, A] = \int d^4x\, \bar{\psi}(i\not{D})\psi, \tag{2.2.14}$$

where

$$\not{D} = \gamma^\mu(\partial_\mu + ieA_\mu). \tag{2.2.15}$$

The generating functional is

$$Z[A] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi\, e^{-S_E[\bar{\psi},\psi,A]}. \tag{2.2.16}$$

Consider an infinitesimal local chiral rotation:

$$\psi(x) \to \psi'(x) = e^{i\alpha(x)\gamma^5}\psi(x), \quad \bar{\psi}(x) \to \bar{\psi}'(x) = \bar{\psi}(x)e^{i\alpha(x)\gamma^5}. \tag{2.2.17}$$

The classical action changes as:

$$S_E[\bar{\psi}', \psi', A] = S_E[\bar{\psi}, \psi, A] - \int d^4x\, \alpha(x)\, \partial_\mu(\bar{\psi}\gamma^\mu\gamma^5\psi), \tag{2.2.18}$$

so classically, the axial current is conserved.

However, the path integral measure $\mathcal{D}\bar{\psi}\mathcal{D}\psi$ is not invariant under this transformation:

$$\mathcal{D}\bar{\psi}'\mathcal{D}\psi' = J[\alpha]\, \mathcal{D}\bar{\psi}\mathcal{D}\psi, \tag{2.2.19}$$

where $J[\alpha]$ is the Jacobian.

Let $\{\phi_n(x)\}$ be a complete set of eigenfunctions of the Dirac operator $i\not{D}$:

$$i\not{D}\phi_n(x) = \lambda_n\phi_n(x), \quad \int d^4x\, \phi_n^\dagger(x)\phi_m(x) = \delta_{nm}. \tag{2.2.20}$$

Expand the fermion fields as

$$\psi(x) = \sum_n a_n\phi_n(x), \quad \bar{\psi}(x) = \sum_n \bar{b}_n\phi_n^\dagger(x), \tag{2.2.21}$$

so that the measure is

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi = \prod_n d\bar{b}_n da_n. \tag{2.2.22}$$

Under the chiral rotation,

$$\psi' = (1 + i\alpha\gamma^5)\psi \implies a_n' = \sum_m (\delta_{nm} + i\alpha\langle\phi_n|\gamma^5|\phi_m\rangle)a_m. \tag{2.2.23}$$

Thus the Jacobian is

$$J[\alpha] = \det\left(1 + i\alpha\gamma^5\right)^{-2} = \exp\left(-2i\int d^4x\, \alpha(x)\sum_n \phi_n^\dagger(x)\gamma^5\phi_n(x)\right), \tag{2.2.24}$$

where the factor of 2 comes from both ψ and $\bar{\psi}$.

The sum $\sum_n \phi_n^\dagger(x)\gamma^5\phi_n(x)$ is formally divergent and must be regularized. Introduce a gauge-invariant regulator:

$$\sum_n \phi_n^\dagger(x)\gamma^5\phi_n(x) \to \lim_{M\to\infty}\sum_n \phi_n^\dagger(x)\gamma^5 e^{-\lambda_n^2/M^2}\phi_n(x) = \lim_{M\to\infty} \mathrm{Tr}\left[\gamma^5 e^{-(i\not{D})^2/M^2}\delta(x-x)\right]. \tag{2.2.25}$$

$$(i\not{D})^2 = -D_\mu D^\mu - \frac{e}{2}\sigma^{\mu\nu}F_{\mu\nu}, \quad \sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]. \tag{2.2.26}$$

The key contribution comes from the term containing $F_{\mu\nu}F_{\rho\sigma}$ when expanding the exponential:

$$\mathrm{Tr}\left[\gamma^5 e^{-(i\not{D})^2/M^2}\right] \sim \frac{1}{32\pi^2}\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{2.2.27}$$

Therefore, the divergence of the axial current is

$$\partial_\mu J_5^\mu = \frac{e^2}{16\pi^2}\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{2.2.28}$$

- The Fujikawa method shows that the anomaly originates from the non-invariance of the path integral measure under chiral transformations.
- The use of a gauge-invariant regulator ensures that vector gauge symmetry is preserved while the axial symmetry is broken.
- This derivation is equivalent to the AVV triangle diagram calculation.

This anomaly has a nice mathematical interpretation: it is manifestation of Atiyah—Singer index Theorem.

$i\not{D}$ is the Dirac operator in a gauge field background and M is a regulator mass, plays a central role. Due to the anti-commutation relation $\{\gamma^5, \not{D}\} = 0$, the nonzero eigenvalues $\lambda_n \neq 0$ appear in pairs of opposite sign: if $i\not{D}\phi_n = \lambda_n\phi_n$, then $i\not{D}(\gamma^5\phi_n) = -\lambda_n(\gamma^5\phi_n)$. Consequently, the contributions of nonzero modes to the trace of γ^5 cancel each other:

$$\phi_n^\dagger\gamma^5\phi_n + (\gamma^5\phi_n)^\dagger\gamma^5(\gamma^5\phi_n) = 0.$$

In contrast, the zero modes $\lambda_n = 0$ are eigenstates of γ^5 with definite chirality, $\gamma^5\phi_0 = \pm\phi_0$, and do not have corresponding partners to cancel their contribution. Therefore, in the limit $M \to \infty$, the regulated trace receives contributions only from the zero modes:

$$\lim_{M\to\infty}\mathrm{Tr}\left[\gamma^5 e^{-(i\not{D})^2/M^2}\right] = n_+ - n_-,$$

where n_+ and n_- are the numbers of positive and negative chirality zero modes, respectively. By definition, the index of the Dirac operator is

$$\mathrm{index}(i\not{D}) = n_+ - n_-,$$

so the trace in the Fujikawa method exactly computes the index. Nonzero modes cancel due to pairing, zero modes survive, and the result is a topological invariant. After regularization and evaluating the trace, in four-dimensional gauge theory this gives

$$\mathrm{Tr}\left[\gamma^5 e^{-(i\not{D})^2/M^2}\right] \to \frac{e^2}{16\pi^2}\int d^4x\, \varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma},$$

demonstrating that the zero mode difference, i.e., the index, is directly related to the topological density $F \wedge F$ and hence to the chiral anomaly.

2.3 't Hooft anomaly(non-perturbative):

Although this is also called an anomaly, it is significantly different from the previous two. It should rather be regarded as a property of the system, rather than an inconsistency. It corresponds to the obstruction encountered when attempting to turn a global symmetry into a local symmetry (that is to say, the obstruction we encounter when we try to gauge a global symmetry). In other words, at the quantum level, the global symmetry still holds, but we cannot consistently gauge this symmetry.

Consider a QFT defined on a d-dimensional manifold M^d. Its 't Hooft anomaly for a symmetry group G is characterized as follows: we introduce a background gauge field B for the group G. Under a gauge transformation $B \to B + d\alpha$, the partition function is not invariant but instead acquires an additional phase factor:

$$Z[B + d\alpha] = \exp(i\varphi[B, \alpha])Z[B].$$

This non-invariance signifies the presence of a 't Hooft anomaly.

2.3.1 Anomaly Inflow

To address this anomaly, one can introduce an anomaly-free theory $\mathcal{T}_1$ living in one higher dimension, on a manifold N^{d+1}. This construction is known as **anomaly inflow**.

Since $\mathcal{T}_1$ is anomaly-free, its partition function $Z_1[B]$ on a manifold N^{d+1} *without* a boundary satisfies:

$$Z_1[B + d\alpha] = Z_1[B].$$

However, if N^{d+1} has a boundary such that $\partial N^{d+1} = M^d$, then the partition function transforms as:

$$Z_1[B + d\alpha] = Z_1[B] \exp[-i\varphi[B, \alpha]].$$

Now, we construct a new combined theory by placing the original anomalous theory $\mathcal{T}$ on the boundary M^d of the higher-dimensional bulk theory $\mathcal{T}_1$. The total partition function for this combined system on N^{d+1} is:

$$Z_{\text{SPT}}[B] = Z_1[B]Z[B].$$

It can be verified that this combined theory $Z_{\text{SPT}}[B]$ is gauge-invariant (i.e., anomaly-free). This process of constructing an anomaly-free theory in one higher dimension, whose boundary hosts the anomalous theory, is the essence of **anomaly inflow**.

2.3.2 Connection to Symmetry-Protected Topological Phases

The subscript SPT in $Z_{\text{SPT}}[B]$ is intentional. From a modern perspective, there is a profound connection between 't Hooft anomalies in QFT and Symmetry-Protected Topological (SPT) orders.

- A d-dimensional QFT with a 't Hooft anomaly for a global symmetry G describes the boundary physics of a (d+1)-dimensional SPT phase protected by the same symmetry G.
- Conversely, if we take a (d+1)-dimensional SPT phase with global symmetry G and place it on a manifold with a d-dimensional boundary, the boundary QFT necessarily possesses a 't Hooft anomaly for G.

In summary, there is a correspondence:

A d-dimensional QFT with a 't Hooft anomaly for symmetry G $\leftrightarrow$ A (d+1)-dimensional SPT phase with global symmet

Thus, anomaly inflow can be viewed as the process of "canceling" the anomaly of a boundary theory by coupling it to a bulk SPT phase.

2.3.3 't Hooft Anomaly Matching Condition

The 't Hooft anomaly matching condition is a fundamental constraint on Renormalization Group (RG) flows. It states that the 't Hooft anomaly must be the same in the Ultraviolet (UV, high-energy) theory and the Infrared (IR, low-energy) theory.

This implies that one cannot connect a UV theory with a non-trivial 't Hooft anomaly via RG flow to a trivial IR theory (which has no anomaly). In the context of the anomaly/SPT correspondence discussed above, this means that the boundary theory of the corresponding SPT phase must be non-trivial. Specifically, the boundary theory is either:

- **Gapless**, or
- **Gapped but with topological order** (i.e., possessing ground state degeneracy).

section gravatational anomaly weyl anomaly conformal anomaly

3 Generalized symmetry

In generalized symmetry language, a symmetry is defined as a topological operator(defects) supported on a codimension-$(p+1)$ manifold, which commutes with the local operators away from its support. This generalizes the usual notion of global symmetries to higher-form symmetries.

3.1 Extending the Landau Paradigm

We need to classify material states into two categories: those with energy gaps and those without energy gaps. Material states with energy gaps, including insulators, quantum Hall states, and others, require the injection of a finite amount of energy to generate an excitation. Materials states without energy gaps, such as superconductors, superfluids, and critical points of quantum phase transitions, may require infinitesimally small amounts of energy to generate an excitation. Both types of material states exhibit organizational forms that transcend symmetry (while also possessing organizational forms described by symmetry).

3.2 Gapped Phases

In systems with energy gaps, this novel form of many-body organization beyond symmetry is termed "topological order" (topological order is essentially a pattern of long-range entanglement), which opened up a new direction in theoretical research on condensed matter physics. The theory of higher-order fusion categories may well be the mathematical framework for describing long-range entanglement.

3.3 Spontaneous symmetry breaking

For a p-form symmetry, the unbroken phase is also when correlations of charged operators are short-ranged, and decay when the charged object grows. For a 1-form symmetry, this is when the charged loop operator exhibits an area law:

$$\langle W(C)\rangle \sim e^{-T_{p+1}\,\mathrm{Area}(C)},$$

where $\mathrm{Area}(C)$ is the area of the minimal surface bounded by the curve C. In the case of electricity and magnetism, an area law for $W_E(C)$ corresponds to the superconducting phase. The broken phase for a p-form symmetry is signaled by a failure of the expectation value of the charged operator to decay with size. For a 1-form symmetry, this is when the charged loop operator exhibits a perimeter law:

$$\langle W(C)\rangle = e^{-T_p\,\mathrm{Perimeter}(C)} + \cdots .$$

The coefficient T_p can be set to zero by modifying the definition of $W(C)$ with counterterms local to C, so Eq. (10) says that a large loop has a non-zero expectation value.

3.4 Topological Order as Symmetry Breaking

The central claim of this section is that **topological order can be understood as spontaneous breaking of higher-form symmetries**. Thus, phases previously thought to lack symmetry breaking or local order parameters can be reinterpreted as exhibiting symmetry breaking in a generalized symmetry framework. This naturally accounts for hallmark features such as topological ground-state degeneracy and anyonic excitations.

3.5 Definition of Topological Order

The article defines topological order through the following features:

1. **Locally indistinguishable degenerate ground states:** on manifolds with nontrivial topology (e.g. a torus), multiple ground states exist. Any local operator has vanishing off-diagonal matrix elements between them.
2. **Anyonic excitations:** point-like quasiparticles that cannot be created or annihilated by local operators, but only at the endpoints of extended string operators.
3. **Long-range entanglement:** the ground state wavefunction contains nonlocal entanglement patterns.

3.5.1 Example: $\mathbb{Z}_p$ Gauge Theory (Toric Code)

Setup and Symmetries.

- Degrees of freedom: a p-level system on each lattice link.
- Operators: generalized Pauli operators Z and X.
- Higher-form symmetries:
  - 1-form symmetry operator $U(M)$ acting on a closed $(D-2)$-dimensional submanifold M:
  $$U(M) = \prod_{\ell \perp M} Z_\ell.$$
  - Charged loop operator $V(C)$ acting on a closed curve C:
  $$V(C) = \prod_{\ell \in C} X_\ell.$$

Algebra and Physical Meaning. The operators satisfy

$$U^m(M)V^n(C) = e^{\frac{2\pi i}{p} mn\, \#(C,M)} V^n(C)U^m(M),$$

where $\#(C, M)$ is the intersection number of C and M. This encodes the electric–magnetic algebra of $\mathbb{Z}_p$ gauge theory and reflects the Aharonov–Bohm phase acquired when a charge winds around a flux.

Symmetry Breaking and Topological Order.

- Diagnosis: spontaneous breaking of a p-form symmetry is signaled by long-range order of charged operators. For 1-form symmetry,
  $$\langle V(C)\rangle \sim e^{-T\cdot \mathrm{Perimeter}(C)},$$
  which in the deconfined phase reduces to a perimeter law ($T = 0$).
- Consequences:
  1. **Ground-state degeneracy:** symmetry operators $U(M)$ act nontrivially on ground states, generating distinct topological sectors.
  2. **Local indistinguishability:** local operators commute with sufficiently large $U(M)$, ensuring identical expectation values in each ground state and vanishing off-diagonal elements.

Extensions

1. **Fractional quantum Hall states:** higher-form symmetry operators obey projective (anomalous) commutation relations, e.g.

$$U^m(C)U^n(C') = e^{\frac{2\pi i}{k}mn\,\#(C,C')}U^n(C')U^m(C),$$

   which enforce topological degeneracy.

2. **Non-Abelian topological order:** higher symmetries generalize to categorical (non-invertible) symmetries, with spontaneous breaking leading to non-Abelian anyons.

4 SUSY

4.1 Core Motivation: The Coleman-Mandula Theorem

We know that physical systems exhibit various symmetries, which correspond to conserved quantities (e.g., time translation symmetry corresponds to energy conservation, spatial rotation symmetry to angular momentum conservation). These symmetries collectively form the so-called *Poincaré group*.

The natural question is: beyond the Poincaré group and those "internal symmetries" that are independent of spacetime (such as charge conservation), can we have a more general type of symmetry that also involves spacetime?

In 1967, Coleman and Mandula proved that, in an interacting quantum field theory in spacetime dimensions greater than $1+1$, the answer is negative.

Any additional spacetime-related symmetry would lead to an absurd conclusion: particles would not scatter at all — the S-matrix would be the identity matrix. This is clearly inconsistent with the physical world, where interactions and scattering processes are fundamental.

Therefore, spacetime symmetry seems to be "maxed out" by the Poincaré group.

It only considers symmetries generated by **bosonic operators**, i.e., generators with integer spin (such as the momentum P^μ and the angular momentum $M^{\mu\nu}$, which transform as vectors and tensors, respectively).

The Birth of Supersymmetry

Physicists soon realized that if there exists a symmetry whose generators are **fermionic** (with half-integer spin), called *spinor charges* (denoted Q), then the Coleman-Mandula theorem no longer applies.

Supersymmetry is precisely such a symmetry — one that connects bosons and fermions. It not only evades the limitations of the Coleman-Mandula theorem, but further research (specifically the *Haag-Łopuszański-Sohnius theorem*) later showed that supersymmetry is the *only* nontrivial extension of spacetime symmetries beyond the Poincaré group that is consistent with quantum field theory.

4.2 Supersymmetric QM

The supersymmetry algebra imposes strict constraints on the structure of the energy spectrum.

Positive Energy States ($E > 0$)

For any eigenstate with energy $E_n > 0$, the supersymmetry algebra $\{Q, Q^\dagger\} = 2E_n$ reduces to the standard fermionic creation and annihilation operator algebra.

This implies that the operators Q and $Q^\dagger$ act like ladder operators: they map a bosonic state with energy E_n to a fermionic state with the same energy E_n, and vice versa.

Conclusion: In supersymmetric theories, all states with positive energy must appear in pairs: one bosonic and one fermionic state with exactly the same energy. This is known as a *supersymmetric multiplet* (or *supermultiplet*).

Zero Energy States ($E = 0$)

Zero energy states (i.e., vacuum states) are special. For these states, the algebra becomes $\{Q, Q^\dagger\} = 0$, which requires that

$$Q|0\rangle = 0 \quad \text{and} \quad Q^\dagger|0\rangle = 0.$$

Conclusion: The vacuum state is annihilated by all supersymmetry charges and is therefore a *supersymmetric singlet*. This means the vacuum does not necessarily have a supersymmetric partner—it can be either bosonic or fermionic.

Spontaneous Supersymmetry Breaking

The concept of supersymmetry breaking is key to understanding how supersymmetry may relate to the real world.

Definition:

- If the ground state (lowest energy state) has exactly zero energy ($E_{\text{ground}} = 0$), supersymmetry is said to be *unbroken*.
- If the ground state has strictly positive energy ($E_{\text{ground}} > 0$), supersymmetry is said to be *spontaneously broken*.

This definition follows directly from the non-negativity of energy. The ground state energy E_{ground} serves as the *order parameter* for supersymmetry breaking.

Motivation for Grassmann variables. In ordinary quantum mechanics, the configuration space of a particle with spin is described by wavefunctions with bosonic and fermionic components. In SUSY QM, we want a single unified description where these two types of degrees of freedom are treated on the same footing and related by supersymmetry transformations. To achieve this, we introduce *anticommuting numbers*, or *Grassmann variables*, which mimic the algebraic behavior of fermionic operators at the classical level.

Grassmann numbers θ and $\bar{\theta}$ satisfy

$$\theta^2 = \bar{\theta}^2 = 0, \qquad \theta\bar{\theta} = -\bar{\theta}\theta. \tag{4.2.1}$$

They anticommute rather than commute, and thus any function of them has only finitely many terms.

Superspace and superfields. We extend ordinary time t to a "superspace" by adding Grassmann coordinates:

$$(t, \theta, \bar{\theta}).$$

A *superfield* is a function on this superspace. For a real superfield $X(t, \theta, \bar{\theta})$, we expand it in components:

$$X(t, \theta, \bar{\theta}) = x(t) + \theta\psi(t) - \bar{\theta}\bar{\psi}(t) + \bar{\theta}\theta F(t), \tag{4.2.2}$$

where $x(t)$ is a bosonic variable, $\psi(t), \bar{\psi}(t)$ are fermionic partners, and $F(t)$ is an auxiliary field introduced to close the supersymmetry algebra off-shell.

Supercharges and covariant derivatives. Supersymmetry acts as a "translation" in the Grassmann directions. Defining

$$Q = \frac{\partial}{\partial\theta} + i\bar{\theta}\frac{\partial}{\partial t}, \qquad \bar{Q} = -\frac{\partial}{\partial\bar{\theta}} - i\theta\frac{\partial}{\partial t}, \tag{4.2.3}$$

we find the algebra

$$\{Q, \bar{Q}\} = 2i\frac{\partial}{\partial t}. \tag{4.2.4}$$

Similarly, one defines covariant derivatives

$$D = \frac{\partial}{\partial\theta} - i\bar{\theta}\frac{\partial}{\partial t}, \qquad \bar{D} = -\frac{\partial}{\partial\bar{\theta}} + i\theta\frac{\partial}{\partial t}, \tag{4.2.5}$$

which anticommute with the supercharges: $\{D, Q\} = 0 = \{\bar{D}, \bar{Q}\}$.

Supersymmetric actions. A Lagrangian constructed as an integral over full superspace,

$$S = \int dt\, d\theta\, d\bar{\theta}\ \mathcal{L}(X),$$

is automatically invariant under supersymmetry, because Grassmann integration extracts the highest component (the $\bar{\theta}\theta$ term), whose variation is a total derivative.

Superpotential and auxiliary field elimination. To construct interacting SUSY QM, we include a *superpotential* $W(X)$, a holomorphic function of the superfield. The general supersymmetric action is

$$S = \int dt\, d\theta\, d\bar{\theta}\ \left[\frac{1}{2}DX\,\bar{D}X + W(X)\right]. \tag{4.2.6}$$

Integrating over $\theta, \bar{\theta}$ gives the component action

$$S = \int dt\left[\frac{1}{2}\dot{x}^2 + i\bar{\psi}\dot{\psi} - \frac{1}{2}(W'(x))^2 - W''(x)\bar{\psi}\psi + \frac{1}{2}F^2 - FW'(x)\right]. \tag{4.2.7}$$

The auxiliary field F has no dynamics and can be eliminated via its algebraic equation of motion $F = W'(x)$, yielding the on-shell Lagrangian

$$L = \frac{1}{2}\dot{x}^2 + i\bar{\psi}\dot{\psi} - \frac{1}{2}(W'(x))^2 - W''(x)\bar{\psi}\psi. \tag{4.2.8}$$

This describes a bosonic coordinate $x(t)$ moving in the potential $V(x) = \frac{1}{2}(W'(x))^2$ coupled to a fermion $\psi(t)$ with Yukawa interaction $W''(x)\bar{\psi}\psi$.

4.2.1 Instantons in Supersymmetric Quantum Mechanics

Physical picture. An *instanton* is a classical solution in *Euclidean time* that connects two distinct classical vacua of the potential $V(x)$. In SUSY QM, these instantons correspond to tunnelling events between classically degenerate vacua, and they generate nonperturbative energy splittings.

We analytically continue to Euclidean time $t \to -i\tau$, so that the Euclidean Lagrangian becomes

$$L_E = \frac{1}{2}\dot{x}^2 + \frac{1}{2}(W'(x))^2 - \bar{\psi}\dot{\psi} - W''(x)\bar{\psi}\psi, \tag{4.2.9}$$

where the overdot now means $\frac{d}{d\tau}$.

Instanton solution. The classical Euclidean equation of motion is

$$\ddot{x} - W'(x)W''(x) = 0. \tag{4.2.10}$$

This admits a first-order ("BPS") solution satisfying

$$\dot{x}_{\rm cl} = +W'(x_{\rm cl}), \tag{4.2.11}$$

interpolating between two minima x_1, x_2 of $W'(x)$. The Euclidean action for this solution is

$$S_{\rm inst} = \int d\tau\, \dot{x}_{\rm cl} W'(x_{\rm cl}) = W(x_2) - W(x_1) \equiv \Delta W. \tag{4.2.12}$$

Fermionic path integrals and Grassmann integration. The path integral in SUSY QM now includes both bosonic and fermionic fields:

$$Z = \int \mathcal{D}x\, \mathcal{D}\psi\, \mathcal{D}\bar{\psi}\; e^{-S_E[x,\psi,\bar{\psi}]}.$$

Because the fermionic part of the action is quadratic and first order in time derivatives, integrating out fermions gives a determinant rather than its inverse:

$$\int \mathcal{D}\psi\, \mathcal{D}\bar{\psi}\; e^{-\int d\tau\, \bar{\psi} A \psi} = \det A, \tag{4.2.13}$$

where $A = \partial_\tau - W''(x_{\rm cl})$ is the Dirac-like operator governing fermion fluctuations.

For a single Grassmann variable ψ, the basic rules are

$$\int d\psi = 0, \qquad \int d\psi\, \psi = 1, \tag{4.2.14}$$

so that integrating a Gaussian fermionic action produces a determinant, and inserting fermionic fields into the path integral "soaks up" their zero modes.

Zero modes. In the instanton background, the operator A has a *fermionic zero mode*:

$$A\,\psi_0 = (\partial_\tau - W''(x_{\rm cl}))\,\psi_0 = 0 \quad \Rightarrow \quad \psi_0(\tau) \propto \dot{x}_{\rm cl}(\tau). \tag{4.2.15}$$

Physically, this corresponds to the broken supersymmetry generated by Q, since the instanton preserves only half of the supercharges.

Similarly, the bosonic fluctuation operator

$$B = -\partial_\tau^2 + (W''(x))^2 + W'(x)W'''(x)$$

has a bosonic zero mode $\delta x \propto \dot{x}_{\rm cl}$, which corresponds to the freedom of translating the instanton in Euclidean time (i.e. choosing its "center" τ_0).

Treatment of zero modes. In the previous discussion, we assumed that the indices i and j run over the same set, i.e., the number of fermionic variables ψ_i and $\bar{\psi}_j$ is equal. Suppose instead that the number of ψ_i's is greater than the number of $\bar{\psi}_j$'s. Then, by an appropriate change of basis, the operator A can be brought to the form:

$$A \to \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_N \\ & & \\ & & 0 \\ & & \vdots \end{pmatrix},$$

where $\lambda_1, \ldots, \lambda_N$ are the non-zero eigenvalues, and the remaining rows correspond to zero eigenvalues.

In this case, the path integral becomes:

$$\int e^{-S} = \int \left(\prod_{j=1}^{N} d\bar{\psi}_j d\psi_j \right) \cdot \left(\prod_{i'} d\psi_{i'} \right) e^{-S} = \det\{\}' A \cdot \left(\int \prod_{i'} d\psi_{i'} \cdot 1 \right) = 0. \tag{4.2.16}$$

The presence of unpaired fermionic zero modes causes the path integral to vanish. These unpaired ψ_i's are referred to as the *zero modes of A*.

To handle them, one introduces collective coordinates (for bosons) and inserts fermionic operators (for fermions) to absorb the zero modes. For instance, inserting the supercharge operator Q into the path integral provides the necessary Grassmann variable to integrate over:

$$\epsilon \;=\; \langle 0_- | Q | 0_+ \rangle \;\sim\; \int d\tau_0 \, d\eta_0 \; e^{-S_{\text{inst}}} \, Q_{\text{cl}}, \tag{4.2.17}$$

where η_0 is the Grassmann collective coordinate associated with the fermionic zero mode.

The integral over τ_0 accounts for the translational zero mode, and the integral over η_0 implements the fermionic zero mode. The latter vanishes unless we include exactly one insertion of a fermionic operator such as Q.

Instanton contribution to the ground state energy. After properly accounting for the zero modes and higher-order fluctuations, the leading instanton contribution to the tunnelling amplitude is

$$\epsilon \sim e^{-\Delta W/\hbar}, \qquad E_0 \sim |\epsilon|^2 \sim e^{-2\Delta W/\hbar}. \tag{4.2.18}$$

Thus, although supersymmetry forbids perturbative corrections to the ground-state energy, non-perturbative instanton effects lift the degeneracy of vacua, leading to an exponentially small supersymmetry breaking effect.

4.2.2 Why the boson and fermion determinants cancel (detailed)

We expand the Euclidean action around an instanton solution $x_{\text{cl}}(\tau)$. Write $x(\tau) = x_{\text{cl}}(\tau) + \delta x(\tau)$ and keep quadratic fluctuations. The fluctuation part of the Euclidean action (schematically) separates into bosonic and fermionic quadratic forms:

$$S_E^{(2)} = \frac{1}{2} \int d\tau \, \delta x \, B \, \delta x \;+\; \int d\tau \, \delta\bar{\psi} \, F \, \delta\psi, \tag{4.2.19}$$

where (for the SUSY QM model with $W(x)$)

$$F = \partial_\tau - W''(x_{\text{cl}}(\tau)), \tag{4.2.20}$$

$$B = -\partial_\tau^2 + \big(W''(x_{\text{cl}})\big)^2 + W'(x_{\text{cl}}) \, W'''(x_{\text{cl}}). \tag{4.2.21}$$

A useful factorization identity holds (verify by direct multiplication):

$$B = F^\dagger F, \tag{4.2.22}$$

where $F^\dagger = -\partial_\tau - W''(x_{\text{cl}}(\tau))$ is the adjoint of F with respect to the standard L^2 inner product on functions of τ. (One checks $F^\dagger F = (-\partial_\tau - W'')(\partial_\tau - W'') = -\partial_\tau^2 + (W'')^2 + W'W'''$.)

Now perform Gaussian path integrals for fluctuations (formally):

- The fermionic (Grassmann) Gaussian integral gives

$$\int \mathcal{D}\delta\psi \, \mathcal{D}\delta\bar{\psi} \; e^{-\int \delta\bar{\psi} F \delta\psi} \;=\; \det F.$$

- The bosonic Gaussian integral gives

$$\int \mathcal{D}\delta x \; e^{-\frac{1}{2}\int \delta x \, B \, \delta x} \;=\; (\det B)^{-1/2}.$$

Using the factorisation (4.2.22) we obtain (formally, ignoring zero modes for now)

$$\det B = \det\big(F^\dagger F\big) = \det F^\dagger \cdot \det F. \tag{4.2.23}$$

Therefore

$$\frac{\det F}{\sqrt{\det B}} \;=\; \frac{\det F}{\sqrt{\det F^\dagger \det F}} \;=\; \frac{\det F}{\sqrt{\det F^\dagger}\sqrt{\det F}} \;=\; \frac{\sqrt{\det F}}{\sqrt{\det F^\dagger}}. \tag{4.2.24}$$

If F has a real, positive spectrum or if we choose a conjugation/regularization such that $\det F$ and $\det F^\dagger$ have the same absolute value, then the absolute value of the prefactor is unity:

$$\left|\frac{\det F}{\sqrt{\det B}}\right| = 1,$$

i.e. bosonic and fermionic fluctuations cancel in magnitude. Any remaining mismatch is a phase (or sign) coming from careful regularization of determinants; in supersymmetric regularizations that preserve the matching between boson and fermion modes this phase can be shown to be trivial (or cancels when summing instanton + anti-instanton contributions appropriately).

A more transparent spectral argument is the following. Suppose $\{\xi_n\}$ is an orthonormal basis of eigenfunctions of B with eigenvalues $\lambda_n > 0$:

$$B\,\xi_n = \lambda_n\,\xi_n.$$

Define

$$\eta_n \equiv \frac{1}{\sqrt{\lambda_n}}\,F\,\xi_n.$$

Then (using $F^\dagger F = B$)

$$F\,\xi_n = \sqrt{\lambda_n}\,\eta_n, \qquad F^\dagger \eta_n = \sqrt{\lambda_n}\,\xi_n,$$

so η_n is (up to normalization) an eigenfunction of $FF^\dagger$ with the same eigenvalue λ_n. Thus the nonzero eigenvalues of B are squares of the nonzero singular values of F, and every nonzero bosonic eigenvalue λ_n corresponds to a pair of fermionic eigenvalues $\pm\sqrt{\lambda_n}$ in the first-order fermionic problem. This one-to-one pairing of nonzero modes is the spectral origin of the cancellation.

Zero modes and the prime determinants. The previous manipulations assumed the operators have no zero eigenvalues. In an instanton background both B and F possess zero modes:

- Bosonic zero mode: $\delta x_0(\tau) \propto \dot{x}_{\rm cl}(\tau)$ (translation of the instanton center).
- Fermionic zero mode: $\psi_0(\tau) \propto \dot{x}_{\rm cl}(\tau)$ (broken supersymmetry mode).

When zero modes exist we must remove them from the determinants and treat them separately. Denote by $\det'(\cdot)$ the determinant with zero eigenvalues omitted. Then the fluctuation contribution becomes

$$\frac{\det' F}{\sqrt{\det' B}}.$$

Using $B = F^\dagger F$ restricted to the nonzero-mode subspace one again finds

$$\det{}' B = \det{}'(F^\dagger F) = \det{}' F^\dagger \cdot \det{}' F,$$

so the nonzero-mode determinants cancel as before:

$$\frac{\det' F}{\sqrt{\det' B}} = \frac{\det' F}{\sqrt{\det' F^\dagger \det' F}} = \frac{\sqrt{\det' F}}{\sqrt{\det' F^\dagger}} \quad \Rightarrow \quad \big|\text{fluctuation prefactor}\big| = 1.$$

The remaining integrals are the explicit integrals over the collective coordinates associated with the zero modes:

- For the bosonic zero mode we introduce the collective coordinate τ_0 (the instanton center), and change variables in the path integral so that the δx_0 direction is replaced by an integral $\int d\tau_0$ with a Jacobian (the so-called zero-mode normalization factor).
- For the fermionic zero mode we must supply a Grassmann insertion to get a nonzero result: without inserting a fermionic operator the Grassmann integral over the zero mode vanishes. Typically one inserts the supercharge Q or a fermionic operator whose leading component overlaps the zero mode, so the Grassmann integral $\int d\eta_0\, \eta_0$ is nonzero.

Putting everything together (nonzero-mode determinants cancelling, zero modes treated explicitly), the leading instanton contribution to an amplitude with the correct fermionic insertions is proportional to

$$e^{-S_{\text{inst}}} \times \underbrace{\left(\text{zero-mode normalizations and } \int d\tau_0\, d\eta_0\right)}_{\text{finite and computable}}.$$

Summary.

- Grassmann variables allow fermions to be treated as classical fields in superspace.
- Supersymmetry transformations act as translations in this superspace.
- The superpotential $W(X)$ encodes all interactions in a manifestly supersymmetric form.
- Instantons are finite-action Euclidean solutions connecting distinct vacua.
- Bosonic and fermionic zero modes correspond to broken symmetries (translation and Q-supersymmetry).
- Integration over Grassmann zero modes requires inserting fermionic operators; this yields nonzero amplitudes.
- The instanton contribution $E_0 \sim e^{-2\Delta W/\hbar}$ demonstrates nonperturbative SUSY breaking.

4.3 Perturbative SUSY

Core Difference: Particles vs. Fields

Understanding the distinction between **particles** and **fields** is the first step in representation theory.

Particles

Particles are the asymptotic, free states we detect in experiments. They live in a Hilbert space. Classification is given by **Wigner's method of little groups**:

- **Massive particles:** We can go to their rest frame. The remaining symmetry is spatial rotations, $SO(3)$. Particles are classified by their **spin** j. $p^\mu = (m, 0, 0, 0)$
- **Massless particles:** We can never reach their rest frame; only rotations around the momentum direction remain, $SO(2)$. Particles are classified by their **helicity** λ. $p^\mu = (E, 0, 0, E)$

Fields

Fields are mathematical objects used to write Lagrangians. They are functions of spacetime, e.g. $\phi(x)$. Classification is via **finite-dimensional representations of the Lorentz group** $SO(3,1)$.

A useful decomposition:

$$SO(3,1) \cong SU(2)_L \times SU(2)_R$$

A field is labeled by (j_L, j_R), the "spins" under $SU(2)_L$ and $SU(2)_R$.

- Scalar: $(0, 0)$
- Left-handed spinor ψ_α: $(1/2, 0)$
- Right-handed spinor $\bar{\psi}_{\dot{\alpha}}$: $(0, 1/2)$
- Vector A_μ: $(1/2, 1/2)$

Spinor Algebra: The "Language" of Supersymmetry

Spinor indices

- $\alpha, \beta = 1, 2$ for $SU(2)_L$ (left-handed spinors)
- $\dot{\alpha}, \dot{\beta} = 1, 2$ for $SU(2)_R$ (right-handed spinors)

Conjugation

Complex conjugation maps left-handed spinors to right-handed spinors:

$$(\psi_\alpha)^* = \bar{\psi}_{\dot{\alpha}}$$

Constructing Lorentz invariants

- Two left-handed spinors ψ, ξ: can form a scalar $\psi\xi$
- Left-handed and right-handed spinor $\psi, \bar{\chi}$: can form a vector

σ-matrices

The σ matrices $\sigma^\mu_{\alpha\dot{\alpha}}$ act as a dictionary between $(1/2, 1/2)$ spinors and vectors A_μ. Example in the SUSY algebra:

$$\{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} \propto \sigma^\mu_{\alpha\dot{\alpha}} P_\mu$$

Constructing Supersymmetric Multiplets

Massive multiplets

In the rest frame, the SUSY algebra simplifies to a set of creation/annihilation operators. Starting from a particle of spin j, one can generate up to three new partner states. The resulting multiplet contains four states with spins:

$$j, j, j + 1/2, j - 1/2$$

Example: Starting from a scalar $j = 0$, we obtain a multiplet containing two scalars and one spin-1/2 fermion. This is called a **massive chiral multiplet**.

Massless multiplets

In a massless reference frame, the SUSY algebra reduces to a single creation/annihilation operator. From a given particle state, one can generate only one new partner state, with helicity shifted by 1/2.

Examples:

- Scalar ($\lambda = 0$) $\rightarrow$ spin-1/2 fermion: chiral multiplet $(0, 1/2)$
- Spin-1/2 fermion ($\lambda = 1/2$) $\rightarrow$ spin-1 boson: vector multiplet $(1/2, 1)$

  The fermionic superpartner of a gauge boson is called a gaugino,
  and the fermionic superpartner of a Higgs boson is called a Higgsino.

4.3.1 $\mathcal{N} = 1$ Superspace and Chiral Superfields

This lecture extends supersymmetry from quantum mechanics to four-dimensional field theory, introducing two essential tools: **Kähler potential** and **superpotential**.

Superspace

Fields form representations of the supersymmetry algebra, and it is most convenient to handle them using superspace.

It should be noted that superspace works very well for 4D $\mathcal{N} = 1$ supersymmetry and "lower" supersymmetry, but is less effective for "extended" supersymmetry in four dimensions or for supersymmetry in higher dimensions.

We will mostly treat superspace as a convenient tool, and whenever component notation is easier, we will use components directly.

We extend spacetime by including Grassmann spinor coordinates, one for each supersymmetry charge:

$$x^\mu \to (x^\mu, \theta_\alpha, \bar{\theta}_{\dot{\alpha}}).$$

Supertranslations are defined as

$$(x^\mu, \theta_\alpha, \bar{\theta}_{\dot{\alpha}}) \to \Big(x^\mu + i\theta\sigma^\mu\bar{\xi} - i\xi\sigma^\mu\bar{\theta},\ \theta_\alpha + \xi_\alpha,\ \bar{\theta}_{\dot{\alpha}} + \bar{\xi}_{\dot{\alpha}}\Big).$$

The differential operators generating supertranslations on superspace are

$$Q_\alpha = +\frac{\partial}{\partial\theta^\alpha} - i(\sigma^\mu\bar{\theta})_\alpha\partial_\mu, \quad \bar{Q}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} + i(\theta\sigma^\mu)_{\dot{\alpha}}\partial_\mu.$$

$$\{Q_\alpha, Q_\beta\} = 0, \quad \{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} = 2i(\sigma^\mu\partial_\mu)_{\alpha\dot{\alpha}}.$$

We also define **covariant derivatives**:

$$D_\alpha = +\frac{\partial}{\partial\theta^\alpha} + i(\sigma^\mu\bar{\theta})_\alpha\partial_\mu, \quad \bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} - i(\theta\sigma^\mu)_{\dot{\alpha}}\partial_\mu.$$

The difference from Q is a relative sign. They satisfy the opposite supersymmetry algebra:

$$\{D_\alpha, \bar{D}_{\dot{\alpha}}\} = -2i(\sigma^\mu\partial_\mu)_{\alpha\dot{\alpha}},$$

and they anticommute with the supersymmetry generators:

$$\{D_\alpha, Q_\beta\} = \{\bar{D}_{\dot{\alpha}}, Q_\beta\} = 0,$$

just like in supersymmetric quantum mechanics.

Chiral Superfields

A **superfield** is a function on superspace, by definition supersymmetric-covariant.

A general superfield has too many component fields and forms a reducible representation of the supersymmetry algebra:

$$\Phi(x, \theta, \bar{\theta}) = \phi + \theta\psi + \bar{\theta}\bar{\chi} + \theta^2 F + \bar{\theta}^2 G + \theta\sigma^\mu\bar{\theta}A_\mu + \dots$$

To obtain an **irreducible** representation of the superalgebra, we must impose constraints on the superfield that (anti)commute with the supersymmetry algebra.

One important constraint is the **chiral superfield** constraint:

$$\bar{D}_{\dot{\alpha}}\Phi = 0.$$

This is consistent because $\bar{D}$ anticommutes with Q and $\bar{Q}$. One can explicitly solve this constraint to show it gives an irreducible representation of the supersymmetry algebra.

Therefore, a general chiral superfield can be written as

$$\Phi = \Phi(y, \theta) = \phi(y) + \sqrt{2}\theta\psi(y) + \theta^2 F(y),$$

with

$$y^\mu \equiv x^\mu + i\theta\sigma^\mu\bar{\theta}.$$

Now it contains only fields of spin j and $j \pm 1/2$, matching an irreducible supersymmetric multiplet. The F field is always non-propagating.

Anti-chiral superfield is similarly defined as

$$D_\alpha \bar{\Phi} = 0.$$

Kähler Potential

A supersymmetric-invariant Lagrangian can be written as

$$\mathcal{L} = \int d^4\theta \; K(\Phi^i, \bar{\Phi}^j, \dots),$$

which is automatically supersymmetric because $\delta_Q \mathcal{L}$ is a total derivative, just as in supersymmetric quantum mechanics.

Note: a term consisting only of chiral superfields,

$$\int d^4\theta \; f(\Phi)$$

is automatically a total derivative. Therefore, K is defined only up to

$$K \sim K + f(\Phi) + \bar{f}(\bar{\Phi}).$$

The simplest nontrivial example is

$$K = \bar{\Phi}\Phi,$$

which gives the free Lagrangian

$$\mathcal{L} = \int d^4\theta \; \bar{\Phi}\Phi = \bar{F}F - \partial_\mu\bar{\phi}\partial^\mu\phi + i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi.$$

This describes a free complex boson and a free Weyl fermion. More generally, $K = K(\Phi^i, \bar{\Phi}^j)$ defines a supersymmetric nonlinear sigma model, the target space is a curved Kähler manifold. A metric is defined on the "field space":

$$g_{i\bar{j}} \equiv \partial_i \partial_{\bar{j}} K(\phi, \bar{\phi}), \quad \partial_i \equiv \frac{\partial}{\partial \phi^i},$$

so that the complex scalar fields play the role of holomorphic coordinates on the target space. The geometry of this space is Kähler.

Constructing Supersymmetric Lagrangians

Two complementary methods:

1. **Kähler potential $K(\Phi, \bar{\Phi})$ (D-term)**
   - Take a real function $K(\Phi, \bar{\Phi})$ and integrate over full superspace:

$$\mathcal{L} = \int d^4\theta \; K(\Phi, \bar{\Phi}).$$

- Produces kinetic terms for scalar and fermion fields.
- Simple example: $K = \bar{\Phi}\Phi$ gives

$$\mathcal{L} = \bar{F}F - \partial_\mu \bar{\phi}\partial^\mu \phi + i\bar{\psi}\bar{\sigma}^\mu \partial_\mu \psi.$$

- **Geometric interpretation:** K defines the geometry of the scalar field space, a Kähler manifold. The kinetic terms reflect the metric

$$g_{i\bar{j}} \equiv \partial_i \partial_{\bar{j}} K(\phi, \bar{\phi}), \quad \partial_i = \frac{\partial}{\partial \phi^i}.$$

2. **Superpotential $W(\Phi)$ (F-term)**
   - Take a holomorphic function $W(\Phi)$ and integrate over half of superspace:

$$\mathcal{L}_W = \int d^2\theta \; W(\Phi) + h.c.$$

   - Generates non-derivative interactions: scalar potential, fermion masses, Yukawa couplings.
   - Component form:

$$\mathcal{L}_W = \frac{\partial W}{\partial \phi^i} F^i - \frac{1}{2} \frac{\partial^2 W}{\partial \phi^i \partial \phi^j} \psi^i \psi^j + h.c.$$

   - Example: $W = m\Phi^2 + y\Phi^3$ produces
     - Fermion mass term: $m\psi\psi$,
     - Yukawa coupling: $y\phi\psi\psi$,
     - Scalar potential: $V = |m\phi + y\phi^2|^2$.

Scalar Potential and Supersymmetric Vacua

- **Auxiliary field F:** Solved from its equation of motion:

$$F^i = -g^{i\bar{j}} \frac{\partial \bar{W}}{\partial \bar{\phi}^{\bar{j}}}.$$

- **Scalar potential:**

$$V(\phi, \bar{\phi}) = g^{i\bar{j}} \frac{\partial W}{\partial \phi^i} \frac{\partial \bar{W}}{\partial \bar{\phi}^{\bar{j}}} \geq 0.$$

- **Supersymmetric vacuum:** $V = 0$ iff

$$\frac{\partial W}{\partial \phi^i} = 0 \quad \forall i.$$

  Finding SUSY vacua reduces to solving simple algebraic equations for ϕ^i.

- **F-term contribution:** The part of the scalar potential originating from W is called the F-term. (D-term contributions arise when vector multiplets are included.)

Summary

- Chiral superfields Φ elegantly package a complex scalar, a Weyl fermion, and an auxiliary field into an irreducible $\mathcal{N} = 1$ SUSY multiplet.
- Supersymmetric Lagrangians can be constructed using:
  - **Kähler potential K (D-term)** $\rightarrow$ kinetic terms, geometry of field space.
  - **Superpotential W (F-term)** $\rightarrow$ scalar potential and interactions.
- Scalar potential is non-negative and SUSY vacua are determined by $\partial_i W = 0$.
- Together, K and W provide the complete framework for constructing $\mathcal{N} = 1$ SUSY models.

Wess-Zumino Model

The Wess-Zumino (WZ) model is the simplest interacting and renormalizable $\mathcal{N} = 1$ SUSY theory built from chiral superfields Φ^i. It consists of two main ingredients:

1. **Canonical Kähler Potential:**
$$K = \sum_i \bar{\Phi}^i \Phi^i$$
This gives the free kinetic terms for all chiral superfields. It corresponds to a "flat" field space geometry and sets the standard normalization of Φ^i.

2. **Most General Renormalizable Superpotential:**
$$W(\Phi) = \nu_i \Phi^i + \mu_{ij} \Phi^i \Phi^j + \lambda_{ijk} \Phi^i \Phi^j \Phi^k$$
   - "Renormalizable" means only operators with mass dimension ≤ 4 are allowed. Since $\int d^2\theta$ has dimension 1 and $[\Phi] = 1$, $W(\Phi)$ can be at most cubic in Φ.
   - ν_i: linear term (sometimes removable by shifting Φ).
   - μ_{ij}: quadratic term, generates mass terms for scalars and fermions.
   - λ_{ijk}: cubic term, generates Yukawa couplings ($\lambda\phi\psi\psi$) and scalar potential terms (cubic and quartic).
   - ν, μ, λ are free complex coupling constants.

Note: The dimension analysis in the later part compares the renormalizable WZ model with more general non-linear sigma models (nlσm). The WZ model describes fundamental interactions at high energies, while nlσm serves as an effective low-energy theory.

4.3.2 Selection rules and normalization theorem

The core point of this lecture is to prove a remarkable result: in supersymmetric theories, the superpotential W is not affected by any quantum corrections (at least perturbatively, and in some cases even non-perturbatively). This is known as the **non-renormalization theorem**.

4.3.3 R-symmetry: a special global symmetry

Definition: In addition to ordinary global symmetries (e.g., changing the phase of a field without affecting spacetime), supersymmetric theories allow for a special $U(1)$ symmetry called **R-symmetry**. Its special feature is that it does not commute with the supercharges Q.

Charges:

- The supercharge Q carries R-charge -1.
- The Grassmann coordinate θ carries R-charge $+1$.

This implies that different components of a superfield Φ (i.e., ϕ, ψ, F) have different R-charges.

Constraint on the superpotential: Since $\int d^2\theta$ itself carries R-charge -2 (because $d\theta$ has charge -1), in order for $\int d^2\theta\, W(\Phi)$ to be R-invariant, the superpotential $W(\Phi)$ must carry R-charge $+2$. This simple rule is extremely useful in practice.

Core argument: holomorphy of the superpotential = selection rule

This is the key step in understanding the non-renormalization theorem.

Treat couplings as background fields: The lecture introduces a brilliant idea: treat all parameters appearing in the superpotential W (such as masses m and couplings λ) as background chiral superfields M and Λ. For example,

$$W = \frac{1}{2}M\Phi^2 + \frac{1}{3}\Lambda\Phi^3.$$

Requirement from supersymmetry: We know that the superpotential W must be holomorphic; that is, it can only depend on the chiral superfields Φ and not on their anti-chiral conjugates $\bar{\Phi}$.

Extension to couplings: Since we now treat m and λ as part of chiral superfields M and Λ, any quantum corrections to the effective superpotential W_{eff} must also depend holomorphically on m and λ, and cannot depend on their complex conjugates m^* or λ^*.

Analogy: selection rules: This is similar to selection rules in quantum mechanics. For example, in the Stark effect (splitting of atomic energy levels in an electric field), the energy corrections cannot arbitrarily depend on the electric field E, because E is a vector and only specific combinations (like E^2, $(\vec{L}\cdot\vec{E})^2$) are allowed by rotational symmetry Similarly, the holomorphy of the superpotential is a selection rule coming from supersymmetry: quantum corrections can only depend on the couplings in a holomorphic way.

4.3.4 Proof of the non-renormalization theorem

Now, by combining the holomorphy selection rule with ordinary symmetries and the behavior of the theory in specific limits, the power of the argument becomes clear.

Example: Consider

$$W = \frac{1}{2}m\Phi^2 + \frac{1}{3}\lambda\Phi^3.$$

Holomorphy: The effective superpotential W_{eff} must be a holomorphic function of Φ, m, and λ, say $f(\Phi, m, \lambda)$.

Ordinary symmetries $U(1)$ and R-symmetry: By assigning appropriate charges to m and λ (so that W has R-charge $+2$), one can constrain the form of W_{eff} to be

$$W_{\text{eff}} = m\Phi^2 f\Big(\frac{\lambda\Phi}{m}\Big).$$

Weak coupling limit (smoothness):

- When $\lambda \to 0$, the theory becomes free, and we know W_{eff} should reduce to $\frac{1}{2}m\Phi^2$. This requires that $f(x)$ near $x = 0$ cannot have negative powers (i.e., $f(x) = a_0 + a_1 x + \dots$).
- When $m \to 0$ (while keeping λ fixed or $\lambda\Phi/m \to 0$), the theory should also be smooth (no divergences). This further requires that $f(x)$ cannot contain x^2 or higher powers, i.e., $f(x)$ can be at most $a_0 + a_1 x$.

Conclusion: Combining all constraints, the only possible functional form is

$$W_{\text{eff}} = a_0 m\Phi^2 + a_1\lambda\Phi^3.$$

By matching to the tree-level result, we determine $a_0 = 1/2$, $a_1 = 1/3$. Hence,

$$W_{\text{eff}} = W_{\text{micro}}.$$

The superpotential receives **no quantum corrections**!

4.3.5 vector superfield and super QED

Abelian vector superfield

A vector superfield $V(x, \theta, \bar{\theta})$ is a real scalar superfield,

$$V = V^\dagger,$$

whose component expansion (general gauge) reads

$$V = B + \theta\chi + \bar{\theta}\bar{\chi} + \theta^2 C + \bar{\theta}^2 C^\dagger - \theta\sigma^\mu\bar{\theta}\, A_\mu$$
$$+ i\theta^2\bar{\theta}\big(\bar{\lambda} + \tfrac{1}{2}\bar{\sigma}^\mu\partial_\mu\chi\big) - i\bar{\theta}^2\theta\big(\lambda + \tfrac{1}{2}\sigma^\mu\partial_\mu\bar{\chi}\big) + \tfrac{1}{2}\theta^2\bar{\theta}^2(D + \partial^2 B),$$

where A_μ (real) is the gauge field, λ_α the gaugino (photino), D an auxiliary real scalar, and B, χ, C are gauge artifacts in the extended superspace gauge symmetry.

The superspace gauge transformation is

$$V \;\longrightarrow\; V + i(\widehat{\Lambda} - \widehat{\Lambda}^\dagger), \tag{4.3.1}$$

with $\widehat{\Lambda}$ an arbitrary chiral superfield gauge parameter. Under (4.3.1) the vector component transforms as

$$\delta A_\mu = \partial_\mu(\Lambda + \Lambda^\dagger),$$

so the real part of Λ is the usual U(1) gauge parameter. The larger complexified gauge freedom implies that B, χ, C are gauge degrees of freedom while A_μ, λ, D are gauge-invariant (physical) components.

Consider a charged field $\phi(x)$ with charge q and a gauge field $A_\mu(x)$ (e.g. the electromagnetic potential). A standard $U(1)$ gauge transformation is parameterized by a real function $\alpha(x)$:

$$\phi(x) \;\to\; e^{-iq\alpha(x)}\,\phi(x), \qquad A_\mu(x) \;\to\; A_\mu(x) + \partial_\mu\alpha(x).$$

The key point is that the parameter $\alpha(x)$ is real. For non-Abelian gauge groups (e.g. $SU(N)$), the parameters are real linear combinations of Lie algebra generators. Thus, these transformations correspond to a *real* Lie group G such as $U(1)$ or $SU(N)$.

Supersymmetric Gauge Transformations

In a supersymmetric theory, the gauge field A_μ is embedded in a vector superfield V, while matter fields are described by chiral superfields Φ. The supersymmetric gauge transformation takes the form

$$V \;\to\; V + i(\Lambda - \overline{\Lambda}),$$

where Λ is a *chiral superfield*. The lowest component of Λ is a complex scalar field,

$$\Lambda_0(x) = \alpha(x) + i\beta(x),$$

with $\alpha(x)$ and $\beta(x)$ real.

Let us now examine how the components of V transform under this gauge transformation:

- The physical gauge field A_μ transforms as

  $$\delta A_\mu = \partial_\mu(\Lambda_0 + \overline{\Lambda}_0) = \partial_\mu\big((\alpha + i\beta) + (\alpha - i\beta)\big) = \partial_\mu(2\alpha).$$

  Thus, A_μ depends only on the *real part* $\alpha(x)$, just like an ordinary gauge transformation with parameter $2\alpha(x)$.

- Other components (such as the auxiliary field B) transform as

  $$\delta B = i(\Lambda_0 - \overline{\Lambda}_0) = i\big((\alpha + i\beta) - (\alpha - i\beta)\big) = -2\beta.$$

  Hence, fields like B transform under the *imaginary part* $\beta(x)$.

Complexified Gauge Group $G_{\mathbb{C}}$

For an ordinary real Lie group G (e.g. $U(1)$ or $SU(N)$), the gauge transformation is generated by real parameters $\alpha^a(x)$:

$$g(x) = \exp\big(i\,\alpha^a(x)T^a\big),$$

where T^a are the generators of the Lie algebra $\mathfrak{g}$.

The *complexified* gauge group $G_{\mathbb{C}}$ (e.g. $U(1)_{\mathbb{C}} = GL(1,\mathbb{C})$ or $SU(N)_{\mathbb{C}} = SL(N,\mathbb{C})$) allows complex parameters:

$$g(x) = \exp\big(i(\alpha^a(x) + i\beta^a(x))T^a\big).$$

Therefore, $G_{\mathbb{C}}$ effectively doubles the number of real parameters, including both the transformations generated by α^a and those generated by β^a.

Since the supersymmetric gauge transformation

$$V \;\to\; V + i(\Lambda - \overline{\Lambda})$$

is governed by a complex parameter $\Lambda_0 = \alpha + i\beta$, and different components of V respond separately to α and β, the full structure of the transformation exhibits the characteristics of a *complexified gauge group* $G_{\mathbb{C}}$, rather than merely the real gauge group G.

In other words, the supersymmetric framework naturally extends the gauge symmetry from the real group G to its complexification $G_{\mathbb{C}}$.

Wess–Zumino (WZ) gauge (practical choice)

One often fixes the superspace gauge to Wess–Zumino gauge:

$$B = \chi = C = 0,$$

so that

$$V_{\mathrm{WZ}} = -\theta\sigma^\mu\bar{\theta}\,A_\mu + i\theta^2\bar{\theta}\,\bar{\lambda} - i\bar{\theta}^2\theta\,\lambda + \tfrac{1}{2}\theta^2\bar{\theta}^2 D. \tag{4.3.2}$$

The Wess–Zumino (WZ) gauge is an *artificially chosen gauge-fixing condition.* Its purpose is to use the "extra" degrees of freedom available in the complexified gauge group $G_{\mathbb{C}}$—in particular, those associated with the imaginary part of Λ—to set the non-physical component fields B, χ, and C to zero by hand. Finally, the *Wess–Zumino (WZ) gauge* plays a crucial role: thanks to the larger $G_{\mathbb{C}}$ symmetry, one can use the extra transformations (parameterized by β) to eliminate non-physical components of V (such as B, χ, C), leaving precisely the physical gauge freedom corresponding to the real group G (parameterized by α).

Field-strength chiral superfield W_α

Define the (Abelian) field-strength chiral superfield

$$W_\alpha \;=\; -\tfrac{1}{4}\,\bar{D}^2\,D_\alpha V, \tag{4.3.3}$$

which satisfies $\bar{D}_{\dot{\alpha}}W_\alpha = 0$ (i.e. W_α is chiral). One checks that W_α is invariant under the gauge variation (4.3.1). In WZ gauge its component expansion is

$$W_\alpha = -i\,\lambda_\alpha(y) + \big[\delta_\alpha{}^\beta\,D - \tfrac{i}{2}(\sigma^{\mu\nu})_\alpha{}^\beta F_{\mu\nu}\big]\,\theta_\beta + (\sigma^\mu\partial_\mu\bar{\lambda})_\alpha\,\theta^2, \tag{4.3.4}$$

so the lowest components encode the gaugino λ, the auxiliary D, and the field strength $F_{\mu\nu}$.

The superspace Bianchi identity reads

$$D^\alpha W_\alpha - \bar{D}_{\dot{\alpha}}\bar{W}^{\dot{\alpha}} = 0.$$

Pure U(1) gauge action (kinetic term and θ-term)

The supersymmetric gauge kinetic action is written as a chiral superspace integral:

$$\mathcal{L}_{\mathrm{gauge}} \;=\; \int d^2\theta\,\Big(-\tfrac{i\tau}{16\pi}\Big)W^\alpha W_\alpha \;+\; \text{h.c.}, \qquad \tau \equiv \frac{\vartheta}{2\pi} + i\frac{4\pi}{g^2}. \tag{4.3.5}$$

Expanding to components (in WZ gauge) reproduces the usual Maxwell term, gaugino kinetic term, the auxiliary D^2 term, and the topological ϑ-term:

$$\mathcal{L}_{\text{gauge}} \supset -\tfrac{1}{4g^2} F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{2g^2} D^2 - \tfrac{i}{g^2}\, \bar{\lambda}\bar{\sigma}^\mu \partial_\mu \lambda + \frac{\vartheta}{32\pi^2} F_{\mu\nu}\tilde{F}^{\mu\nu}.$$

Important point: the gauge kinetic function is holomorphic (encoded in τ), and writing the kinetic term as an F-term (chiral integral) leads to powerful holomorphy constraints in quantum theory.

Fayet–Iliopoulos term (Abelian only)

For an Abelian vector multiplet one may add the Fayet–Iliopoulos (FI) term:

$$\mathcal{L}_{\text{FI}} = \int d^4\theta\, (2\kappa\, V) \;=\; \kappa\, D, \tag{4.3.6}$$

which shifts the D-term by a constant and can be a source of D-term supersymmetry breaking. Equivalently (using the Bianchi identity) the FI term can be written (up to total derivatives) as a chiral integral proportional to $\int d^2\theta\, \kappa\, W^\alpha$. Note: FI terms are gauge-invariant only for Abelian factors.

Coupling to chiral superfields — superQED

Now we can combine the chiral superfields (matter fields Φ) and the vector superfield (gauge field V) to construct the full supersymmetric Lagrangian.

Gauge Kinetic Term

$$\mathcal{L}_{\text{kin}} = \int d^2\theta\, \frac{\tau}{16\pi i}\, W^\alpha W_\alpha + \text{h.c.}$$

Here, the complex coupling τ contains both the usual gauge coupling constant g and the topological θ-angle:

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2}.$$

Matter Kinetic Term (Minimal Coupling)

$$\mathcal{L}_{\text{matter}} = \sum_i \int d^4\theta\, \overline{\Phi}_i\, e^{q_i V}\, \Phi^i.$$

The factor $e^{q_i V}$ is the supersymmetric analogue of the gauge-covariant derivative. It ensures gauge invariance of the kinetic term under supersymmetric gauge transformations.

Superpotential Term

$$\mathcal{L}_W = \int d^2\theta\, \mathcal{W}(\Phi^i) + \text{h.c.}$$

The superpotential $\mathcal{W}(\Phi)$ must be gauge invariant, which implies that in every term of $\mathcal{W}$, the sum of the charges q_i of all the involved fields must vanish.

Fayet–Iliopoulos (FI) Term (Abelian Case Only)

$$\mathcal{L}_{\text{FI}} = \int d^4\theta\, (2\xi V) = \xi D.$$

This term is allowed only in $U(1)$ gauge theories, where ξ is a real constant parameter.

D-term Potential: New Vacuum Constraint

This is the most important new feature that arises after introducing gauge interactions.

Auxiliary Field D

The vector superfield V contains a real auxiliary field D.

Eliminating D

Just as we eliminate the auxiliary field F in chiral superfields, we can eliminate D by solving its (algebraic) equation of motion. The equation of motion for D receives contributions from:

- the gauge kinetic term, which gives a term proportional to g^2/D;
- the matter coupling term, which gives $\sum_i q_i|\phi_i|^2$;
- the Fayet–Iliopoulos (FI) term, which contributes ξ.

D-term Potential

After solving for D and substituting back into the Lagrangian, we obtain an additional positive-definite contribution to the scalar potential, known as the *D-term potential*:

$$V_D = \frac{g^2}{2}\left(\xi + \sum_i q_i|\phi_i|^2\right)^2 .$$

Total Scalar Potential

The total scalar potential is the sum of the F-term and D-term contributions:

$$V = V_F + V_D = \sum_i \left|\frac{\partial W}{\partial \phi_i}\right|^2 + \frac{g^2}{2}\left(\xi + \sum_i q_i|\phi_i|^2\right)^2 .$$

New Vacuum Conditions

To find a supersymmetric vacuum with zero energy, both the F-term and D-term equations must be satisfied simultaneously:

$$\text{F-term:} \quad \frac{\partial W}{\partial \phi_i} = 0, \qquad \text{D-term:} \quad \xi + \sum_i q_i|\phi_i|^2 = 0.$$

After introducing gauge interactions (via the vector superfield V), a supersymmetric vacuum must satisfy not only the F-term constraints arising from the superpotential W, but also the D-term constraints coming from the gauge interactions. The D-term constraint plays a crucial role in determining the vacuum structure (moduli space) of the theory, and the presence of the FI term provides a possible mechanism for supersymmetry breaking.

Non-Abelian vector multiplets (brief)

For a non-Abelian gauge group G with hermitian generators T^a, the vector superfield is $V = T^a V^a$ and transforms as

$$e^V \longrightarrow e^{-i\widehat{\Lambda}^\dagger}\, e^V\, e^{i\widehat{\Lambda}},$$

so the natural covariantization of (4.3.3) is

$$W_\alpha \;=\; -\tfrac{1}{4}\,\bar{D}^2\big(e^{-V}D_\alpha e^{V}\big),$$

which transforms in the adjoint: $W_\alpha \to e^{-i\widehat{\Lambda}}W_\alpha e^{i\widehat{\Lambda}}$. The gauge kinetic term is $\int d^2\theta\,\mathrm{tr}\big(W^\alpha W_\alpha\big)$. Matter in representation R couples via $\int d^4\theta\,\Phi^\dagger e^{V_R}\Phi$. Non-Abelian structure introduces covariant derivatives, structure constants in component interactions, and the usual gauge self-interactions for A_μ.

In superspace the vector multiplet is packaged in a single real superfield V. Gauge-invariant supersymmetric actions are constructed using W_α (chiral field-strength) and $\Phi^\dagger e^{qV}\Phi$ (gauge-covariant matter kinetic terms). The D-component controls D-term potentials and FI terms. The superspace formulation makes manifest the interplay between gauge symmetry, supersymmetry, and holomorphy—this is the basis for many non-perturbative and anomaly-based arguments in SUSY QFT.

4.3.6 Supersymmetry and Gauge Symmetry Breaking

Purpose and scope. In this section we analyze how spontaneous breaking of supersymmetry (SUSY) and gauge symmetry arises from the structure of the scalar potential in $\mathcal{N}=1$ supersymmetric gauge theories. The key objects are the auxiliary fields F_i (from chiral multiplets) and D^a (from vector multiplets). Their vacuum expectation values determine whether SUSY and gauge symmetry remain unbroken.

Scalar potential and vacuum equations

After eliminating auxiliary fields, the scalar potential in an $\mathcal{N}=1$ gauge theory takes the form

$$V(\phi,\bar{\phi}) = \sum_i |F_i|^2 + \frac{1}{2}\sum_a (D^a)^2, \qquad F_i = -\frac{\partial W}{\partial \phi_i}, \qquad D^a = -g\,\phi^\dagger T^a\phi - \kappa^a, \tag{4.3.7}$$

where:

- $W(\phi)$ is the superpotential,
- T^a are the generators of the gauge group acting on ϕ,
- κ^a are possible Fayet–Iliopoulos (FI) parameters for abelian factors.

Why start from V? The vacuum structure follows directly from the minima of V; supersymmetric vacua correspond to configurations with $V=0$.

The supersymmetry-preserving vacuum conditions are

$$F_i = 0 \quad \text{for all } i, \qquad D^a = 0 \quad \text{for all } a. \tag{4.3.8}$$

If and only if such a configuration exists, SUSY is unbroken.

Logic. Because V is a sum of squares, $V=0$ can only be achieved when each F_i and D^a vanish. If this set of equations has no solution, SUSY must be spontaneously broken.

Gauge symmetry breaking vs. SUSY breaking

Gauge symmetry and supersymmetry may break independently.

- **Gauge symmetry breaking:** occurs when the vacuum expectation value (vev) of some charged field $\langle\phi_i\rangle$ is nonzero, so that the corresponding gauge generator T^a acts nontrivially:

  $$\langle\phi^\dagger T^a\phi\rangle \neq 0.$$

  This leads to a mass term for the gauge boson via the Higgs mechanism.

- **Supersymmetry breaking:** occurs when at least one auxiliary field develops a nonzero vev:

$$\langle F_i \rangle \neq 0 \quad \text{or} \quad \langle D^a \rangle \neq 0.$$

Observation: Gauge symmetry breaking and SUSY breaking can occur simultaneously or separately. SUSY may remain exact while gauge symmetry is broken (as in the supersymmetric Higgs mechanism), or SUSY may be broken while the gauge symmetry remains unbroken (as in O'Raifeartaigh or Fayet models).

Supersymmetric Higgs mechanism

When gauge symmetry breaks but SUSY remains unbroken, the scalar vev satisfies the F- and D-flatness conditions in Eq. (4.3.8). For example, in an abelian theory with two chiral fields of opposite charges $q_1 = -q_2 = 1$ and superpotential $W = 0$, the D-term is

$$D = g\left(|\phi_1|^2 - |\phi_2|^2\right) + \kappa.$$

The $D = 0$ condition defines the vacuum manifold:

$$|\phi_1|^2 - |\phi_2|^2 = -\frac{\kappa}{g}.$$

If $\kappa \neq 0$, both ϕ_1 and ϕ_2 acquire nonzero vevs that break the $U(1)$ gauge symmetry, but the vacuum still satisfies $F_i = 0$ and hence preserves supersymmetry.

D-term SUSY breaking (Fayet model)

Consider again an abelian gauge theory with chiral fields ϕ_i of charges q_i and an FI constant κ. Take $W(\phi) = 0$, so that all F-terms vanish identically. The scalar potential is

$$V(\phi, \bar{\phi}) = \frac{1}{2} D^2 = \frac{1}{2}\big(g \sum_i q_i |\phi_i|^2 + \kappa\big)^2. \tag{4.3.9}$$

The D-flatness condition is

$$g \sum_i q_i |\phi_i|^2 + \kappa = 0.$$

If all q_i have the same sign and $\kappa \neq 0$, this equation has no solution, so $\langle D \rangle = -\kappa \neq 0$ and

$$V_{\min} = \frac{1}{2}\kappa^2 > 0.$$

Thus SUSY is spontaneously broken while the gauge symmetry remains unbroken.

Logical chain: We started with $W = 0$ (so F-terms cannot break SUSY). Then we saw that D-flatness cannot be achieved because of the sign mismatch between κ and the charges. Therefore, SUSY is broken purely by the D-term expectation value.

F-term SUSY breaking (O'Raifeartaigh model)

A minimal example of F-term breaking is given by the superpotential

$$W = fX + m\phi_1\phi_2 + \lambda X \phi_1^2,$$

with f, m, λ constant parameters. The F-term equations are

$$\begin{aligned} F_X &= -(f + \lambda\phi_1^2), \\ F_{\phi_1} &= -(m\phi_2 + 2\lambda X \phi_1), \\ F_{\phi_2} &= -m\phi_1. \end{aligned}$$

The SUSY vacuum conditions $F_i = 0$ require

$$m\phi_1 = 0 \Rightarrow \phi_1 = 0, \quad f + \lambda\phi_1^2 = f = 0,$$

which cannot hold if $f \neq 0$. Hence no supersymmetric vacuum exists, and SUSY is spontaneously broken with

$$\langle F_X \rangle = -f, \qquad V_{\min} = |f|^2.$$

Gauge symmetry remains unbroken (all ϕ_i are neutral here).

This is the standard algebraic demonstration of tree-level F-term SUSY breaking, analogous to the Fayet model but with F-term as the order parameter.

Identifying the Goldstino

When SUSY is spontaneously broken, one linear combination of the fermions becomes massless: the Goldstino. It is given by

$$\eta \propto \sum_i \langle F_i \rangle \psi_i + \sum_a \langle D^a \rangle \lambda^a,$$

where ψ_i are the chiral fermions and λ^a are the gaugini. Its transformation under supersymmetry has a constant shift

$$\delta_\epsilon \eta \sim \sqrt{2}\, \epsilon\big(\langle F_i \rangle, \langle D^a \rangle\big),$$

indicating the spontaneous breaking of SUSY. This massless fermion will be "eaten" by the gravitino if supergravity is included (super-Higgs mechanism).

Thus, the algebraic vacuum conditions $F_i = 0$ and $D^a = 0$ encode both supersymmetry and gauge symmetry breaking in a unified way, allowing a purely algebraic classification of possible vacua in $\mathcal{N} = 1$ theories.

4.3.7 D-term and the Higgs Branch

Purpose. In this section we examine the role of D-terms in determining the vacuum structure of $\mathcal{N} = 1$ and $\mathcal{N} = 2$ supersymmetric gauge theories. In particular, we study how the conditions

$$F_i = 0, \qquad D^a = 0$$

Definition of the Higgs Branch

The Higgs branch arises when scalar fields in chiral (or hypermultiplet) matter fields acquire vacuum expectation values (VEVs)(chiral multiplet broken), while the vector multiplet scalars remain unexcited:

$$\langle Q \rangle \neq 0, \quad \langle \tilde{Q} \rangle \neq 0, \quad \langle \phi \rangle = 0.$$

This leads to complete breaking of the gauge symmetry and a vacuum manifold described by a hyper-Kähler (or Kähler) quotient. The central idea is that the D-term potential enforces constraints that remove redundant gauge directions, so that the physical moduli space is a quotient of the D-flat configurations by gauge transformations.

Scalar potential and D-flatness revisited

Recall that the scalar potential in an $\mathcal{N} = 1$ gauge theory is

$$V(\phi, \bar{\phi}) = \sum_i |F_i|^2 + \frac{1}{2} \sum_a (D^a)^2, \qquad D^a = g\, \phi^\dagger T^a \phi + \kappa^a. \tag{4.3.10}$$

Why focus on D-terms now? Because in many cases (for example, in $\mathcal{N} = 2$ theories or in Higgs phases) the superpotential W vanishes or is constrained by extended supersymmetry, so the vacuum equations reduce entirely to D-flatness conditions.

When all $F_i = 0$, the potential reduces to

$$V_D = \frac{1}{2}\sum_a (D^a)^2,$$

and the vacua satisfy

$$D^a = 0. \tag{4.3.11}$$

Geometric interpretation. The equations $D^a = 0$ define a real submanifold of field space (the "D-flat manifold"). However, points related by gauge transformations correspond to physically equivalent configurations. Therefore, the physical vacuum moduli space is the space of gauge-inequivalent D-flat configurations:

$$\mathcal{M}_{\text{vac}} = \{\, \phi \mid D^a(\phi) = 0 \,\}/G_{\mathbb{R}}, \tag{4.3.12}$$

where $G_{\mathbb{R}}$ denotes the real gauge group.

D-flatness removes "unbalanced" charge directions, and the quotient by $G_{\mathbb{R}}$ removes gauge redundancy. This combination yields a smooth physical moduli space of inequivalent supersymmetric vacua.

Complex formulation and holomorphic quotient

If we add a superpotential to the theory, our analysis of the D-term equations does not change. If there exists a moduli space $\mathcal{M}'$ of solutions to the F-term constraints coming from the superpotential, then this entire space will also be a solution of the D-term equations. Therefore, the D-flat moduli space can equivalently be described as a holomorphic (or algebraic) quotient:

$$\mathcal{M} = \mathcal{M}'/G_{\mathbb{C}} = \{\, \phi_i \mid F_j = 0 \,\} \mathbin{/\!\!/} G_{\mathbb{C}} \tag{4.3.13}$$

where the double slash denotes the *Kähler quotient* (or symplectic quotient) by $G_{\mathbb{C}}$, which is equivalent to solving $D^a = 0$ and dividing by $G_{\mathbb{R}}$.

Why use $G_{\mathbb{C}}$? Because the D-flatness equations $D^a = 0$ correspond to moment-map constraints for the Hamiltonian action of $G_{\mathbb{R}}$ on the Kähler manifold of scalar fields. Taking the quotient by the complexified group $G_{\mathbb{C}}$ automatically imposes both the D-flat constraint and gauge equivalence in a holomorphically natural way.

Thus, the Higgs branch can be understood as a Kähler (or symplectic) quotient:

$$\mathcal{M}_{\text{Higgs}} = \mu^{-1}(0)/G_{\mathbb{R}} \;\simeq\; \operatorname{Spec}\big(\mathbb{C}[\phi]^{G_{\mathbb{C}}}\big), \tag{4.3.14}$$

where $\mu^a(\phi) = D^a$ is the moment map associated with the gauge symmetry.

Example: $U(1)$ gauge theory with N chiral fields

Consider a $U(1)$ gauge theory with chiral multiplets Φ_i of charges q_i and no superpotential. Then the D-term potential is

$$V_D = \frac{g^2}{2}\Big(\sum_i q_i|\phi_i|^2 + \kappa\Big)^2.$$

The D-flatness condition $D = 0$ gives

$$\sum_i q_i|\phi_i|^2 = -\frac{\kappa}{g}. \tag{4.3.15}$$

Case 1: $\kappa = 0$. Then the D-flat locus is the cone

$$\sum_i q_i|\phi_i|^2 = 0.$$

For example, with two fields of opposite charge, $q_1 = 1$, $q_2 = -1$, this gives $|\phi_1| = |\phi_2|$. After dividing by the $U(1)$ gauge action (which removes one phase), moduli space is a complex one-dimensional manifold, isomorphic to $\mathbb{C}$.

Case 2: $\kappa \neq 0$**.** Then Eq. (4.3.15) shifts the allowed values, and typically both fields acquire nonzero vevs, leading to spontaneous breaking of the $U(1)$ gauge symmetry. However, since $F_i = 0$, SUSY remains unbroken.

Physical meaning: In both cases, the Higgs branch is parametrized by gauge-invariant combinations such as $\phi_1\phi_2$. The D-flatness condition ensures balance between positively and negatively charged fields, and the quotient removes unphysical phases.

General Higgs branch geometry

In general, for a gauge group G acting linearly on a set of chiral multiplets ϕ^i, the D-flatness equations read

$$D^a = g\,\phi^\dagger T^a \phi + \kappa^a = 0.$$

These are the real moment-map equations associated with the action of $G_\mathbb{R}$ on the Kähler manifold of scalars.

The set $\mu^{-1}(0) = \{\phi \mid D^a = 0\}$ is invariant under $G_\mathbb{R}$, and the quotient $\mu^{-1}(0)/G_\mathbb{R}$ inherits a Kähler structure. This quotient space is the *Higgs branch* of the theory:

$$\mathcal{M}_{\text{Higgs}} = \mu^{-1}(0)/G_\mathbb{R}. \tag{4.3.16}$$

Advantages of this construction:

- It provides a purely geometric method to describe the moduli space of vacua.
- It automatically ensures supersymmetry, since D-flatness implies $V_D = 0$.
- The quotient removes gauge redundancy, yielding the space of physically inequivalent vacua.

4.3.8 Coulomb Branch in $N = 2$

For a pure $N = 2$ gauge theory (without hypermultiplets), the moduli space is the Coulomb branch. After adding hypermultiplets, the Coulomb branch still exists.

In supersymmetric gauge theories, Coulomb branch is a branch of the moduli space opened by a scalar field acquiring a VEV under a adjoint representation(gauge multiplet broken). defined as follows:

- **Driven by vector multiplet scalars:** The CB arises when the complex scalar ϕ in the $N = 2$ vector multiplet acquires a non-zero VEV.
- **Matter fields vanish:** On the CB, the scalar fields from $N = 2$ hypermultiplets satisfy

$$\langle Q \rangle = \langle \tilde{Q} \rangle = 0.$$

- **Supersymmetry preserved:** Full $N = 2$ supersymmetry is preserved along the branch.

The Coulomb branch corresponds to an "electric phase" of the theory. Here the gauge symmetry is partially broken, leaving a residual Abelian subgroup. The low-energy effective theory is an Abelian gauge theory with massless photons.

How Does the Coulomb Branch Arise?

Consider the simplest nontrivial example: $N = 2$ SU(2) pure gauge theory (no hypermultiplets).

- **Field content:** Only one $N = 2$ vector multiplet.
- **Vector multiplet components:** Gauge boson A_μ, two Weyl fermions λ, ψ (forming a Dirac fermion), and a complex scalar ϕ.

- **Key point:** ϕ transforms in the adjoint representation of the gauge group SU(2):

$$\phi = \phi^a T^a, \quad T^a = \frac{\sigma^a}{2}.$$

The scalar potential in $N = 2$ supersymmetric gauge theory consists of D-terms and F-terms:

$$V = \frac{1}{2g^2}(D^a)^2 + |F|^2.$$

For pure gauge theory, there is no superpotential, so $F = 0$. The D-term is

$$D^a = if^{abc}\phi^{b*}\phi^c.$$

For SU(2), $f^{abc} = \epsilon^{abc}$, so the vacuum condition $V = 0$ requires:

$$D^a = 0 \quad \Rightarrow \quad [\phi^\dagger, \phi] = 0.$$

This is the origin of the Coulomb branch: ϕ and $\phi^\dagger$ must commute, which means they can be simultaneously diagonalized.

Construction and Mathematical Realization of the Coulomb Branch

1. Choice of Vacuum

The most general solution to $[\phi^\dagger, \phi] = 0$ is to take ϕ in the **Cartan subalgebra** of the gauge group. For SU(2), the Cartan subalgebra is one-dimensional, generated by T^3:

$$\langle\phi\rangle = \frac{a}{2}\sigma^3 = \begin{pmatrix} \frac{a}{2} & 0 \\ 0 & -\frac{a}{2} \end{pmatrix},$$

where a is a complex parameter labeling different vacua on the CB.

2. Modding out Gauge Symmetry

Although we fix the gauge by choosing $\langle\phi\rangle \propto \sigma^3$, the gauge group still contains a discrete **Weyl group** that identifies equivalent vacua. For SU(2), the Weyl group is $\mathbb{Z}_2$, which identifies $a \sim -a$.

Hence a gauge-invariant coordinate is

$$u = \langle \mathrm{Tr}\, \phi^2 \rangle = \frac{1}{2}a^2,$$

which serves as the true coordinate of the Coulomb branch $\mathcal{M}_C$. For SU(2) pure gauge theory, the CB is a one-dimensional complex plane $\mathbb{C}$ parameterized by u.

3. Dimension Formula

In general, the complex dimension of the Coulomb branch equals the rank of the gauge group G:

$$\dim_{\mathbb{C}} \mathcal{M}_C = \mathrm{rank}(G).$$

For SU(N), $\mathrm{rank}(G) = N - 1$, so the CB is $(N-1)$-dimensional.

Residual Gauge Symmetry: Why Only $U(1)$ Remains

To understand the "not fully gauge-fixed" mechanism, consider the vacuum:

$$\langle\phi\rangle = \frac{a}{2}\sigma^3.$$

A general SU(2) transformation $U = \exp(i\theta^a T^a)$ acts as

$$\phi \to U\phi U^{-1}.$$

Consider $U = \exp\big(i\alpha T^3\big)$. Since $[T^3, T^3] = 0$, $[\langle\phi\rangle, T^3] = 0$,

$$U\langle\phi\rangle U^{-1} = \langle\phi\rangle.$$

All T^3-generated $U(1)$ transformations leave the vacuum invariant, giving the residual $U(1)$ gauge symmetry.

Broken Symmetry $SU(2)/U(1)$

For $U = \exp\big(i\beta T^1\big)$, $[\mathrm{T}^3, T^1] = iT^2 \neq 0, \quad [T_1, \langle\phi\rangle] \neq 0, \quad [T_2, \langle\phi\rangle] \neq 0,$

thus

$$U\langle\phi\rangle U^{-1} \neq \langle\phi\rangle$$

so this rotation changes the vacuum. T^1 and T^2 directions are broken, generating massive $W_\mu^\pm$ gauge bosons by absorbing the corresponding Goldstone modes in ϕ^1, ϕ^2:

$$M_W \propto g|a|.$$

The unbroken T^3 gauge boson A_μ^3 remains massless, representing the photon of the residual $U(1)$.

Low-Energy Effective Theory

At energies much lower than M_W, the low-energy theory is an $N = 2$ $U(1)$ Abelian gauge theory, described by a prepotential $\mathcal{F}(a)$:

$$\tau(a) = \frac{\partial^2 \mathcal{F}}{\partial a^2},$$

where $\tau(a)$ is the effective gauge coupling.

Coulomb Branch vs. Higgs Branch

Feature	Coulomb Branch (CB)	Higgs Branch (HB)
Driving Fields	Vector multiplet scalar ϕ	Hypermultiplet scalars Q,
Field Representation	Adjoint	Usually fundamental or other repr
Other Field VEVs	$\langle Q\rangle = \langle\tilde{Q}\rangle = 0$	$\langle\phi\rangle = 0$
Gauge Symmetry Breaking	Partial: $G \to U(1)^{\mathrm{rank}(G)}$	Complete: $G \to$ trivial subg
Low-Energy Theory	Abelian gauge theory (massless photons)	Non-linear sigma model (no gaug
Mathematical Construction	Parametrized by $u_k = \langle \mathrm{Tr}\,\phi^k\rangle$	Geometric quotient $\mathcal{M}_H = \{V$
Dimension	$\dim_{\mathbb{C}} = \mathrm{rank}(G)$	$\dim_{\mathbb{C}} = 2N_f - \dim(G)$ (depends o
Physical Picture	"Not fully gauge-fixed", residual $U(1)$ photon	"Fully gauge-fixed", gauge symmetry co
Existence in $N = 1$	Rare; requires adjoint chiral multiplet	Common; standard SUSY Higgs r
Existence in $N = 2$	Generic; appears even in pure gauge theory	Common; appears with hyperm

Why distinguish them? Because on the Coulomb branch, $D^a = 0$ automatically due to vanishing matter vevs, while on the Higgs branch, $D^a = 0$ imposes nontrivial constraints on matter vevs. In extended supersymmetric theories ($\mathcal{N} = 2$ or $\mathcal{N} = 4$), these two branches intersect only at the origin and are protected by non-renormalization theorems.

4.3.9 Perturbative (loop) effects

Higgs branch (perturbative).

- In $\mathcal{N} = 2$ (and higher) theories the *complex structure* and chiral ring of the Higgs branch are protected from perturbative corrections; only non-holomorphic data (e.g. Kähler metric) can receive loop corrections. Intuitively, holomorphy and extended SUSY are strong constraints. (See reviews and examples in which the hyper-Kähler quotient description remains valid perturbatively.)
- In $\mathcal{N} = 1$ the protection is weaker: the Kähler potential and D-terms can receive loop corrections, so the Higgs-branch metric / D-flatness locus can be renormalized (though holomorphic gauge-invariants remain meaningful coordinates for the chiral ring). This implies in practice that the shape (metric) of $\mathcal{M}_{\text{Higgs}}$ can change perturbatively in $\mathcal{N} = 1$.

Coulomb branch (perturbative).

- The Coulomb branch effective action is controlled by holomorphic functions (e.g. the prepotential in $\mathcal{N} = 2$); perturbative corrections (e.g. one-loop running) modify the effective gauge coupling $\tau_{\text{eff}}(a)$ and hence the metric on $\mathcal{M}_{\text{Coulomb}}$. For example, at weak coupling one-loop gives a logarithmic correction $\tau \sim (b_0/2\pi i) \log a + \cdots$. These perturbative effects are well captured by renormalization-group methods. :contentReference[oaicite:5]index=5

4.4 Non pertubative SUSY

4.4.1 Non-perturbative effects of HB and CB

General remark. Because of supersymmetric non-renormalization theorems, the superpotential $W(\Phi)$ receives *no perturbative* corrections, but it can receive *non-perturbative* contributions (instantons, monopoles, gaugino condensation, etc.). Such non-perturbative terms can qualitatively change the vacuum structure by (i) generating an effective W_{dyn} that lifts classical flat directions; or (ii) making previously massive BPS states massless at special points on the Coulomb branch, which can then condense and change the IR physics (dual Higgs / confinement).

(A) Affleck–Dine–Seiberg (ADS) type dynamical superpotential (how Higgs branch can be lifted). Consider $\mathcal{N} = 1$ supersymmetric QCD (SQCD) with gauge group $SU(N_c)$ and N_f flavours of quark chiral multiplets Q^i and $\tilde{Q}_j$ $(i, j = 1, \ldots, N_f)$, with canonical Kähler potential and (for now) vanishing classical superpotential $W_{\text{cl}} = 0$. Let $M^i{}_j = Q^i \tilde{Q}_j$ denote the meson gauge-invariant. Affleck–Dine–Seiberg argued that for $N_f < N_c$ a dynamical superpotential is generated of the form

$$W_{\text{ADS}} = (N_c - N_f) \left(\frac{\Lambda^{3N_c - N_f}}{\det M} \right)^{\frac{1}{N_c - N_f}} , \tag{4.4.1}$$

where Λ is the strong-coupling scale (dynamical scale) of the gauge theory. This functional form is fixed by (i) holomorphy, (ii) symmetry (flavor and $U(1)_R$), and (iii) dimensional analysis. When present, W_{ADS} generically lifts classical flat directions parametrized by M (no finite solution of $\partial W_{\text{eff}}/\partial M = 0$).

Heuristic origin: zero-mode counting for semiclassical contributions. One standard way to see how non-perturbative effects can generate a holomorphic W_{dyn} is to examine semiclassical configurations (instantons or monopole-instantons after compactification) and count fermionic zero modes. The basic rules are:

- A Euclidean instanton (or monopole-instanton) contributes to the path integral a factor $\sim e^{-S_{\text{cl}}}$, where $S_{\text{cl}} = \frac{8\pi^2}{g^2(\mu)}$ (plus phases).

- For a nonzero contribution to a superpotential term, the semiclassical amplitude must soak up exactly two superspace θ integrals (i.e. produce an F-term). Equivalently, the instanton must produce precisely two fermionic zero modes corresponding to the broken supersymmetries (or, after saturating extra zero modes by operator insertions, yield an effective W).
- Each fermionic zero mode must be soaked up by either operator insertions in the correlation function or by fermion bilinears produced in expanding the path integral measure; unsoaked Grassmann zero modes cause the amplitude to vanish.

Zero-mode counting in $SU(N_c)$ instanton: For a single (BPST) instanton in $SU(N_c)$ with N_f fundamental flavours, the number of fermionic zero modes is:

$$\text{gaugino zero modes: } 2N_c, \qquad \text{quark zero modes: } 2N_f.$$

(Each Weyl fermion in the adjoint contributes $2N_c$ zero modes; each fundamental Weyl fermion contributes two zero modes.) Thus the total number of fermionic zero modes is $2N_c + 2N_f$.

To produce an F-term (a superpotential contribution), we require a single insertion of the chiral superspace measure $d^2\theta$, which corresponds to integrating over two Grassmann (supercharge) zero modes. Therefore a naive one-instanton contribution can generate a superpotential only if after contracting with operator insertions exactly 2 fermionic zero modes remain unsaturated (i.e. the semiclassical amplitude reduces to an F-term). Practically, starting from $2N_c + 2N_f$ zero modes, one can use insertions of $2N_f$ quark fields (i.e. produce the meson determinants) to soak up the quark zero modes, leaving $2N_c$ gaugino zero modes: these gaugino zero modes can pair up to produce the factor $\Lambda^{3N_c-N_f}$ (via gaugino condensate structure) or be soaked up by further effects. The detailed mechanism depends on N_f relative to N_c; the ADS result (4.4.1) can be obtained by combining symmetry/holomorphy arguments with semiclassical checks in special cases.

Two regimes and how ADS arises concretely. (i) The case $N_f = N_c - 1$: - For $N_f = N_c - 1$, there exists a one-instanton semiclassical contribution that can generate W_{dyn} directly. The instanton has $2N_c$ gaugino and $2N_f = 2(N_c - 1)$ quark zero modes; by inserting $2(N_c - 1)$ quark legs (i.e. constructing the gauge invariant $\det M$ factor in the denominator), the remaining two gaugino zero modes are precisely those corresponding to the superspace measure $d^2\theta$, and thus the instanton amplitude produces an F-term with the schematic dependence

$$W_{\text{dyn}} \sim \frac{\Lambda^{3N_c-N_f}}{\det M}.$$

Detailed evaluation of the instanton integral fixes the numerical coefficient (computable in principle); this is the simplest semiclassical derivation of the ADS form for this special case.

(ii) The case $N_f < N_c - 1$: - For smaller N_f the one-instanton does not by itself produce the correct mass-dimension dependence; the ADS superpotential can still be inferred by holomorphy and will be generated by more subtle nonperturbative dynamics (gaugino condensation in a low-energy pure SYM subsector or multi-instanton/monopole effects after appropriate compactification). Arguments using symmetry, anomalies, and continuity (integrating out massive flavours) determine the exact form (4.4.1). The derivation in these cases often proceeds by: 1. Start from the $N_f = N_c - 1$ case where the instanton result applies; 2. Add masses to some flavours and integrate them out semiclassically, matching scales via decoupling relations $\Lambda^{3N_c-N_f} \to m\Lambda^{3N_c-(N_f-1)}$; 3. Iterate to general $N_f < N_c$. This matching plus holomorphy fixes the ADS form unambiguously.

How W_{ADS} lifts the classical Higgs branch: explicit algebraic demonstration. Assume a classical Higgs branch coordinate M (an $N_f \times N_f$ meson matrix) parameterizes a flat direction with $\det M \neq 0$ generically. If the effective superpotential contains (4.4.1), then the F-term equations are

$$\frac{\partial W_{\text{ADS}}}{\partial M} \propto -\frac{1}{N_c - N_f}\left(\frac{\Lambda^{3N_c-N_f}}{\det M}\right)^{\frac{1}{N_c-N_f}} (\det M)^{-1} \cdot (\text{adj } M) = 0,$$

where (adj M) denotes the adjugate matrix of M. This equation has no solution at finite M because the right-hand side is nonzero for any finite $\det M$. Thus the classical moduli direction parametrized by M is

removed (lifted) — there is no supersymmetric vacuum at finite field values. One may find instead that the true vacuum lies at infinite field or that SUSY is dynamically broken depending on global structure and other terms. This is the algebraic statement of "lifting" by a dynamical superpotential.

Key consequence (lifting): When W_{ADS} is generated, the F-term equations $\partial W_{\text{eff}}/\partial M = 0$ typically have no solution at finite M, i.e. classical flat directions parametrized by M are lifted (no SUSY vacuum at finite field space). Concretely, $\partial_M W_{\text{ADS}} \propto -(\det M)^{-1/(N_c-N_f)-1} \neq 0$ prevents $\partial_M W = 0$ for generic parameters, so classical Higgs branch directions are removed (lifted) by the non-perturbative superpotential. Thus ADS is a canonical mechanism whereby a classical Higgs moduli space can be destroyed by non-perturbative effects.

ADS from monopole-instantons on $\mathbb{R}^3 \times S^1$ (modern semiclassical viewpoint). A modern controlled derivation uses compactification on a small circle S^1 (with periodic boundary conditions for fermions to preserve SUSY), reducing $4d$ theory to an effective 3d theory with monopole-instanton contributions; these monopole-instantons can generate terms that match the ADS structure when decompactified. The core steps:

1. Compactify on S^1 of radius R and go to a regime where semiclassical methods (weak 3d coupling) apply.
2. The 3d theory has monopole-instanton configurations labeled by simple roots; each monopole has a definite number of fermion zero modes (counted by index theorems / Callias index).
3. Summing monopole contributions and matching to 4d dynamics reproduces the dynamical superpotential; for certain ranges of N_f these monopoles directly produce the ADS terms. This technique makes explicit how non-perturbative contributions that are not ordinary 4d instantons can generate holomorphic W_{dyn}.

Holomorphy and symmetry fix the functional dependence of W_{ADS} but not always the overall numerical coefficient (which may be scheme-dependent). For $N_f = N_c - 1$ the one-instanton computation gives the coefficient; for other N_f it can be fixed by decoupling limits and matching. Careful semiclassical computations (including the instanton measure, determinants of nonzero modes, and collective coordinate integrals) reproduce the expected prefactors in controlled cases. See references for full measure computations.

(B) Seiberg–Witten mechanism: massless monopoles/dyons on the Coulomb branch and dual Higgsing. The Seiberg–Witten solution for $\mathcal{N} = 2$ $SU(2)$ gauge theory provides a paradigmatic example of non-perturbative physics that affects the Coulomb branch: the SW curve is

$$y^2 = (x^2 - u)^2 - \Lambda^4 = (x^2 - u + \Lambda^2)(x^2 - u - \Lambda^2), \tag{4.4.2}$$

where $u = \langle \operatorname{Tr} \Phi^2 \rangle$ parametrizes the Coulomb branch and Λ is the dynamical scale. The Seiberg–Witten (SW) meromorphic differential λ_{SW} can be chosen (up to total derivatives) as

$$\lambda_{\text{SW}} = \frac{\sqrt{2}}{2\pi} \frac{x^2 - u}{y} \, dx,$$

or other equivalent normalizations differing by exact differentials (the normalization choice affects an overall factor in a, a_D but not monodromies). The physical periods are

$$a(u) = \oint_\alpha \lambda_{\text{SW}}, \qquad a_D(u) = \oint_\beta \lambda_{\text{SW}},$$

where $\{\alpha, \beta\}$ is a canonical basis of 1-cycles on the genus-1 curve (elliptic curve) defined by (4.4.2).

Branch points and cycle choice — explicit parametrization. The quartic polynomial in x has four branch points at

$$x = \pm e_+, \quad x = \pm e_-, \qquad e_\pm = \sqrt{u \pm \Lambda^2}.$$

Thus the curve is an elliptic curve obtained by gluing two sheets of the x-plane with branch cuts between $-e_+$ and $-e_-$, and between e_- and e_+. A convenient choice is to take the α-cycle encircling the cut between e_- and e_+ and the β-cycle encircling the cut between $-e_+$ and e_-. The periods are elliptic integrals of the first and second kind depending on the modulus k given by

$$k^2 = \frac{e_+^2 - e_-^2}{e_+^2 - (-e_+)^2} = \frac{(u+\Lambda^2) - (u-\Lambda^2)}{(u+\Lambda^2) - (-(u+\Lambda^2))} = \frac{2\Lambda^2}{2(u+\Lambda^2)} = \frac{\Lambda^2}{u+\Lambda^2},$$

Expressing the periods as elliptic integrals (schematic). One can change variables to write the periods in terms of standard complete elliptic integrals $K(k)$ and $E(k)$. A common schematic form is

$$a(u) = \frac{\sqrt{2}}{\pi} \int_{e_-}^{e_+} \frac{x^2 - u}{\sqrt{(x^2 - e_-^2)(e_+^2 - x^2)}}\, dx,$$

and similarly for a_D with integration over the complementary cycle. After an algebraic change of variables $x = e_- \cos\theta$ (or similar), these integrals reduce to combinations of $K(k)$ and $E(k)$. The important structural point is that $a_D(u)$ is proportional to the β-period and has monodromies around the branch points; analytic continuation around singular points changes (a, a_D) by $SL(2,\mathbb{Z})$ transformations.

Locations where BPS masses vanish (singularities). BPS masses of states with charges (n_e, n_m) are

$$M_{n_e,n_m}(u) = |\, n_e a(u) + n_m a_D(u)\,|\,.$$

Zeros of these masses correspond to values of u such that the appropriate linear combination of periods vanishes. In particular for pure SU(2) there are two singularities at

$$u = \pm\Lambda^2,$$

where either $a_D(u_*) = 0$ (monopole becoming massless) or a dyon becomes massless (depending on which singularity). This follows from explicit evaluation/analytic continuation of the elliptic periods (see SW). Concretely, near $u = \Lambda^2$ one finds $a_D(u) \propto (u - \Lambda^2)$ up to nonzero constant factor, hence $a_D(\Lambda^2) = 0$, i.e. the monopole mass vanishes at that point.

Local effective superpotential near a monopole point and condensation. Introduce a soft $\mathcal{N} = 1$ perturbation by adding a mass term for the adjoint (breaking $\mathcal{N} = 2 \to \mathcal{N} = 1$):

$$W_{\text{pert}} = m\,\mathrm{Tr}\,\Phi^2 = m\,u.$$

Near a monopole singularity u_0 where a monopole hypermultiplet $(M, \tilde{M})$ becomes massless, the effective superpotential for the light fields takes the schematic form

$$W_{\text{eff}} = \sqrt{2}\, a_D(u)\, M\tilde{M} + m\,u + \cdots\,.$$

Expanding $a_D(u) \simeq c\,(u - u_0)$ near u_0, we have

$$W_{\text{eff}} \simeq \sqrt{2}c\,(u - u_0) M\tilde{M} + m\,u.$$

Stationarity conditions are

$$\frac{\partial W_{\text{eff}}}{\partial u} = \sqrt{2}c\, M\tilde{M} + m = 0, \qquad \frac{\partial W_{\text{eff}}}{\partial M} = \sqrt{2}c\,(u - u_0)\tilde{M} = 0, \qquad \frac{\partial W_{\text{eff}}}{\partial \tilde{M}} = \sqrt{2}c\,(u - u_0) M = 0.$$

A solution is given by

$$u = u_0, \qquad M\tilde{M} = -\frac{m}{\sqrt{2}c}.$$

If we tune parameters so that u minimizes W_{eff} at u_0, the monopole fields condense, $\langle M \rangle \neq 0$, leading to a Higgsing of the dual magnetic $U(1)$ gauge symmetry ("dual Higgs") and leading to confinement of electric charges in the original variables. This algebraic demonstration captures the physical picture: monopole

masslessness $\Rightarrow$ light monopole fields in the IR $\Rightarrow$ perturbation can trigger monopole condensation $\Rightarrow$ dual Higgsing / confinement.

Key formula (BPS mass) and its vanishing condition:

$$M_{1,0} = |a(u)|, \qquad M_{0,1} = |a_D(u)|,$$

so that the monopole is massless when $a_D(u_0) = 0$, which occurs at the SW singularity $u_0 = \Lambda^2$ for SU(2). The full non-perturbative structure of $\tau(u)$, monodromies and BPS spectrum is encoded in the SW curve.

Period integrals: more explicit expressions (optional detail). If one wants an explicit expression for $a(u)$ and $a_D(u)$, one can reduce the integrals to elliptic standard forms. For example define

$$x = \sqrt{u + \Lambda^2}\,\sin\phi, \quad \text{or} \quad x = \sqrt{u - \Lambda^2}\,\cos\phi,$$

and the integral reduces to a linear combination of complete elliptic integrals $K(k), E(k)$ with modulus $k^2 = \dfrac{2\Lambda^2}{u + \Lambda^2}$ (or equivalent expressions in other parameterizations). Then one can analyze the behavior near $u = \pm\Lambda^2$ by using the known expansions of $K(k)$ as $k \to 1$ or $k \to 0$. The vanishing of a_D at $u = \Lambda^2$ follows from the leading behavior of these elliptic integrals. For a full exposé and explicit coefficient determination, consult standard references and reviews

5 Sym TFT

5.1 Symmetry TFT for Duality Defects

The starting point is that the symmetry data of a d-dimensional QFT with a fusion (higher-)category $\mathcal{C}$ can be encoded by a $(d+1)$-dimensional topological field theory (the Symmetry TFT, or SymTFT). Abstractly,

$$\text{SymTFT}(\mathcal{C}) \;\simeq\; \text{Turaev–Viro}[\mathcal{C}] \;\simeq\; \text{Reshetikhin–Turaev}[Z(\mathcal{C})],$$

so the bulk TQFT is controlled by the Drinfeld center $Z(\mathcal{C})$ (the "quantum double" of $\mathcal{C}$). For group-like symmetries $\mathcal{C} = \mathrm{Vec}_\alpha(G)$ this reproduces Dijkgraaf–Witten (DW) gauge theory with cocycle α.

A basic $(1+1)$d example of a non-invertible symmetry is the Tambara–Yamagami category $\mathrm{TY}(G)$ for G abelian. It contains invertible lines $\{L_g\}_{g\in G}$ and one non-invertible line N with fusion

$$L_{g_1} \times L_{g_2} = L_{g_1 g_2}, \qquad L_g \times N = N \times L_g = N, \qquad N \times N = \sum_{g\in G} L_g. \tag{5.1.1}$$

In higher dimensions there are higher-categorical analogues (e.g. with one-form symmetries $G = \mathbb{Z}_N^{(1)}$ in $(3+1)$d).

A concrete, Lagrangian route to these non-invertible structures is to start from a suitable DW gauge theory and *gauge an electro-magnetic (EM) exchange symmetry* that permutes electric and magnetic operators. This construction not only produces the right spectrum of objects in $Z(\mathcal{C})$ but also gives computable fusion and associator data.

Conventional data of $\mathrm{TY}(G)$. Specifying $\mathrm{TY}(G)$ requires, in addition to the fusion rules, a $\mathbb{Z}_2$ Frobenius–Schur indicator ε and a bicharacter $\chi \in H^2(G, U(1))$. In what follows one may take the minimal case with trivial ε and χ unless stated otherwise.

5.2 Organization and Conventions

Organization. First treat $(1+1)$d theories with $\mathrm{TY}(\mathbb{Z}_N)$ symmetry via a $(2+1)$d SymTFT; then extend to $(3+1)$d one-form symmetries via a $(4+1)$d SymTFT. The construction proceeds in two steps in each

dimension: (i) build the DW (or untwisted $\mathbb{Z}_N$) gauge theory SymTFT for the underlying group-like symmetry; (ii) gauge the bulk EM duality to obtain the SymTFT realizing the non-invertible duality symmetry (Tambara–Yamagami-like).

Conventions. Dynamical/background 1-form gauge fields are denoted a/A; 2-form fields b/B. Cup products are usually suppressed ($aA := a \cup A$). X_d denotes spacetime, M_n defect worldvolumes. Global fusion/ direct sum are $\times$ / $+$; local fusion/ direct sum are $\otimes$ / $\oplus$. A notation like $O_c \subset O_a \otimes O_b$ indicates a junction from incoming O_a, O_b to outgoing O_c.

5.3 Duality Interfaces in $(1+1)$d

5.3.1 Duality Interfaces from Half-Space Gauging

Let X be a non-spin (1+1)d QFT with an anomaly-free $\mathbb{Z}_N^{(0)}$ symmetry and background $A \in H^1(X_2, \widehat{\mathbb{Z}}_N)$. Gauging on a closed surface X_2 gives

$$Z_{X/\mathbb{Z}_N}[X_2, A] = \frac{1}{|H^0(X_2, \mathbb{Z}_N)|} \sum_{a \in H^1(X_2, \mathbb{Z}_N)} Z_X[X_2, a] \, \exp\Big(\tfrac{2\pi i}{N} \int_{X_2} aA\Big). \tag{5.3.1}$$

The quantum symmetry after gauging is generated by the Wilson line

$$\eta(\gamma) = \exp\left(\tfrac{2\pi i}{N} \oint_\gamma a\right). \tag{5.3.2}$$

Gauging twice maps $X \mapsto X$ up to a universal Euler counterterm

$$\chi[X_2, \mathbb{Z}_N] = \frac{|H^0(X_2, \mathbb{Z}_N)| \, |H^2(X_2, \mathbb{Z}_N)|}{|H^1(X_2, \mathbb{Z}_N)|}. \tag{5.3.3}$$

One may re-normalize (5.3.1) by a power χ^κ; the symmetric choice $\kappa = \frac{1}{2}$ makes "gauge twice" exact, but we keep the canonical normalization to see χ explicitly in interface formulas.

Half-space gauging and the interface. Decompose $X_2 = X_2^{<0} \cup X_2^{\geq 0}$ with common boundary $M_1 = \partial X_2^{\geq 0}$ at $x = 0$. Gauge only on the right half $X_2^{\geq 0}$ with *Dirichlet* boundary condition for the dynamical a at M_1. The right-half partition function is

$$Z_{X/\mathbb{Z}_N}[X_2^{\geq 0}, A] = \frac{1}{|H^0(X_2^{\geq 0}, M_1, \mathbb{Z}_N)|} \sum_{a \in H^1(X_2^{\geq 0}, M_1, \mathbb{Z}_N)} Z_X[X_2^{\geq 0}, a] \, \exp\Big(\tfrac{2\pi i}{N} \int_{X_2^{\geq 0}} aA\Big). \tag{5.3.4}$$

The Dirichlet condition $a|_{M_1} = 0$ ensures well-defined gluing to the ungauged side. The codimension-one locus at M_1 is the *duality interface* $\mathcal{N}$ between X (left) and $X/\mathbb{Z}_N$ (right).

Orientation reversal and the Euler factor. Exchanging the sides produces the reversed interface. Because "gauge twice" differs by χ, the reversed interface differs by precisely this Euler factor:

$$\mathcal{N}^{\text{rev}} = \chi[X_2^{\geq 0}, \mathbb{Z}_N] \, \mathcal{N}.$$

Thus $\mathcal{N}$ is not strictly invertible; it is invariant up to a universal counterterm.

5.3.2 Fusion Rules of Duality Interfaces

Trivial fusion with symmetry lines. Since $a|_{M_1} = 0$ on the boundary, the $\mathbb{Z}_N$ symmetry line η is trivial there, hence

$$\eta \times \mathcal{N} = \mathcal{N}. \tag{5.3.5}$$

Non-invertible fusion $\mathcal{N}\times\mathcal{N}$. Place two interfaces at $x=0$ and $x=\epsilon$, decompose

$$X_2 = X_2^{<0} \;\cup\; X_2^{[0,\epsilon)} \;\cup\; X_2^{\geq\epsilon},$$

and gauge on the right two pieces with Dirichlet on $M_1|_{x=0}$ and $M_1|_{x=\epsilon}$. A direct path-integral derivation proceeds as follows.

Step 1: Cohomology $\to$ cochains. Rewrite each relative cohomology sum as a sum over cochains with Lagrange multipliers enforcing closure and boundary conditions. Use that $H^1 = Z^1/B^1$, $|B^1| = |C^0|/|Z^0|$, $|B^0| = 1$ to convert

$$\frac{1}{|H^0|}\sum_{a\in H^1} \quad\longrightarrow\quad \frac{1}{|C^0|}\sum_{a\in Z^1} \quad\longrightarrow\quad \frac{1}{|C^0|\,|C^0|}\sum_{a\in C^1}\sum_{\phi\in C^0}\exp\Big(\tfrac{2\pi i}{N}\int\phi\,\delta a\Big),$$

and similarly for the second gauging with $(\tilde{a},\tilde{\phi})$ on $X_2^{\geq\epsilon}$, plus boundary terms $\int_{M_1}\phi\,a$ etc. to impose relative conditions.

Step 2: Sum over $\tilde{a}$. The exponent contains a term $\frac{2\pi i}{N}\int_{X_2^{\geq\epsilon}}(a-A)\,\tilde{a} - \frac{2\pi i}{N}\int_{X_2^{\geq\epsilon}}\delta\tilde{\phi}\,\tilde{a}$, so summing over $\tilde{a}\in C^1(X_2^{\geq\epsilon},\mathbb{Z}_N)$ enforces

$$(a-A-\delta\tilde{\phi})\big|_{X_2^{\geq\epsilon}} = 0,$$

i.e. $a=A$ on the far-right region up to a pure gauge absorbed by $\tilde{\phi}$.

Step 3: Sum over ϕ. Integrating over ϕ imposes $a\in Z^1(X_2^{\geq 0}, M_1|_{x=0};\mathbb{Z}_N)$ and collapses the bulk Dirichlet constraint at $x=0$.

Step 4: Factorization of cochain cardinalities and Euler counterterm. Use the product decomposition of cochains for the right half

$$|C^n(X_2^{\geq 0}, M_1|_0)| = |C^n(X_2^{[0,\epsilon)}, M_1|_0\cup M_1|_\epsilon)|\cdot|C^n(X_2^{\geq\epsilon})|, \tag{5.3.6}$$

(which *does not* hold at the level of cocycles because flatness is violated at $M_1|_\epsilon$), and $|C^2(X_2^{\geq\epsilon})| = |C^0(X_2^{\geq\epsilon}, M_1|_\epsilon)|$ by Fourier duality. Collecting prefactors yields precisely the inverse Euler term

$$\chi[X_2^{\geq\epsilon},\mathbb{Z}_N]^{-1} = \frac{|H^1(X_2^{\geq\epsilon},\mathbb{Z}_N)|}{|H^0(X_2^{\geq\epsilon},\mathbb{Z}_N)|\,|H^2(X_2^{\geq\epsilon},\mathbb{Z}_N)|} = \frac{|C^1|}{|C^0|\,|C^2|}. \tag{5.3.7}$$

Step 5: Residual gauging on the thin slab. After the overlapping region "double-gauging" annihilates to χ^{-1}, the only nontrivial path integral remains on the thin slab $X_2^{[0,\epsilon)}$ with *Dirichlet at both ends.* In the $\epsilon\to 0^+$ limit, this reduces to a sum over flat a on the strip, i.e. over $H^1(X_2^{[0,\epsilon)}, M_1|_0\cup M_1|_\epsilon;\mathbb{Z}_N)$, which Lefschetz-dualizes to $H^1(M_1,\mathbb{Z}_N)$ on the interface.

Putting everything together one finds

$$\mathcal{N}\times\mathcal{N} = \chi[X_2^{\geq\epsilon},\mathbb{Z}_N]^{-1}\sum_{\gamma\in H^1(M_1,\mathbb{Z}_N)}\eta(\gamma). \tag{5.3.8}$$

Using the orientation-reversal relation discussed above one can absorb the Euler factor into a convention choice (or eliminate it with the symmetric normalization of gauging), yielding the canonical Tambara–Yamagami-type fusion rule on the interface line:

$$\boxed{\mathcal{N}\times\mathcal{N} = \sum_{\gamma\in H^1(M_1,\mathbb{Z}_N)}\eta(\gamma), \qquad \eta\times\mathcal{N} = \mathcal{N}.} \tag{5.3.9}$$

Remarks on normalization. The freedom to multiply gauging by χ^κ shifts whether (5.3.8) carries χ^{-1} or not. Choosing $\kappa=\frac{1}{2}$ makes "gauge twice" exact and removes Euler remnants in all such compositions. Keeping the canonical normalization exposes how the overlap of two half-space gaugings consolidates into a universal topological factor.

Self-dual case and non-invertible symmetry. If X is self-dual under gauging ($X = X/\mathbb{Z}_N$), the interface $\mathcal{N}$ is a genuine defect *inside* X. Since $\mathcal{N} \times \mathcal{N}$ is a sum of symmetry lines rather than the identity, $\mathcal{N}$ generates a non-invertible symmetry in the $(1+1)$d theory with Tambara–Yamagami fusion.

5.4 (2+1)d Symmetry TFT for a (0)-form Z_N symmetry

Z_N gauge theory as the Symmetry TFT

Setup (slab construction). Let X be a (1+1)d QFT with an anomaly-free $Z_N^{(0)}$ symmetry and background $A \in H^1(X_2, \mathbb{Z}_N)$. Place X on the right boundary of a $(2+1)$d slab $X_3 = X_2 \times I_{[0,\varepsilon]}$ and endow the left boundary with a *topological* boundary condition. The bulk SymTFT is a $(2+1)$d Z_N gauge theory with action

$$S \;=\; \frac{2\pi}{N} \int_{X_3} \hat{a}\, \delta a, \tag{5.4.1}$$

where $a, \hat{a} \in C^1(X_3, \mathbb{Z}_N)$ are dynamical 1-cochains. Right boundary (at $x = \varepsilon$): an "enriched Neumann" state encoding X,

$$\big|X\big\rangle \;=\; \sum_{a \in H^1(X_2, \mathbb{Z}_N)} Z_X[X_2, a]\, \big|a\big\rangle. \tag{5.4.2}$$

Left boundary (at $x = 0$): a Dirichlet state fixing $a = A$,

$$\langle D(A)| \;=\; \sum_{a \in H^1(X_2, \mathbb{Z}_N)} \langle a|\, \delta(a - A). \tag{5.4.3}$$

Shrinking the slab ($\varepsilon \to 0$) reproduces X:

$$\langle D(A)|X\rangle \;=\; Z_X[X_2, A]. \quad \text{(Dirichlet picks } a = A) \tag{5.4.4}$$

Gauging as a boundary swap. Change the left boundary to Neumann

$$\langle N(A)| \;=\; \frac{1}{|H^0(X_2, \mathbb{Z}_N)|} \sum_{a \in H^1(X_2, \mathbb{Z}_N)} \langle a|\, \exp\Big(\frac{2\pi i}{N} \int_{X_2} a\, A\Big), \tag{5.4.5}$$

then

$$\langle N(A)|X\rangle \;=\; Z_{X/Z_N}[X_2, A], \tag{5.4.6}$$

i.e. changing the topological boundary implements Z_N gauging of the (1+1)d theory. (*Key moral:* "topological boundary condition = topological manipulation".)

5.4.1 Lines and surfaces in $(2+1)$d Z_N gauge theory

Genuine topological lines. For a closed loop γ, define

$$L_{(e,m)}(\gamma) \;=\; \exp\Big(\frac{2\pi i}{N} \oint_\gamma e\, a\Big)\, \exp\Big(\frac{2\pi i}{N} \oint_\gamma m\, \hat{a}\Big), \qquad (e,m) \in \mathbb{Z}_N^2. \tag{5.4.7}$$

Fusion is group-like:

$$L_{(e,m)} \times L_{(e',m')} \;=\; L_{(e+e',\, m+m')}. \tag{5.4.8}$$

Braiding phase (Hopf link or equal-time crossing on a plane):

$$L_{(e,m)}(\gamma)\, L_{(e',m')}(\gamma') \;=\; \exp\Big(-\frac{2\pi i}{N}\big(em' + me'\big)\, \langle \gamma, \gamma' \rangle\Big)\, L_{(e',m')}(\gamma')\, L_{(e,m)}(\gamma). \tag{5.4.9}$$

From this one obtains the *quantum torus algebra* for two segments of the *same* type:

$$L_{(e,m)}(\gamma)\, L_{(e,m)}(\gamma') \;=\; \exp\Big(-\frac{2\pi i}{N}\, em\, \langle \gamma, \gamma' \rangle\Big)\, L_{(e,m)}(\gamma + \gamma'). \tag{5.4.10}$$

Electromagnetic exchange Z_2^{EM} and a condensation defect. The action is invariant under $a \leftrightarrow \hat{a}$ (a 0-form $\mathbb{Z}_2$ symmetry). It is realized by a codimension-1 *symmetry defect* D_{EM} (a surface) that "converts" $L_{(e,m)}$ crossing it into $L_{(m,e)}$. Physically and algebraically, D_{EM} can be built as a *condensation defect* by gauging on a surface M_2 the 1-form symmetry generated by $L_{(1,-1)}$:

$$D_{\mathrm{EM}}(M_2) \;=\; \frac{1}{|H^0(M_2,\mathbb{Z}_N)|} \sum_{\gamma\in H^1(M_2,\mathbb{Z}_N)} L_{(1,-1)}(\gamma). \tag{5.4.11}$$

Using the quantum torus algebra one checks

(i) Piercing property: $L_{(e,m)}$ from the left equals D_{EM} from the right with $(e,m) \leftrightarrow (m,e)$, (5.4.12)

(ii) Invertibility up to an Euler counterterm: $D_{\mathrm{EM}}(M_2) \times D_{\mathrm{EM}}(M_2) \;=\; \chi[M_2,\mathbb{Z}_N]^{-1}$. (5.4.13)

Hence D_{EM} is (topologically) invertible in the TQFT sense.

5.4.2 Twist defects as higher duality interfaces

Definition (with boundary). Give the condensation surface M_2 a boundary $M_1 = \partial M_2$ and impose Dirichlet on M_1. The resulting operator is a *twist defect*—a non-genuine topological *line* on M_1 with an attached condensation surface M_2:

$$\Sigma^{(0)}(M_1,M_2) \;=\; \frac{1}{|H^0(M_2,M_1,\mathbb{Z}_N)|} \sum_{\gamma\in H^1(M_2,\mathbb{Z}_N)} L_{(1,-1)}(\gamma), \quad \text{(Dirichlet at } M_1). \tag{5.4.14}$$

Because of the boundary Dirichlet condition, $\Sigma^{(0)}$ *absorbs* any $L_{(n,-n)}$ along M_1 but not a general $L_{(e,m)}$ unless $e+m \equiv 0 \bmod N$.

A family of N distinct twist defects. Fuse an electric line $L_{(n+e,-n)}$ along the boundary of $\Sigma^{(0)}$:

$$\Sigma^{(e)}(M_1,M_2) \;:=\; L_{(n+e,-n)}(M_1) \times \Sigma^{(0)}(M_1,M_2), \qquad e = 0,\dots,N-1, \tag{5.4.15}$$

which depends only on $e+m \bmod N$ (hence N species). Each $\Sigma^{(e)}$ is invariant under Z_2^{EM}.

Interpretation as higher duality interfaces. Twist defects arise from *gauging on half of a codimension-1 submanifold* (the surface M_2 with boundary M_1), paralleling the idea of "half-space gauging" which in (1+1)d produces duality *interfaces*. Therefore twist defects are precisely *higher duality interfaces*.

Fusion rules.

- With genuine lines:
$$\Sigma^{(e)} \times L_{(e',m')} \;=\; \Sigma^{(e+e'+m')}. \quad \text{(labels shift)} \tag{5.4.16}$$
- Twist×twist (non-invertible): placing $\Sigma^{(0)}$ at $x=0$ and another at $x=\varepsilon>0$ and letting $\varepsilon\to 0$, one finds
$$\Sigma^{(0)} \times \Sigma^{(0)} \;\simeq\; \chi^{-1} \times \frac{1}{|H^0(\cdot)|} \sum_{\gamma\in H^1(M_2,\mathbb{Z}_N)} L_{(1,-1)}(\gamma) \;\Longrightarrow\; \Sigma^{(0)} \times \Sigma^{(0)} \;\sim\; \sum_{\gamma\in H^1(M_1,\mathbb{Z}_N)} L_{(1,-1)}(\gamma), \tag{5.4.17}$$
i.e. a *direct sum over lines*, not an inverse—hence non-invertibility. (Cochain manipulations and Euler factors as in Sec. 2.2 generalize here.)

3.4 From $(2+1)$d topological defects to $(1+1)$d symmetry interfaces

DEM turns Dirichlet into Neumann at the left boundary. Insert $D_{\mathrm{EM}}(M_2)$ parallel to the left boundary. When it collides with the Dirichlet boundary, it produces the Neumann boundary:

$$\langle D(A)| \xrightarrow{D_{\mathrm{EM}}} \langle N(A)|, \tag{5.4.18}$$

which is a Fourier transform in the EM-dual bases.

Twist $\to$ duality interface upon shrinking. Place a twist defect $\Sigma^{(e)}(M_1, M_2)$ in the slab parallel to the left boundary, then slide it to the boundary and shrink. The lower strip is effectively gauged, so $\Sigma^{(e)}$ descends to the $(1+1)$d duality *interface* N between X and X/Z_N. Moreover, because $L_{(e,0)}$ is absorbed by Dirichlet, all $\Sigma^{(e)}$ reduce to a *single* interface:

$$N(M_1) \;=\; \Sigma^{(e)}(M_1, M_2)\big|_{x\to 0}, \quad \forall\, e \in \mathbb{Z}_N. \tag{5.4.19}$$

Fusion rules descend.

$$\Sigma^{(e)} \times L_{(e',m')} \;\rightsquigarrow\; N \times \eta^{m'} \;=\; N, \tag{5.4.20}$$

$$\Sigma^{(0)} \times \Sigma^{(0)} \;\rightsquigarrow\; N \times N \;=\; \chi^{-1} \sum_{\gamma \in H^1(M_1, \mathbb{Z}_N)} \eta(\gamma), \tag{5.4.21}$$

exactly the TY-type non-invertible fusion of duality interfaces.

5.4.3 F-symbols of twist defects and duality interfaces

Choice of basis for genuine lines. One can work in a basis in which the F-symbols of genuine lines $L_{(e,m)}$ are trivial, while their half-braidings reproduce the braiding phases above (Fig. 14 in the paper).

Junctions, framing dependence, and the A-mesh. A trivalent junction relevant for twist defects consists of two twists $\Sigma^{(e)}, \Sigma^{(e')}$ and one genuine line $L_{(e+e'+n,-n)}$. The junction depends on the angle θ between that line and the D_{EM} surface ("framing"): we label $\theta = \pi$ by a triangle and $\theta = 0^+$ by a square. Introduce the algebra object

$$A \;=\; \bigoplus_{n=0}^{N-1} L_{(n,-n)}, \tag{5.4.22}$$

so that D_{EM} can be resolved into an A-mesh. Requiring independence of the anchor point along Σ gives a set of consistency equations. Given a solution for the triangular junction, one *defines* the square junction by the A-mesh prescription.

Two genuine lines and two twists. From the fusion rules, the internal leg must be a *twist*. Compute the F-symbol by: (i) expressing the square via the triangle; (ii) decomposing the twist boundary by

$$A \otimes L_{(e,0)} \;=\; \bigoplus_{n=0}^{N-1} L_{(e+n,-n)}; \tag{5.4.23}$$

(iii) using the half-braiding and fusion of genuine lines; (iv) reassembling the twist. The resulting phase structure is read off diagrammatically.

Four twists (internal leg is genuine). Take three incoming twists $\Sigma(e_1), \Sigma(e_2), \Sigma(e_3)$; place the A-mesh as in Fig. 19. Relate a configuration with internal $L_{(e,m)}$ to one with $L_{(\tilde{e},\tilde{m})}$; the parameters obey

$$e_1 + e_2 = e + m, \qquad e_2 + e_3 = \tilde{e} + \tilde{m}. \tag{5.4.24}$$

Keeping only phases (normalizations omitted) one finds the coefficients shown in Figs. 19–20; for $N = 2$ these match known results.

Passing to $(1+1)$d: F-symbols for the duality interface N. Upon shrinking the slab, $\Sigma(e) \mapsto N$. In the configuration descending from Fig. 18, observe: a line $L_{(e_2,m_2)}$ that collides with the Dirichlet boundary loses its electric part and reduces to the ordinary Z_N generator η_{m_2}; a line $L_{(e_1,m_1)}$ on top of D_{EM} collides with the Neumann boundary and retains its electric part, becoming the *quantum* Z_N generator $\hat{\eta}_{e_1}$. Therefore the mixed F-symbol is

$$\hat{\eta}_{e_1}\, N\, \eta_{m_2} \;=\; \exp\Big(\frac{2\pi i}{N}\, e_1 m_2\Big)\, \hat{\eta}_{e_1}\, N\, \eta_{m_2}, \tag{5.4.25}$$

i.e. a nontrivial bicharacter between ordinary and quantum Z_N across the duality interface.

From Fig. 19 one similarly obtains a three-N F-move:

$$N\,N\,N\,\hat{\eta}_e \;=\; \sum_{\tilde{m}=0}^{N-1} \exp\Big(-\,\frac{2\pi i}{N}\, e\,\tilde{m}\Big)\; N\,N\,N\,\eta_{\tilde{m}}, \tag{5.4.26}$$

whose nontrivial phase implies (after gauging Z_2^{EM} to make N a genuine duality *defect*) that N is *self-anomalous*: it obstructs a trivially gapped phase.

Conceptual summary (for quick recall).

- SymTFT for $(1+1)$d Z_N symmetry is $(2+1)$d Z_N gauge theory; swapping Dirichlet/Neumann on the left boundary implements no-gauging/gauging in $(1+1)$d.
- EM exchange Z_2^{EM} is realized by a condensation surface D_{EM} that absorbs $L_{(1,-1)}$ and is invertible up to Euler.
- Putting D_{EM} on a surface with boundary (Dirichlet on the boundary) yields a *twist defect* $\Sigma^{(e)}$ (a higher duality interface). Its fusion is non-invertible.
- Colliding $\Sigma^{(e)}$ with the Dirichlet boundary of the slab produces the $(1+1)$d duality interface N; all e give the same N. Fusion and F-symbols of Σ descend to those of N (TY-type).

5.5 (2+1)d Symmetry TFT for Duality Defects

we constructed the $(2+1)$d $\mathbb{Z}_N$ gauge theory as the Symmetry TFT encoding the ordinary $\mathbb{Z}_N^{(0)}$ symmetry of a $(1+1)$d QFT. We found that the electromagnetic exchange defect D_{EM} produced *twist defects* $\Sigma^{(e)}$ whose fusion and F-symbols descend, upon shrinking the slab, to those of the $(1+1)$d duality interface N. The fusion algebra of N is non-invertible and of *Tambara–Yamagami type.* Section 4 now asks the inverse question: *Can we build a $(2+1)d$ SymTFT whose excitations precisely realize this non-invertible $TY(\mathbb{Z}_N)$ structure?* That SymTFT will serve as the "universal topological background" for duality defects in two-dimensional systems.

Constructing the SymTFT realizing $TY(\mathbb{Z}_N)$

Review of the TY fusion category. Mathematically, a Tambara–Yamagami category $TY(G,\chi,\tau)$ is a fusion category built from an Abelian group G (here $G=\mathbb{Z}_N$), a nondegenerate symmetric bicharacter $\chi : G\times G \to U(1)$, and a scalar τ satisfying $\tau^2 = |G|^{-1}\sum_{g\in G}\chi(g,g)$. Its simple objects are

$$\{\, g \in G \,\} \;\cup\; \{N\},$$

where each g is *invertible* (forming the group G) and N is a *non-invertible* object obeying the fusion rules

$$g \times g' = g + g', \tag{5.5.1}$$

$$g \times N = N \times g = N, \tag{5.5.2}$$

$$N \times N = \sum_{g\in G} g. \tag{5.5.3}$$

Physically, g represents the ordinary $\mathbb{Z}_N$ symmetry lines, while N represents the *duality line*. The F-symbols contain the phase data $\chi(g,g') = e^{\frac{2\pi i}{N} gg'}$ and a sign from τ. Our task is to reproduce these fusion and F-symbols from a (2+1)d topological field theory. Since Section 3 showed that the (2+1)d $\mathbb{Z}_N$ gauge theory with an electromagnetic-exchange surface D_{EM} already contains objects with TY-like fusion, we now *promote* the EM exchange $\mathbb{Z}_2^{\mathrm{EM}}$ from a global duality to a *gauged* symmetry in the bulk. Gauging this $\mathbb{Z}_2$ creates a new $(2+1)$d topological theory— the desired SymTFT for $TY(\mathbb{Z}_N)$. Intuitively, it "doubles" the $\mathbb{Z}_N$ gauge theory and identifies configurations related by $a \leftrightarrow \hat{a}$, thus making the duality line N a genuine (non-invertible) topological excitation.

Field-theory realization. Introduce discrete 1-cochains $a, \hat{a} \in C^1(X_3, \mathbb{Z}_N)$ and a $\mathbb{Z}_2$-valued 1-cochain ρ gauging the EM exchange. Under ρ, the fields swap:

$$(a, \hat{a}) \longmapsto (\hat{a}, a) \quad \text{when crossed by a } \rho\text{-domain wall.}$$

The effective action combining both $\mathbb{Z}_N$ gauge fields and the $\mathbb{Z}_2$ twisting is

$$S_{TY} = \frac{2\pi}{N}\int_{X_3} \hat{a}\,\delta a + \pi\int_{X_3} \rho\,\delta\rho + \frac{2\pi}{N}\int_{X_3} \rho \smile (a - \hat{a}), \tag{5.5.4}$$

where the last term couples ρ to the EM-odd combination. When $\rho = 0$ we recover the original $\mathbb{Z}_N$ gauge theory; turning on ρ along a surface enacts the duality exchange. This theory is invertible when restricted to the ρ-even sector, but globally it includes the non-invertible "duality surface" corresponding to N.

Physical meaning. The three terms have clear interpretations:

1. $\frac{2\pi}{N}\int \hat{a}\,\delta a$: the standard $\mathbb{Z}_N$ BF term for the genuine lines $L_{(e,m)}$;
2. $\pi\int \rho\,\delta\rho$: a Dijkgraaf–Witten term for the $\mathbb{Z}_2$ background implementing the $\mathbb{Z}_2^{\mathrm{EM}}$ self-duality;
3. $\frac{2\pi}{N}\int \rho \smile (a - \hat{a})$: couples the $\mathbb{Z}_2$ domain walls to the difference of electric/magnetic gauge fields, producing the non-invertible duality surface.

Gauging the $\mathbb{Z}_2^{\mathrm{EM}}$ effectively identifies $L_{(e,m)}$ and $L_{(m,e)}$, so that only their symmetric combinations remain genuine, while antisymmetric combinations become absorbed by N. This matches exactly the $TY(\mathbb{Z}_N)$ data.

5.5.1 Cohomological viewpoint

6 higher form symmetry

6.1 Ordinary (0-form) Symmetries

Classical Noether's Theorem Consider a continuous *global* symmetry whose transformation parameters ϵ_a are constants. This is the familiar setting of a 0-form symmetry.

Localization trick. To derive the conserved current, we *pretend* that the symmetry parameters become space–time dependent, $\epsilon_a \to \epsilon_a(x)$. Since the original symmetry implies $\delta S = 0$ for constant ϵ_a, the variation of the action must now be proportional to the derivative of $\epsilon_a(x)$:

$$\delta S \propto \partial_\mu \epsilon_a(x).$$

Form of the variation. We can therefore express the variation of the action as

$$\delta S = \int d^D x\, J_a^\mu\, \partial_\mu \epsilon_a(x), \tag{6.1.1}$$

where J_a^μ is identified as the *Noether current* associated with the symmetry.

Conservation law. Using the equations of motion ($\delta S = 0$) and the arbitrariness of $\epsilon_a(x)$, we perform an integration by parts on Eq. (6.1.1) to obtain

$$\partial_\mu J_a^\mu = 0. \tag{6.1.2}$$

This is the local conservation law expressing the symmetry of the action.

Conserved charge. The conserved *Noether charge* corresponding to J_a^μ is defined as

$$Q_a \equiv \int d^{D-1}x\, J_a^0, \tag{6.1.3}$$

which satisfies $\dot{Q}_a = 0$ as a direct consequence of the conservation law (6.1.2).

6.2 Symmetries in Quantum Field Theory

Canonical formulation. In quantum mechanics, a symmetry is represented by a unitary operator U. For a continuous symmetry, the operator can be written as

$$U = e^{i\epsilon_a Q_a}, \tag{6.2.1}$$

where Q_a is the *Noether charge*, which serves as the generator of the symmetry. The corresponding infinitesimal transformation of a field operator ϕ is given by the commutator

$$\delta_a \phi = i[Q_a, \phi], \tag{6.2.2}$$

which is the quantum version of the classical Poisson bracket relation.

Path integral and Ward identities. In the path-integral formulation, the classical conservation law $\partial_\mu J_a^\mu = 0$ is promoted to an operator identity known as the *Ward identity*. This can be derived by performing a local transformation $\epsilon_a(x)$ on the integration variables (the fields ϕ) in the functional integral.

In the absence of anomalies, the Ward identity states that the divergence of the current in correlation functions vanishes everywhere except at the positions of other operator insertions (so-called "contact terms"):

$$\partial_\mu \langle J_a^\mu(x) X\rangle = -i \sum_j \delta^{(D)}(x - x_j)\langle \dots \delta_a \phi(x_j) \dots \rangle. \tag{6.2.3}$$

This expresses the conservation of the current at the quantum level.

6.3 Spontaneous Symmetry Breaking (SSB) and Goldstone's Theorem

Definition of SSB. A symmetry is said to be *spontaneously broken* if the Lagrangian (or action) of the theory is invariant under the symmetry, but the vacuum (ground) state is not. This manifests as a nonzero vacuum expectation value (VEV) of some order parameter:

$$\langle \delta_a \phi \rangle \neq 0. \tag{6.3.1}$$

Derivation of Goldstone's theorem. Starting from the Ward identity (6.2.3), take its Fourier transform to momentum space:

$$p_\mu \langle J_a^\mu(p)\, \phi(-p)\rangle = \langle \delta_a \phi(p = 0)\rangle. \tag{6.3.2}$$

Key argument:

- In the symmetric phase, $\langle \delta_a \phi \rangle = 0$, so no singular behavior arises.
- In the broken phase, $\langle \delta_a \phi \rangle \neq 0$ (a nonzero constant). To satisfy Eq. (6.3.2) as $p \to 0$, the correlator $\langle J_a^\mu(p)\, \phi(-p)\rangle$ must develop a pole at $p = 0$:

$$\langle J_a^\mu(p)\, \phi(-p)\rangle \sim \frac{p^\mu}{p^2}.$$

Physical interpretation. In quantum field theory, a pole at $p^2 = 0$ corresponds to the propagator of a *massless particle*. Thus, the presence of this pole implies the existence of a massless excitation — the **Goldstone boson** associated with the spontaneously broken continuous symmetry.

6.4 Reformulating Ordinary Symmetries in Topological Language

From vectors to forms. The Noether current J_μ can be represented as a differential 1-form:

$$J = J_\mu \, dx^\mu. \tag{6.4.1}$$

Its Hodge dual $*J$ is then a $(D-1)$-form, whose integral over a $(D-1)$-dimensional spatial hypersurface gives the conserved charge.

From divergence to exterior derivative. The conservation law $\partial_\mu J^\mu = 0$ is elegantly rewritten as

$$d*J = 0, \tag{6.4.2}$$

where d is the exterior derivative. Thus, conservation corresponds to the closedness of the $(D-1)$-form $*J$.

The conserved Noether charge can now be expressed geometrically as an integral over a closed $(D-1)$-dimensional submanifold Σ:

$$Q(\Sigma) = \int_\Sigma *J. \tag{6.4.3}$$

To connect this with the Ward identity, we move to Euclidean spacetime and integrate the Ward identity in differential-form notation,

$$d\langle *J(x)\, \phi(y)\rangle \sim \delta(x-y)\, \langle \delta\phi(y)\rangle, \tag{6.4.4}$$

over a D-dimensional region Ω_Σ whose boundary satisfies $\partial\Omega_\Sigma = \Sigma$.

Applying Stokes' theorem gives

$$\int_{\Omega_\Sigma} d\langle *J\, \phi(y)\rangle = \int_{\partial\Omega_\Sigma} \langle *J\, \phi(y)\rangle = \int_\Sigma \langle *J\, \phi(y)\rangle = \langle Q(\Sigma)\, \phi(y)\rangle. \tag{6.4.5}$$

6.4.1 The Linking Number

The right-hand side of Eq. (6.4.4), when integrated over Ω_Σ, becomes

$$\int_{\Omega_\Sigma} d^D x\, \delta(x-y), \tag{6.4.6}$$

which defines the *linking number* between Σ and the point y:

$$Link(\Sigma, y) = \begin{cases} 1, & y \text{ lies inside } \Omega_\Sigma, \\ 0, & y \text{ lies outside.} \end{cases} \tag{6.4.7}$$

This is a purely topological quantity — it depends only on whether Σ encloses the point y, not on the detailed geometry.

6.4.2 Summary: A New Paradigm for 0-form Symmetry

Combining the above relations, the Ward identity can now be written in a topological form:

$$\langle Q(\Sigma)\, \phi(y)\rangle = -i\, Link(\Sigma, y)\, \langle \delta\phi(y)\rangle. \tag{6.4.8}$$

This reveals a deep geometric picture:

- The symmetry generator $Q(\Sigma)$ is itself a *topological operator*. Its expectation value is invariant under smooth deformations of Σ, as long as Σ does not cross the insertion point y.
- Whether $Q(\Sigma)$ acts nontrivially on the charged operator $\phi(y)$ depends solely on their topological relationship — whether Σ links with y (i.e., whether $Link(\Sigma, y) \neq 0$).

Definition. Thus, an ordinary (**0-form**) symmetry can be defined as:

A $(D-1)$-dimensional topological operator $Q(\Sigma)$ acting on a 0-dimensional charged object $\phi(y)$.

This topological reformulation sets the stage for the generalization to higher-form symmetries.

6.5 Higher-Form Symmetries

This section represents the second major conceptual leap. It generalizes the topological picture of ordinary (0-form) symmetries from Section III to define *higher-form symmetries.*

6.5.1 From 0-form to 1-form Symmetry

0-form case. An ordinary (0-form) symmetry is characterized by a constant (closed) 0-form parameter ξ satisfying $d\xi = 0$.

1-form generalization. We now assume the existence of a new type of symmetry whose parameter is a closed 1-form ξ_1, satisfying

$$d\xi_1 = 0. \tag{6.5.1}$$

This defines a **1-form symmetry**.

Localization trick. Using the same "localization" idea as in Section II, we let $d\xi_1 \neq 0$ and compute the variation of the action:

$$\delta S = \int *J \wedge d\xi_1. \tag{6.5.2}$$

For this to be well-defined, since $d\xi_1$ is a 2-form, $*J$ must be a $(D-2)$-form. Therefore, the conserved current J itself is a *2-form* (an antisymmetric tensor $J^{\mu\nu}$).

Conservation law. The conservation condition becomes

$$d * J = 0 \quad \Longleftrightarrow \quad \partial_\mu J^{\mu\nu} = 0. \tag{6.5.3}$$

Conserved charge. The conserved charge is now defined as the integral of $*J$ over a closed $(D-2)$-dimensional submanifold Σ_{D-2}:

$$Q(\Sigma_{D-2}) = \int_{\Sigma_{D-2}} *J. \tag{6.5.4}$$

In the 0-form case, a $(D-1)$-dimensional topological operator acts on 0-dimensional charged operators (local fields). What objects does a $(D-2)$-dimensional operator $Q(\Sigma_{D-2})$ act on?

Dimensional complementarity (Poincaré duality). In $(D-1)$-dimensional space, for a $(D-2)$-dimensional surface Σ_{D-2} to link nontrivially with another object, that object must be *1-dimensional.* Hence, 1-form symmetries act on **line operators** $W[\mathcal{C}]$.

Action of the symmetry. The 1-form symmetry parameter ξ_1 naturally couples to such line operators via

$$\delta W[\mathcal{C}] \sim \int_{\mathcal{C}} \xi_1. \tag{6.5.5}$$

6.5.2 1-form Ward Identity and Examples

Ward identity. The 1-form Ward identity takes the form

$$\langle \partial_\mu J^{\mu\nu}(x)\, W[\mathcal{C}]\rangle = -i \int_{\mathcal{C}} dy^\nu\, \delta^{(D)}(x-y)\, \langle \delta W[\mathcal{C}]\rangle. \tag{6.5.6}$$

This shows that the divergence of the current is supported only on the worldline $\mathcal{C}$.

Topological linkage. The relationship between $Q(\Sigma_{D-2})$ and $W[\mathcal{C}]$ is purely topological: the operator $Q(\Sigma_{D-2})$ acts nontrivially only when Σ_{D-2} and $\mathcal{C}$ are *linked*, i.e., $Link(\Sigma_{D-2}, \mathcal{C}) \neq 0$.

Examples.

- **D = 2:** $(D-2) = 0$. $Q(S^0)$ consists of two points, and the charged object $W[\mathcal{C}]$ is a line. The line passing between the two points gives $Link = 1$.
- **D = 3:** $(D-2) = 1$. $Q(S^1)$ is a loop, and $W[\mathcal{C}]$ is another loop. The symmetry acts nontrivially when the two loops are linked once.
- **D = 4:** $(D-2) = 2$. $Q(S^2)$ is a sphere, and $W[\mathcal{C}]$ is a line operator. When the line pierces the sphere, $Link = 1$.

6.5.3 Generalization: q-form Symmetries

Definition. A **q-form symmetry** is defined by a conserved $(q+1)$-form current J satisfying

$$d * J = 0. \tag{6.5.7}$$

Its conserved charge is defined on a $(D-q-1)$-dimensional manifold Σ_{D-q-1}:

$$Q(\Sigma_{D-q-1}) = \int_{\Sigma_{D-q-1}} *J. \tag{6.5.8}$$

The charged object is a q-dimensional operator $W[\mathcal{C}_q]$, such that the two satisfy the dimensional complementarity condition

$$q + (D-q-1) = D-1. \tag{6.5.9}$$

Summary. In general, a *q-form symmetry* is characterized by:

- a conserved $(q+1)$-form current J,
- a topological operator $Q(\Sigma_{D-q-1})$ of dimension $(D-q-1)$,
- and a q-dimensional charged operator $W[\mathcal{C}_q]$.

Their mutual topological linking encodes the symmetry action, generalizing the 0-form and 1-form cases to arbitrary form degree.

6.6 Higher-Form Symmetries in Maxwell Theory

In this section, we apply the abstract framework of higher-form symmetries to the concrete example of free Maxwell theory, which serves as the prototypical realization of higher-form symmetries.

Action. The Maxwell action in D-dimensional spacetime is

$$S = \int -\frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu}, \tag{6.6.1}$$

where $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$ is the $U(1)$ field strength 2-form $f = da$.

Compactness assumption. We assume the gauge group is *compact* $U(1)$. This means the gauge field a is an angular variable with periodicity $a \sim a + 2\pi$, and gauge transformations are defined modulo 2π:

$$\Lambda \sim \Lambda + 2\pi.$$

Physical meaning of compactness. Compactness allows the existence of large gauge transformations (LGTs). For example, on a compact time circle S^1 of length L_0, one can define

$$\Lambda(x^0) = \frac{2\pi x^0}{L_0}.$$

This satisfies $\Lambda(L_0) - \Lambda(0) = 2\pi$, which corresponds to the same point on the $U(1)$ circle, so Λ is a legitimate gauge transformation.

Charge quantization. Under an LGT, a Wilson line of charge q_e,

$$W_{q_e} = \exp\left(iq_e \oint a\right), \tag{6.6.2}$$

transforms as

$$W'_{q_e} = W_{q_e} \exp\left(iq_e \oint \partial\Lambda\right) = W_{q_e}\, e^{iq_e 2\pi}.$$

Gauge invariance ($W'_{q_e} = W_{q_e}$) requires $e^{iq_e 2\pi} = 1$, which holds if and only if $q_e \in \mathbb{Z}$. Hence, compactness of the gauge group implies *quantization of electric charge.*

The Wilson line

$$W_{q_e}[\mathcal{C}] = \exp\left(iq_e \int_{\mathcal{C}} a\right) \tag{6.6.3}$$

is a one-dimensional extended operator representing the worldline of a static probe particle with electric charge q_e. It has no intrinsic dynamics—it serves as a probe of the gauge field configuration.

6.6.1 Two Higher-Form Symmetries of Maxwell Theory

The vacuum Maxwell equations are

$$d * f = 0, \qquad \text{(equations of motion)} \tag{6.6.4}$$

$$df = 0, \qquad \text{(Bianchi identity)} \tag{6.6.5}$$

Each of these equations expresses a conserved higher-form current and therefore defines a higher-form symmetry.

6.6.2 Classification of the Two Symmetries

Electric 1-form symmetry $U(1)_e^{(1)}$.

- **Conserved current:**

$$J_e \equiv \frac{1}{e^2} f,$$

  which is a 2-form.

- **Conservation law:**

$$d * J_e = 0,$$

  corresponding to $d * f = 0$.

- **Conserved charge:**

$$Q_e(\Sigma_{D-2}) = \int_{\Sigma_{D-2}} *J_e = \frac{1}{e^2} \int_{\Sigma_{D-2}} *f. \tag{6.6.6}$$

- **Charged objects:** Wilson line operators $W_{q_e}[\mathcal{C}]$.

Magnetic $(D-3)$-form symmetry $U(1)_m^{(D-3)}$.

- **Conserved current:**
$$J_m \equiv \frac{1}{2\pi} * f,$$
which is a $(D-2)$-form.

- **Conservation law:**
$$d * J_m = 0,$$
equivalent to the Bianchi identity $df = 0$.

- **Conserved charge:**
$$Q_m(\Sigma_2) = \int_{\Sigma_2} *J_m = \frac{1}{2\pi} \int_{\Sigma_2} f. \tag{6.6.7}$$

- **Charged objects:** 't Hooft operators $T_{q_m}[\Gamma_{D-3}]$.

6.6.3 The 't Hooft Operator and the Dual Gauge Field

To describe the magnetic symmetry more naturally, we introduce the *dual gauge field* $\tilde{a}$, a $(D-3)$-form defined by
$$*f = d\tilde{a}. \tag{6.6.8}$$
The 't Hooft operator $T_{q_m}[\Gamma_{D-3}]$ is defined as the exponential of the integral of $\tilde{a}$ over a $(D-3)$-dimensional submanifold Γ_{D-3}:
$$T_{q_m}[\Gamma_{D-3}] = \exp\left(i q_m \int_{\Gamma_{D-3}} \tilde{a} \right), \tag{6.6.9}$$
representing the worldline (or worldvolume) of a magnetic charge q_m.

6.6.4 Symmetry Breaking and Screening

If the theory contains dynamical electric charges of integer value n, the continuous electric 1-form symmetry $U(1)_e^{(1)}$ is broken to a discrete subgroup:
$$U(1)_e^{(1)} \longrightarrow \mathbb{Z}_n^{(1)}. \tag{6.6.10}$$

Physical picture. In the vacuum, pairs of virtual charges $+n$ and $-n$ can spontaneously appear and annihilate. A surface operator $Q_e(\Sigma)$ that measures electric flux may enclose only one member of such a pair, making the measured value ambiguous up to multiples of n. Consequently, only the flux modulo n is physically well-defined.

Summary.

- Compactness of $U(1)$ implies quantized electric charge.
- Maxwell theory exhibits two higher-form symmetries:

$U(1)_e^{(1)}$:	electric 1-form symmetry acting on Wilson lines,
$U(1)_m^{(D-3)}$:	magnetic $(D-3)$-form symmetry acting on 't Hooft operators.

- Including dynamical charges breaks these continuous symmetries to discrete ones.

6.7 Maxwell Theory in $D = 4$

We now specialize the discussion of Section V to four-dimensional spacetime, where the geometric structure of the higher-form symmetries becomes especially transparent.

6.7.1 Symmetries and Charges in $D = 4$

In four dimensions, $(D - 3) = 1$, so both the electric and magnetic symmetries are *1-form* symmetries. The full symmetry group is

$$U(1)_e^{(1)} \times U(1)_m^{(1)}. \tag{6.7.1}$$

Conserved charges. Both charges are defined on closed two-dimensional surfaces Σ_2:

$$Q_e(\Sigma_2) = \frac{1}{e^2} \int_{\Sigma_2} *f \qquad \leftrightarrow \text{ electric flux } \int \vec{E} \cdot d\vec{S}, \tag{6.7.2}$$

$$Q_m(\Sigma_2) = \frac{1}{2\pi} \int_{\Sigma_2} f \qquad \leftrightarrow \text{ magnetic flux } \int \vec{B} \cdot d\vec{S}. \tag{6.7.3}$$

6.7.2 Charged Line Operators

Because both symmetries are 1-form, their charged objects are 1-dimensional line operators:

$$\text{Electric line:} \quad W_{q_e}[\mathcal{C}] = \exp\left(i q_e \oint_{\mathcal{C}} a \right), \tag{6.7.4}$$

$$\text{Magnetic line:} \quad T_{q_m}[\mathcal{C}] = \exp\left(i q_m \oint_{\mathcal{C}} \tilde{a} \right). \tag{6.7.5}$$

Here W_{q_e} represents an electric probe of charge q_e, and T_{q_m} a magnetic probe of charge q_m.

6.7.3 Hamiltonian Perspective: Canonical Quantization

Let us now switch to the Hamiltonian picture, with spatial coordinates (x^1, x^2, x^3) at fixed time.

Electric symmetry. The symmetry operator is defined on a 2-dimensional surface, e.g.

$$Q_e^3 = \int dx^1 dx^2 \, E^3, \tag{6.7.6}$$

corresponding to a plane normal to the x^3-direction. The associated line operator is

$$W_{q_e}[\mathcal{C}_3] = \exp\left(-i q_e \int dx^3 \, a_3 \right), \tag{6.7.7}$$

a line parallel to the x^3-axis.

The canonical commutation relation

$$[a^i(\vec{x}), E^j(\vec{y})] = -i \, \delta^{ij} \delta^{(3)}(\vec{x} - \vec{y}) \tag{6.7.8}$$

allows one to compute

$$U_e W_e U_e^\dagger = \exp\left\{ i \alpha_e q_e \int_{\mathcal{C}_3} dy^3 \, \delta(x^3 - y^3) \right\} W_e,$$

where the integral $\int dy^3 \, \delta(x^3 - y^3)$ counts the number of intersection points between the line $\mathcal{C}_3$ and the surface supporting Q_e^3. This demonstrates explicitly how a 1-form symmetry operator acts on a 1-dimensional charged object.

Magnetic symmetry. An analogous computation applies to the magnetic 1-form symmetry. Introducing the dual potential $\tilde{a}$ with canonical momentum $\tilde{\Pi}^i = -B^i$, we have

$$[\tilde{a}^i(\vec{x}), B^j(\vec{y})] = -i\,\delta^{ij}\delta^{(3)}(\vec{x}-\vec{y}). \tag{6.7.9}$$

The magnetic charge operator

$$Q_m^3 = \frac{1}{2\pi}\int dx^1 dx^2\, B^3 \tag{6.7.10}$$

acts on the 't Hooft line $T_{q_m}[\mathcal{C}_3]$ in an entirely parallel manner.

6.7.4 The Wilson–'t Hooft Algebra

A central result concerns the algebra between electric and magnetic line operators. For two loops $\mathcal{C}$ and $\mathcal{C}'$, one finds

$$W_{q_e}[\mathcal{C}]\,T_{q_m}[\mathcal{C}'] = e^{\,i2\pi q_e q_m\,\mathrm{Link}(\mathcal{C},\mathcal{C}')}\,T_{q_m}[\mathcal{C}']\,W_{q_e}[\mathcal{C}]. \tag{6.7.11}$$

Here $\mathrm{Link}(\mathcal{C},\mathcal{C}')$ is the topological linking number between the two curves. When Link $= 1$, the operators fail to commute and pick up the phase $e^{i2\pi q_e q_m}$, which is precisely the Aharonov–Bohm phase for an electric charge q_e encircling a magnetic charge q_m.

In free Maxwell theory both q_e and q_m are integers, so $e^{i2\pi q_e q_m} = 1$ and the operators commute. Nevertheless, the nontrivial phase in (6.7.11) captures the fundamental electric–magnetic linking structure.

6.7.5 Path-Integral Perspective

We now rederive the same topological relation in the Lagrangian (Euclidean) picture. Insert the surface operator associated with the electric 1-form symmetry into the path integral:

$$\left\langle e^{\,i\alpha_e Q_e(\Sigma_2)}\cdots\right\rangle. \tag{6.7.12}$$

Using Stokes' theorem,

$$Q_e(\Sigma_2) = \int_{\Sigma_2} *f = \int_{\Omega_3} d*f, \tag{6.7.13}$$

where $\partial\Omega_3 = \Sigma_2$. The $d*f$ term modifies the Maxwell action $S \sim \int f^2$ as a shift of the gauge potential:

$$a_\mu \;\longrightarrow\; a_\mu - \alpha_e J_\mu(\Omega), \tag{6.7.14}$$

where $J_\mu(\Omega)$ localizes on Ω_3.

When evaluating

$$\left\langle e^{\,i\alpha_e Q_e(\Sigma_2)}\,W_{q_e}[\mathcal{C}]\right\rangle,$$

the Wilson line transforms as

$$W_{q_e}[\mathcal{C}] \;\longrightarrow\; W_{q_e}[\mathcal{C}]\exp\left(-iq_e\alpha_e\oint_{\mathcal{C}} J_\mu(\Omega)\right).$$

The contour integral $\oint_{\mathcal{C}} J_\mu(\Omega)$ equals the linking number $\mathrm{Link}(\Sigma_2,\mathcal{C})$ between the surface and the line. Therefore,

$$\left\langle e^{\,i\alpha_e Q_e(\Sigma_2)} e^{\,iq_e\oint_{\mathcal{C}} a}\right\rangle = e^{-iq_e\alpha_e\,\mathrm{Link}(\Sigma_2,\mathcal{C})}\left\langle e^{\,iq_e\oint_{\mathcal{C}} a}\right\rangle. \tag{6.7.15}$$

Interpretation. Equation (6.7.15) provides a rigorous path-integral proof that the 1-form symmetry operator $U_e(\Sigma_2)$ is a *topological* operator: its action on a charged line $W_e[\mathcal{C}]$ depends only on the linking number between Σ_2 and $\mathcal{C}$.

Summary.

- In $D = 4$, both electric and magnetic symmetries are 1-form:

$$U(1)_e^{(1)} \times U(1)_m^{(1)}.$$

- Their charged objects are line operators W_{q_e} and T_{q_m}.
- The Wilson–'t Hooft algebra encodes their mutual linking, reflecting the topological nature of higher-form symmetries.

6.8 Maxwell Theory in $D = 3$

This section highlights the essential difference between Maxwell theory in three and four dimensions. In $D = 3$, the magnetic symmetry ceases to be a higher-form symmetry and becomes an *ordinary* (0-form) global symmetry. Consequently, its charged objects (magnetic monopoles) become *local operators*.

From the general structure in Section V,

$$U(1)_e^{(1)} \times U(1)_m^{(D-3)},$$

we find that for $D = 3$,

$$U(1)_e^{(1)} \times U(1)_m^{(0)}. \tag{6.8.1}$$

Electric 1-form symmetry. The electric symmetry remains a 1-form symmetry with conserved charge

$$Q_e = \frac{1}{e^2} \int_{\Sigma_1} *f, \tag{6.8.2}$$

defined on a one-dimensional curve Σ_1. Its charged object is the Wilson line

$$W_{q_e}[\mathcal{C}] = \exp\left(i q_e \oint_{\mathcal{C}} a \right), \tag{6.8.3}$$

representing an electric probe particle.

Magnetic 0-form symmetry. In contrast, the magnetic symmetry becomes a *0-form* symmetry with charge

$$Q_m = \frac{1}{2\pi} \int_{\Sigma_2} f, \tag{6.8.4}$$

where Σ_2 is a spatial surface (in $D = 3$, the whole space at fixed time). The magnetic flux through space defines the magnetic charge.

6.8.1 From 't Hooft Line to Monopole Operator

In $D = 4$, the magnetic charged object—the 't Hooft operator T_{q_m}—is a 1-dimensional line operator ($D - 3 = 1$). In $D = 3$, however, $D - 3 = 0$: the magnetic charged object becomes a *0-dimensional* operator,

$$T_{q_m}(x), \tag{6.8.5}$$

i.e. a *local* operator.

Definition. This local magnetic operator is called the **monopole operator**. It creates a unit of magnetic flux at the spacetime point x.

6.8.2 Dual Photon Description

To make this structure explicit, we introduce the *dual field* $\tilde{a}$, defined via

$$*f = d\tilde{a}. \tag{6.8.6}$$

In $D = 3$, $*f$ is a 1-form, so $\tilde{a}$ must be a *0-form*, i.e. a scalar field—the **dual photon**.

In this dual formulation, the monopole operator takes the explicit local form

$$T_{q_m}(x) = e^{\,i\,2\pi q_m\,\tilde{a}(x)}. \tag{6.8.7}$$

Hence, the magnetic operator is manifestly local in $D = 3$.

6.8.3 Including Monopole Operators: Confinement Mechanism

Unlike the nonlocal line operator in $D = 4$, the local monopole operator $T_{q_m}(x)$ in $D = 3$ can be directly added to the action. The dual photon action becomes

$$S[\tilde{a}] = \int d^3x \left[\frac{e^2}{2}(\partial_\mu \tilde{a})^2 + \lambda \cos(2\pi q_m \tilde{a})\right]. \tag{6.8.8}$$

The cosine term originates from the monopole operator T_{q_m} and its conjugate.

Physical meaning. The potential term $\cos(2\pi q_m \tilde{a})$ acts as a mass term for the dual photon $\tilde{a}$, giving rise to a *mass gap*. In three dimensions, a massive photon implies that the potential between electric charges grows linearly with distance (rather than $1/r$), leading to **confinement**.

Historical remark. This is the celebrated result of A. Polyakov: in 3D compact QED, monopole (instanton) effects dominate the vacuum, generate a photon mass, and confine electric charges.

6.8.4 Symmetry Breaking

The magnetic 0-form symmetry $U(1)_m^{(0)}$ is generated by

$$Q_m = \frac{1}{2\pi}\int d^2x\, B. \tag{6.8.9}$$

Under this symmetry, the monopole operator transforms as

$$T_{q_m}(x) = e^{\,i\,2\pi q_m\,\tilde{a}(x)} \quad \longrightarrow \quad e^{\,i\,2\pi q_m(\tilde{a}(x)+\alpha_m)} = e^{\,i\,2\pi q_m \alpha_m} T_{q_m}(x), \tag{6.8.10}$$

showing that T_{q_m} carries magnetic charge q_m.

Including the term $\cos(2\pi q_m \tilde{a})$ in the action explicitly breaks the continuous $U(1)_m^{(0)}$ symmetry down to a discrete subgroup:

$$U(1)_m^{(0)} \longrightarrow \mathbb{Z}_{q_m}. \tag{6.8.11}$$

This explicit breaking is due to monopole proliferation in the vacuum.

6.8.5 Particle–Vortex Duality

Finally, let us examine the electric 1-form symmetry in the dual description. Its charge

$$Q_e = \frac{1}{e^2}\oint_{\Sigma_1} *f \tag{6.8.12}$$

can be rewritten using $*f = d\tilde{a}$ as

$$Q_e = \frac{1}{e^2}\oint_{\Sigma_1} d\tilde{a}. \tag{6.8.13}$$

The integral $\oint d\tilde{a}$ measures the *winding number* of the scalar field $\tilde{a}$ around a loop—precisely the topological charge carried by a *vortex*.

Conclusion. Electric charges (the endpoints of Wilson lines) in the original gauge description map to vortices of the dual scalar field in the dual description. This correspondence is known as the **particle–vortex duality** in three dimensions.

Summary.

- Symmetry structure:
$$U(1)_e^{(1)} \times U(1)_m^{(0)}.$$
- Magnetic charged objects become local monopole operators $T_{q_m}(x) = e^{i2\pi q_m \tilde{a}(x)}$.
- The monopole term $\cos(2\pi q_m \tilde{a})$ generates a mass gap and leads to confinement.
- Adding monopole operators breaks $U(1)_m^{(0)} \to \mathbb{Z}_{q_m}$.
- Electric charges $\leftrightarrow$ vortices in the dual picture: particle–vortex duality.

6.9 Chern-Simon Theory in 3D

The three-dimensional (2+1D) $\mathrm{U}(1)_k$ Chern–Simons theory provides a canonical example of a topological quantum field theory (TQFT) realizing a 1-form symmetry and illustrating how such topological terms give rise to anyonic statistics and edge dynamics.

The action is
$$S_{\mathrm{CS}}[a] = \frac{k}{4\pi}\int_M a \wedge \mathrm{d}a = \frac{k}{8\pi}\int_M \mathrm{d}^3x\, \epsilon^{\mu\nu\rho}\, a_\mu \partial_\nu a_\rho, \qquad k \in \mathbb{Z}, \tag{6.9.1}$$
where a is an emergent U(1) gauge field and $f = \mathrm{d}a$ its field strength. No Maxwell term appears, so there are no propagating degrees of freedom: the theory is purely topological.

—

Level quantization. Gauge invariance of the quantum theory requires k to be an integer. On a manifold $M = S^1 \times S^2$, the magnetic flux through S^2 is quantized,
$$\frac{1}{2\pi}\int_{S^2} f = n \in \mathbb{Z}, \tag{6.9.2}$$
and a large gauge transformation winding once around the S^1 direction, $\Lambda = 2\pi x^0/L_0$, shifts $a_0 \to a_0 + 2\pi/L_0$. The Chern–Simons action then changes by
$$\Delta S_{\mathrm{CS}} = 2\pi k n, \tag{6.9.3}$$
so ^{S} is invariant only if $k \in \mathbb{Z}$.

—

Equations of motion and topological nature. Varying the action gives
$$\frac{\delta S_{\mathrm{CS}}}{\delta a_\mu} = 0 \quad \Longrightarrow \quad f_{\mu\nu} = 0. \tag{6.9.4}$$
Thus all classical configurations are locally pure gauge, $a_\mu = \partial_\mu \lambda$, confirming that there are no local bulk excitations. Physical observables come only from global topology, e.g. Wilson loops
$$W_\gamma = \exp\left(\oint_\gamma a\right),$$
whose linking number determines anyonic phases.

—

Coupling to an external electromagnetic field. To describe electric response, couple a to an external U(1) gauge field A via
$$S[a, A] = \frac{k}{4\pi}\int a \wedge \mathrm{d}a + \frac{1}{2\pi}\int A \wedge \mathrm{d}a. \tag{6.9.5}$$

The a–equation of motion gives

$$\frac{k}{2\pi}\mathrm{d}a + \frac{1}{2\pi}\mathrm{d}A = 0 \quad \Rightarrow \quad \mathrm{d}a = -\frac{1}{k}\mathrm{d}A.$$

Plugging this back yields the effective response

$$S_{\text{eff}}[A] = \frac{1}{4\pi k}\int A \wedge \mathrm{d}A, \qquad \sigma_{xy} = \frac{1}{2\pi k}, \tag{6.9.6}$$

the fractional Hall conductivity. The CS term thus captures the universal topological response of a fractional quantum Hall state.

—

Charge–flux attachment and anyons. In 2+1D, inserting a magnetic flux $\Phi = 2\pi q_m$ through a small S^2 defines a monopole operator. Integrating the equation

$$\frac{k}{2\pi}\mathrm{d}a + j = 0$$

over that sphere gives the induced electric charge

$$Q_e = k\, q_m, \tag{6.9.7}$$

so a unit flux carries k units of electric charge. This binding of charge and flux changes particle statistics: moving one such composite around another yields a phase

$$\theta_{12} = \frac{2\pi}{k}\, q_1 q_2. \tag{6.9.8}$$

For $k = 1$, a unit flux turns a boson into a fermion (statistical phase π); for general k the excitations are anyons with fractional spin $s = q^2/(2k)$.

—

Boundary and anomaly inflow. On a manifold with boundary, gauge transformations of the bulk CS term produce a surface variation

$$\delta_\Lambda S_{\text{CS}} = \frac{k}{4\pi}\int_{\partial M} \Lambda\, \mathrm{d}a,$$

so the theory is not gauge-invariant by itself. Consistency demands new boundary degrees of freedom whose anomaly cancels this variation. Writing the bulk field near the boundary $x^2 = 0$ as a pure gauge, $a_i = \partial_i \phi$ $(i = 0, 1)$, and substituting into S_{CS} gives the edge action

$$S_{\text{edge}} = -\frac{k}{4\pi}\int_{\partial M} \mathrm{d}^2x \big(\partial_0\phi\, \partial_1\phi + v\, (\partial_1\phi)^2\big), \tag{6.9.9}$$

where v is a boundary propagation velocity. The field ϕ is a *chiral boson*; its equal-time commutator,

$$[\partial_1\phi(x^1),\, \phi(y^1)] = \frac{2\pi}{k}\, \delta(x^1 - y^1), \tag{6.9.10}$$

implies

$$[\phi(x^1), \phi(y^1)] = \frac{\pi}{k}\, \mathrm{sign}(x^1 - y^1), \tag{6.9.11}$$

and the vertex operators

$$V_n(x) = e^{in\phi(x)}$$

carry fractional statistics $\theta = \pi n^2/k$. The gauge anomaly of the chiral boson precisely cancels the bulk variation—an explicit realization of anomaly inflow.

—

Relation to bosonization. The boundary chiral boson is equivalent to a one-dimensional chiral fermion by the Coleman–Mandelstam correspondence:

$$\psi_R \sim e^{i\phi}.$$

Thus the (2+1)D Chern–Simons theory naturally generates the (1+1)D bosonization structure on its edge.

—

Ground-state degeneracy and 't Hooft anomaly. Placing $_k$ CS theory on a spatial torus T^2 yields two independent Wilson loops

$$W_x = \exp\Big(\oint_{C_x} a\Big), \qquad W_y = \exp\Big(\oint_{C_y} a\Big),$$

obeying

$$W_x W_y = e^{2\pi i/k}\, W_y W_x. \tag{6.9.12}$$

Because these operators do not commute, the Hilbert space has dimension $|k|$; there are k degenerate ground states labeled by eigenvalues of, say, W_x. This projective algebra signals a mixed 't Hooft anomaly between the electric and magnetic 1-form symmetries, explaining the topological ground-state degeneracy.

6.10 1-form Symmetry in Non-Abelian Gauge Theories

In Abelian gauge theory, a conserved 1-form current

$$j^{\mu\nu} = \frac{1}{2\pi}\,\epsilon^{\mu\nu\rho\sigma} f_{\rho\sigma}, \tag{6.10.1}$$

implies a continuous 1-form symmetry associated with the conservation of electric flux, $\partial_\mu j^{\mu\nu} = 0$. However, this structure no longer survives in non-Abelian Yang–Mills theory.

—

Loss of continuous 1-form symmetry

In an $SU(N)$ gauge theory, the field strength

$$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu - i[a_\mu, a_\nu]$$

takes values in the Lie algebra and transforms covariantly, $f_{\mu\nu} \to U f_{\mu\nu} U^\dagger$. Because of this non-Abelian structure, $f_{\mu\nu}$ itself is not gauge-invariant, and the quantity $\int_\Sigma *f$ depends on the gauge choice. Hence, there is no continuous, gauge-invariant 1-form charge as in the Abelian case.

—

The discrete center symmetry $\mathbb{Z}_N^{(1)}$

Although a continuous 1-form symmetry is absent, $SU(N)$ possesses a discrete *center subgroup*

$$Z_N = \{e^{2\pi i m/N}\,\mathbf{1}_N \mid m = 0, 1, \ldots, N-1\}.$$

Elements of Z_N commute with all of $SU(N)$, so they act trivially on adjoint fields (like the gauge field a_μ) but nontrivially on *Wilson lines.*

A Wilson line in representation R is defined as

$$W_R(C) = \mathrm{Tr}_R\, \mathcal{P} \exp\left(i \oint_C a\right).$$

Under a center transformation $h \in Z_N$,

$$W_R(C) \;\longrightarrow\; e^{2\pi i q_R/N}\, W_R(C),$$

where q_R is the N-*ality* (center charge) of R. For example:

- Fundamental representation: $q_R = 1$.
- Adjoint representation: $q_R = 0$.

Thus the center acts as a global $\mathbb{Z}_N^{(1)}$ 1-form symmetry, with Wilson lines as charged operators.

—

't Hooft lines and magnetic 1-form symmetry

The magnetic dual of a Wilson line is the *'t Hooft line*, defined not by an exponential operator but by a boundary condition. Around a 't Hooft line, the gauge field has a monopole-like singularity: as one encircles the line, the gauge transformation winds nontrivially in Z_N. Hence, the 't Hooft line carries the magnetic charge of the $\mathbb{Z}_N^{(1)}$ symmetry.

The Wilson and 't Hooft operators obey a nontrivial commutation relation:

$$W(C)\,H(C') = e^{2\pi i/N}\,H(C')\,W(C) \tag{6.10.2}$$

when the loops C and C' link once. This expresses the topological $\mathbb{Z}_N$ electric–magnetic duality.

—

Electric and magnetic 1-form symmetries

The two $\mathbb{Z}_N^{(1)}$ symmetries can be viewed as:

$$(\mathbb{Z}_N)_{\mathrm{e}}^{(1)} \quad \text{(electric)}, \qquad (\mathbb{Z}_N)_{\mathrm{m}}^{(1)} \quad \text{(magnetic)}.$$

They are exchanged under electric–magnetic duality.

Physical interpretation

- **Wilson lines** probe color-electric charges and carry $\mathbb{Z}_N$ electric charge.
- **'t Hooft lines** probe magnetic monopoles or vortices and carry $\mathbb{Z}_N$ magnetic charge.
- Their noncommutativity encodes the electric–magnetic duality structure of the theory.
- The existence of a discrete 1-form symmetry reveals the global topology of the gauge group, even though the local Yang–Mills Lagrangian is identical.

6.11 Spontaneous Breaking of Higher-Form Symmetries

The spontaneous breaking of higher-form symmetries provides a unifying viewpoint for understanding different phases of gauge theories. For concreteness, consider the case of a *1-form symmetry*, whose charged operators are the Wilson and 't Hooft lines. The expectation value of a large Wilson loop serves as an *order parameter* for the electric 1-form symmetry:

$$\langle W[C]\rangle \sim e^{-\mathrm{Area}[C]} \quad \text{or} \quad \langle W[C]\rangle \sim e^{-\mathrm{Perimeter}[C]}. \tag{6.11.1}$$

The behavior of $\langle W[C]\rangle$ with respect to the geometry of C distinguishes different physical phases. For sufficiently large loops, an area law indicates exponential suppression, while perimeter or Coulomb-like decay signals a phase where the 1-form symmetry is spontaneously broken.

Wilson loops as order parameters

To see why $\langle W[C]\rangle$ acts as an order parameter, consider a rectangular loop C of size $R \times T$ in Euclidean spacetime. It describes the worldlines of a static pair of test charges that are created in the distant past, separated to distance R, and annihilated after Euclidean time T. The corresponding vacuum expectation value is

$$\langle W[C]\rangle = \big\langle \mathrm{Tr}\, \mathcal{P} e^{i \oint_C a} \big\rangle, \tag{6.11.2}$$

which in the axial gauge $a_0 = 0$ reduces to the spatial holonomies along the two edges. Inserting a complete set of energy eigenstates $|n\rangle$ yields

$$\langle W[C]\rangle = \sum_n e^{-TE_n(R)} \, |\langle n|\psi(0)\rangle|^2 \quad \Rightarrow \quad \langle W[C]\rangle \sim e^{-TE_0(R)}. \tag{6.11.3}$$

Hence, the static potential between the charges is extracted as

$$V(R) = -\lim_{T\to\infty} \frac{1}{T} \ln\langle W[C]\rangle. \tag{6.11.4}$$

The functional form of $\langle W[C]\rangle$ therefore directly characterizes the potential $V(R)$ and the corresponding phase of the gauge theory.

—

Phases of gauge theories

There are three characteristic behaviors for large rectangular loops:

(i) Area law.

$$\langle W[C]\rangle \sim e^{-\sigma T R} \quad \Rightarrow \quad V(R) = \sigma R. \tag{6.11.5}$$

The potential grows linearly with distance, indicating confinement: color flux forms a stringlike tube connecting the charges. The 1-form symmetry remains unbroken.

(ii) Perimeter law.

$$\langle W[C]\rangle \sim e^{-\rho(T+R)} \quad \Rightarrow \quad V(R) = \rho. \tag{6.11.6}$$

The energy cost saturates at large separation, corresponding to a *deconfined phase*. The nonzero expectation value of the Wilson loop implies spontaneous breaking of the 1-form symmetry.

(iii) Coulomb (scale-invariant) law.

$$\langle W[C]\rangle \sim \exp\Big[-\alpha \frac{T}{R} - \beta \frac{R}{T}\Big] \quad \Rightarrow \quad V(R) = \frac{\alpha}{R}. \tag{6.11.7}$$

This is the Coulomb phase with long-range interactions and a massless gauge boson. It also corresponds to spontaneous breaking of a continuous 1-form symmetry.

In summary:

Area law	$\Rightarrow$	Confinement (unbroken symmetry),
Perimeter/Coulomb law	$\Rightarrow$	Deconfinement (SSB of 1-form symmetry).

—

Ward identity and the Goldstone theorem for 1-form symmetries

The 1-form version of the Ward identity reads

$$\langle \partial_\mu J^{\mu\nu}(x)\, W[C]\rangle = -\, q_e \int_C dy^\nu\, \delta^{(D)}(x-y)\, \langle W[C]\rangle. \tag{6.11.8}$$

Fourier transforming gives

$$ip_\mu\, \langle J^{\mu\nu}(p)\, W[C]\rangle = q_e f^\nu(p;C)\, \langle W[C]\rangle, \qquad f^\nu(p;C) \equiv \int_C dy^\nu e^{ip\cdot y}. \tag{6.11.9}$$

For a closed loop C, one has

$$p_\nu f^\nu(p;C) = 0, \qquad f^\nu(0;C) = \int_C dy^\nu \neq 0.$$

Taking the limit $p \to 0$ yields

$$\lim_{p\to 0} ip_\mu\, \langle J^{\mu\nu}(p)\, W[C]\rangle = q_e f^\nu(0;C)\, \langle W[C]\rangle. \tag{6.11.10}$$

If $\langle W[C]\rangle \neq 0$, the correlator must develop a pole at $p^2 = 0$:

$$\langle J^{\mu\nu}(p)\, W[C]\rangle \sim \frac{p_\mu f^\nu(p;C) - p_\nu f^\mu(p;C)}{p^2}. \tag{6.11.11}$$

Therefore, spontaneous breaking of a continuous 1-form symmetry implies the existence of a *massless excitation*, the higher-form analogue of the Goldstone boson.

—

The photon as a 1-form Goldstone mode in 4D QED

In four-dimensional free QED, the Wilson loop follows the Coulomb law, hence the 1-form symmetry is spontaneously broken and there exists a Goldstone excitation. Indeed, the conserved current is simply

$$J^{\mu\nu} = f^{\mu\nu} = \partial^\mu a^\nu - \partial^\nu a^\mu.$$

Quantizing the theory in Feynman gauge ($\partial_\mu a^\mu = 0$), the free field expansion reads

$$a_\mu(x) = \frac{1}{(2\pi)^{3/2}} \int \frac{d^3\vec{p}}{2|\vec{p}|} \sum_{\lambda=1}^{4} e_\mu^{(\lambda)}(p) \big[a_\lambda(p) e^{-ip\cdot x} + a_\lambda^\dagger(p) e^{ip\cdot x}\big]. \tag{6.11.12}$$

Physical polarizations correspond to $\lambda = 1, 2$, while $\lambda = 3, 4$ are gauge or zero-norm modes. Single-photon states are

$$|\lambda, \vec{p}\rangle = a_\lambda^\dagger(p)|0\rangle, \qquad [a_\lambda(p), a_{\lambda'}^\dagger(p')] = 2|\vec{p}|\, \delta_{\lambda\lambda'}\, \delta^{(3)}(\vec{p} - \vec{p}\,').$$

Matrix elements of the field strength are

$$\langle 0|f^{\mu\nu}(x)|\lambda, \vec{p}\rangle = \frac{i}{(2\pi)^{3/2}} [\, e_\lambda^\mu(p) p^\nu - e_\lambda^\nu(p) p^\mu\,]\, e^{-ip\cdot x} \neq 0. \tag{6.11.13}$$

The current $J^{\mu\nu} = f^{\mu\nu}$ thus creates single-photon states from the vacuum, showing that the Goldstone excitation of the broken 1-form symmetry *is precisely the photon.*

—

Physical interpretation and summary

- A 1-form symmetry acts on line operators rather than local fields. Its spontaneous breaking is diagnosed by the behavior of $\langle W[C]\rangle$.

- The **area law** corresponds to the symmetric (confining) phase, while **perimeter** and **Coulomb laws** correspond to the broken (deconfined) phases.
- The 1-form Ward identity leads to a pole in the current–Wilson correlator, revealing a massless Goldstone excitation.
- In 4D QED this excitation is the photon, illustrating that a massless gauge field can be viewed as the Goldstone mode of a spontaneously broken continuous 1-form symmetry.
- This perspective recasts the problem of confinement as the question of whether the 1-form symmetry is unbroken or spontaneously broken.

6.12 what new stuff can be given by higher form symmetry

6.12.1 Symmetry can be represented by a topological defect operator $U_g(\Sigma)$

Derivation

Define the symmetry defect operator on a $(d-p-1)$-dimensional manifold Σ_{d-p-1}:

$$U_g(\Sigma_{d-p-1}) := \exp\left\{ i\lambda \int_{\Sigma_{d-p-1}} \star J_{p+1} \right\}, \qquad g = e^{i\lambda}.$$

Now show it is topological. Take two homotopic surfaces Σ and Σ' and a $(d-p)$-dimensional region $\widehat{\Sigma}$ with

$$\partial\widehat{\Sigma} = \Sigma \cup (\Sigma')^{\vee},$$

where $(\Sigma')^{\vee}$ is Σ' with opposite orientation.

Compute

$$U_g(\Sigma)\, U_g(\Sigma')^{-1} = \exp\left\{ i\lambda \left(\int_{\Sigma} \star J_{p+1} - \int_{\Sigma'} \star J_{p+1} \right) \right\} = \exp\left\{ i\lambda \int_{\partial\widehat{\Sigma}} \star J_{p+1} \right\}.$$

Apply Stokes:

$$\int_{\partial\widehat{\Sigma}} \star J_{p+1} = \int_{\widehat{\Sigma}} d \star J_{p+1}.$$

If no charged operator insertions cross $\widehat{\Sigma}$, then $d \star J_{p+1} = 0$, hence

$$U_g(\Sigma)\, U_g(\Sigma')^{-1} = \exp\left\{ i\lambda \int_{\widehat{\Sigma}} d \star J_{p+1} \right\} = \exp\{ i\lambda \cdot 0 \} = 1,$$

so $U_g(\Sigma) = U_g(\Sigma')$. Therefore U_g depends only on the homotopy class: it is topological.

6.12.2 A structural theorem—all $p \geq 1$ higher-form symmetries must be Abelian

Derivation (the transverse 2-plane argument)

Consider two symmetry defect operators

$$U_g(\Sigma_{d-p-1}), \qquad U_h(\Sigma_{d-p-1}),$$

with $p \geq 1$. Because Σ has codimension $p+1$,

$$\operatorname{codim}(\Sigma) = p + 1 \geq 2.$$

Hence there exists (locally) a transverse 2-plane to the defects.

Because the defects are topological, we may smoothly deform them (without crossing charged operators). In the transverse 2-plane we can exchange the positions of the two defects by a smooth deformation. Therefore the product cannot depend on the ordering:

$$U_g(\Sigma)\, U_h(\Sigma) \;=\; U_h(\Sigma)\, U_g(\Sigma).$$

So the group law is Abelian when $p \geq 1$.

6.12.3 Symmetry action becomes a computable linking-number phase

Goal

Derive

$$\big\langle U_g(\Sigma_{d-p-1})\, W_q(\Gamma_p)\big\rangle = e^{iq\lambda\,\mathrm{Link}(\Sigma,\Gamma)}\,\big\langle W_q(\Gamma_p)\, U_g(\Sigma'_{d-p-1})\big\rangle, \qquad g = e^{i\lambda}.$$

Step 1: Ward identity

For a charge-q p-dimensional operator $W_q(\Gamma_p)$, the Ward identity is

$$d\star J_{p+1}(x)\, W_q(\Gamma_p) = q\,\delta^{(d-p)}(x\in\Gamma_p)\, W_q(\Gamma_p).$$

It means $d\star J_{p+1}=0$ except on Γ_p, where there is a delta-form source.

Step 2: Integrate over a filling manifold

Choose Σ_{d-p-1} that links Γ_p, and choose Σ'_{d-p-1} homotopic to Σ but not linking Γ_p. Let $\widehat{\Sigma}$ be a $(d-p)$-manifold such that

$$\partial\widehat{\Sigma} = \Sigma\cup(\Sigma')^{\vee}.$$

Integrate the Ward identity over $\widehat{\Sigma}$:

$$\int_{\widehat{\Sigma}} d\star J_{p+1}(x)\, W_q(\Gamma_p) = q\int_{\widehat{\Sigma}}\delta^{(d-p)}(x\in\Gamma_p)\, W_q(\Gamma_p).$$

The integral of the delta-form counts intersections, giving the linking number:

$$\int_{\widehat{\Sigma}}\delta^{(d-p)}(x\in\Gamma_p) = \mathrm{Link}(\Sigma,\Gamma).$$

Hence

$$\int_{\widehat{\Sigma}} d\star J_{p+1}(x)\, W_q(\Gamma_p) = q\,\mathrm{Link}(\Sigma,\Gamma)\, W_q(\Gamma_p).$$

Step 3: Convert to a phase relation for U_g

By Stokes,

$$\int_{\widehat{\Sigma}} d\star J_{p+1} = \int_{\partial\widehat{\Sigma}}\star J_{p+1} = \int_{\Sigma}\star J_{p+1} - \int_{\Sigma'}\star J_{p+1}.$$

Exponentiate with $i\lambda$:

$$\exp\left\{i\lambda\int_{\Sigma}\star J_{p+1}\right\} = \exp\left\{i\lambda\int_{\Sigma'}\star J_{p+1}\right\}\exp\left\{iq\lambda\,\mathrm{Link}(\Sigma,\Gamma)\right\}.$$

Recognize the exponentials as defect operators:

$$\langle U_g(\Sigma_{d-p-1})W_q(\Gamma_p)\rangle = e^{iq\lambda\mathrm{Link}(\Sigma,\Gamma)}\langle W_q(\Gamma_p)U_g(\Sigma'_{d-p-1})\rangle, \quad g=e^{i\lambda}.$$

6.12.4 A defect insertion is equivalent to a singular background gauge field

New thing: $U_g(\Sigma)$ can be implemented as a concrete background field configuration $B_{p+1} = \lambda\,\delta^{(p+1)}(x\in\Sigma)$.

This converts defect-operator statements into background-field computations.

Goal

Derive

$$\Lambda_p = \lambda\,\Omega_p, \qquad B_{p+1} = d\Lambda_p = \lambda\,\delta^{(p+1)}(x \in \Sigma_{d-p-1}),$$

and explain why it reproduces the linking action.

Step 1: From defect to background gauge transformation

The idea is: inserting $U_g(\Sigma)$ is equivalent to turning on a background gauge transformation whose parameter winds around Σ. One associates

$$\Lambda_p = \lambda\,\Omega_p,$$

where Ω_p is the global angular form of the normal bundle of $\Sigma \subset M_d$.

Step 2: Induced background field strength

Apply $\delta B_{p+1} = d\Lambda_p$:

$$B_{p+1} = d\Lambda_p = \lambda\,d\Omega_p.$$

But $d\Omega_p$ is the Thom class (a delta-form) supported on Σ, so

$$B_{p+1} = \lambda\,\delta^{(p+1)}(x \in \Sigma_{d-p-1}).$$

Step 3: Recover the defect operator and linking phase

Consider the coupling term

$$i\int B_{p+1} \wedge \star J_{p+1}.$$

If B_{p+1} is supported on Σ, then

$$i\int_{M_d} \lambda\,\delta^{(p+1)}(x \in \Sigma) \wedge \star J_{p+1} = i\lambda \int_\Sigma \star J_{p+1}.$$

Therefore

$$\exp\left\{i\int B_{p+1} \wedge \star J_{p+1}\right\} = \exp\left\{i\lambda \int_\Sigma \star J_{p+1}\right\} = U_g(\Sigma).$$

Then the linking phase with $W_q(\Gamma_p)$ follows from the Ward identity in the previous block.

6.12.5 Anomaly detection becomes geometric and RG-invariant (and can forbid gauging)

Step 1: Background-field meaning of anomaly

Introduce background fields for global symmetries. If under a background gauge transformation the partition function changes by a phase that cannot be removed by any local counterterm, we say there is a 't Hooft anomaly.

A typical signal is: the would-be defect operator depends on the choice of Σ (it is not topological) because the current is not conserved. For example, if

$$d \star j_A = \frac{2}{8\pi^2} f \wedge f,$$

then deforming $\Sigma_3 \to \Sigma_3'$ produces

$$U_g(\Sigma_3) = U_g(\Sigma_3') \times \exp\left\{\frac{2i\alpha}{8\pi^2}\int_{\Gamma_4} f \wedge f\right\}.$$

So the defect fails to be topological by a definite phase. That phase is the anomaly data.

Step 2: "Cannot gauge both" as an obstruction

In gauge-type constructions, one checks gauge invariance by computing the shift of the action under gauge transformations. If the shift is in $2\pi i\,\mathbb{Z}$, then e^{-S} is invariant. For instance,

$$\delta S = \frac{iN}{2\pi}\int d\lambda_{p-1}\wedge dB_{d-p-1} \in 2\pi i\,\mathbb{Z}.$$

Then the exponentiated action is gauge invariant.

However, if we turn on two background fields for two higher-form symmetries (such as electric and magnetic 1-form symmetries in Maxwell theory), then a background gauge transformation for one symmetry can produce a nontrivial phase involving the other background field. If that phase cannot be removed for both simultaneously, it is a mixed anomaly. This is the precise sense in which one cannot gauge both symmetries at the same time: gauging requires full gauge invariance of the coupled background theory.

6.12.6 A concrete example of a strong IR constraint from higher-form symmetry (YM at $\theta = \pi$)

New thing: By coupling the 1-form center symmetry to a background B_2, one derives a mixed anomaly with time reversal that forbids a trivially gapped IR at $\theta = \pi$.

The 1-form center symmetry of $SU(N)$ Yang–Mills is $\mathbb{Z}_N^{(1)}$. Coupling a $\mathbb{Z}_N^{(1)}$ symmetry to a background means: we allow gauge bundles that are not honest $SU(N)$ bundles, but rather $PSU(N) = SU(N)/\mathbb{Z}_N$ bundles.

The obstruction to lifting a $PSU(N)$ bundle back to an $SU(N)$ bundle is a cohomology class

$$w_2 \in H^2(M_4;\mathbb{Z}_N).$$

one often writes an integral lift

$$w_2 \in H^2(M_4;\mathbb{Z}), \qquad \frac{B_2}{2\pi} = \frac{1}{N}w_2,$$

so that B_2 is a 2-form background field whose flux is quantized in units of $2\pi/N$. Intuitively: B_2 measures the failure of transition functions to satisfy the $SU(N)$ cocycle condition; the failure is a center element, and w_2 records it.

Step 1: Start from the θ-term and define the topological charge

The Yang–Mills action is

$$S = \int_{M_4}\frac{1}{2g^2}\mathrm{Tr}(F\wedge\star F) + \frac{i\theta}{8\pi^2}\int_{M_4}\mathrm{Tr}(F\wedge F).$$

Define the topological charge

$$Q := \frac{1}{8\pi^2}\int_{M_4}\mathrm{Tr}(F\wedge F).$$

For an honest $SU(N)$ bundle, $Q\in\mathbb{Z}$, and therefore $\theta\sim\theta+2\pi$.

Step 2: Turn on B_2 (equivalently, turn on w_2) and explain why Q becomes fractional

When we turn on the $\mathbb{Z}_N^{(1)}$ background, we must sum over $PSU(N)$ bundles with fixed w_2. For a $PSU(N)$ bundle, the second Chern class is not necessarily an integer. A standard (and key) result is:

$$Q = \frac{1}{8\pi^2}\int_{M_4}\mathrm{Tr}(F\wedge F) \equiv \frac{N-1}{N}\int_{M_4}\frac{w_2\wedge w_2}{2} \pmod{\mathbb{Z}}.$$

So Q can have a fractional part determined by w_2. This is the precise meaning of "instanton number fractionalization."

A $PSU(N)$ bundle can be described locally by $SU(N)$ transition functions that fail the cocycle condition by a center element. This failure is measured by w_2. That failure also shifts the global topology in a way that changes the allowed values of the Chern class, so the instanton number is no longer forced to be an integer.

Step 3: New periodicity of θ in the presence of w_2

Because Q can shift by $1/N$ fractions, the phase from the θ-term,

$$\exp(i\theta Q),$$

is no longer invariant under $\theta \to \theta + 2\pi$ for fixed nonzero w_2. Instead, we need $\theta \to \theta + 2\pi N$ to guarantee invariance for all allowed fractional Q. So the effective periodicity becomes

$$\theta \sim \theta + 2\pi N \qquad \text{(when } w_2 \neq 0 \text{ is allowed).}$$

Step 4: Time reversal and why $\theta = \pi$ becomes subtle

Time reversal T acts as

$$T: \ \theta \mapsto -\theta.$$

If the periodicity were still 2π, then $\theta = \pi$ would be T-invariant because

$$-\pi \equiv \pi \pmod{2\pi}.$$

But with background w_2 present, the periodicity is effectively $2\pi N$, so the same argument is no longer valid. We must compare $\theta = \pi$ and $\theta = -\pi$ *carefully* at fixed w_2.

Write

$$-\pi = \pi - 2\pi.$$

The difference between $\theta = \pi$ and $\theta = -\pi$ is a shift of -2π, which produces an extra factor in the path integral:

$$\exp\big(i(-2\pi)Q\big) = \exp(-2\pi i\, Q).$$

Now use the fractional part of Q from Step 2:

$$Q \equiv \frac{N-1}{N} \int_{M_4} \frac{w_2 \wedge w_2}{2} \pmod{\mathbb{Z}}.$$

Therefore

$$\exp(-2\pi i\, Q) = \exp\left\{-2\pi i\, \frac{N-1}{N} \int_{M_4} \frac{w_2 \wedge w_2}{2}\right\}.$$

Up to a sign convention (depending on how we define $\delta_T Z$), this is the background-dependent phase under T:

$$\delta_T Z[w_2] = \exp\left\{2\pi i\, \frac{N-1}{N} \int_{M_4} \frac{w_2 \wedge w_2}{2}\right\} Z[w_2].$$

The crucial point is: the phase depends only on the background w_2 (equivalently B_2), so it is not a dynamical effect; it is an anomaly-type effect.

Step 5: What is the "discrete angle" (discrete θ-term) here?

When we couple to a background w_2, we are allowed to add local 4d counterterms built only from w_2. A natural one is

$$S_{\text{disc}}[w_2] \ = \ \frac{2\pi i\, p}{N} \int_{M_4} \frac{w_2 \wedge w_2}{2}, \qquad p \in \mathbb{Z}.$$

Because w_2 is $\mathbb{Z}_N$-valued, this term is only well-defined modulo $2\pi i$, so only $p \pmod{N}$ is meaningful. That is why p is a **discrete angle**: it is not continuous; it is a choice in a finite set.

What does S_{disc} do? It changes the partition function by a background-dependent phase:

$$Z[w_2] \;\to\; Z[w_2] \cdot \exp\left\{ \frac{2\pi i\, p}{N} \int_{M_4} \frac{w_2 \wedge w_2}{2} \right\}.$$

So it can shift the T-phase by an allowed quantized amount.

The background-dependent phase under T is not removable in a fully consistent way; this is the mixed 't Hooft anomaly between T and $\mathbb{Z}_N^{(1)}$.

Because 't Hooft anomalies are RG-invariant, the IR theory at $\theta = \pi$ must reproduce the same anomaly. In particular, it cannot be a completely trivial gapped theory (a unique vacuum with no topological order and no symmetry breaking). So higher-form symmetry gives a strong, model-independent constraint:

4d $SU(N)$ Yang–Mills at $\theta = \pi$ cannot flow to a trivially gapped theory.

6.12.7 A higher-form Coleman–Mermin–Wagner theorem (a general no-go for SSB)

New thing: Higher-form symmetry has a universal "no SSB" theorem.
If $d - p \leq 2$, a continuous p-form global symmetry cannot be spontaneously broken.

This is new because it gives a model-independent constraint on phases: it tells you in which spacetime dimensions a broken p-form symmetry phase is impossible.

Goal

Assume a continuous p-form symmetry is spontaneously broken in d dimensions. Show that this assumption leads to an infrared (IR) divergence when $d - p \leq 2$, so the broken phase cannot exist.

Step 1: What "broken p-form symmetry" means

A continuous p-form global symmetry has a topological defect

$$U_g(\Sigma_{d-p-1}) = \exp\left\{ i\lambda \int_{\Sigma_{d-p-1}} \star J_{p+1} \right\}, \qquad g = e^{i\lambda}.$$

A charged operator is a p-dimensional operator $W_q(\Gamma_p)$, and in a broken phase one expects a nonzero "long-distance order" for large Γ_p, i.e. the expectation value does not vanish after suitable renormalization (this is the higher-form analogue of a nonzero order parameter).

The standard effective-field-theory statement is:

$$\text{Broken } p\text{-form symmetry} \implies \text{a massless (Goldstone) } p\text{-form field } B_p$$

with a gauge redundancy

$$B_p \sim B_p + d\lambda_{p-1},$$

and a free IR action of Maxwell type

$$S_{\text{IR}}[B_p] = \frac{1}{2e^2} \int_{M_d} H_{p+1} \wedge \star H_{p+1}, \qquad H_{p+1} := dB_p.$$

(Heuristically: this is the simplest local action consistent with the symmetry and the gauge redundancy.)

Step 2: Use a charged operator as an IR diagnostic

A natural charged probe (in the broken-phase EFT) is a "Wilson-type" operator

$$W(\Gamma_p) := \exp\left\{ i \int_{\Gamma_p} B_p \right\}.$$

For a Gaussian theory, its expectation value is controlled by the two-point function of B_p:

$$\langle W(\Gamma_p)\rangle = \exp\left\{-\frac{1}{2}\Big\langle\Big(\int_{\Gamma_p} B_p\Big)^2\Big\rangle\right\},$$

where

$$\Big\langle\Big(\int_{\Gamma_p} B_p\Big)^2\Big\rangle = \int_{\Gamma_p}\int_{\Gamma_p} \langle B_p(x)\, B_p(y)\rangle.$$

Therefore:

$$\text{If } \Big\langle\Big(\int_{\Gamma_p} B_p\Big)^2\Big\rangle \text{ diverges in the IR, then } \langle W(\Gamma_p)\rangle = 0$$

in the infinite-volume / long-distance limit. That would contradict the assumption of a stable broken phase.

Step 3: Estimate the IR behavior of the Goldstone correlator

For a massless field, the IR behavior is dominated by small momentum $k \to 0$. In momentum space, the propagator of a free massless gauge field behaves schematically as

$$\langle B_p B_p\rangle(k) \;\sim\; \frac{1}{k^2}$$

(up to tensor/projector structure that does not change the IR power counting).

Now the key point is that the operator $\int_{\Gamma_p} B_p$ effectively removes p directions by integrating over a p-dimensional submanifold. So the remaining IR momentum integral is controlled by the $(d-p)$-dimensional transverse momentum:

$$\Big\langle\Big(\int_{\Gamma_p} B_p\Big)^2\Big\rangle_{\text{IR}} \;\sim\; \int_{k\ \text{small}} \frac{d^{\,d-p}k}{k^2}.$$

This is the same power counting that in ordinary CMW ($p = 0$) gives $\int d^d k/k^2$.

Step 4: Determine when the IR integral diverges

Let $n := d - p$. Then the IR integral is

$$I_{\text{IR}}(n) := \int_{k\ \text{small}} \frac{d^{\,n}k}{k^2}.$$

In spherical coordinates in n dimensions, $d^{\,n}k \sim k^{n-1}dk$, so

$$I_{\text{IR}}(n) \;\sim\; \int_0^\Lambda \frac{k^{n-1}}{k^2}\, dk = \int_0^\Lambda k^{n-3}\, dk.$$

Now check the behavior near $k = 0$:

- If $n - 3 > -1$, i.e. $n > 2$, then the integral is finite in the IR.
- If $n - 3 = -1$, i.e. $n = 2$, then

  $$I_{\text{IR}}(2) \sim \int_0^\Lambda \frac{dk}{k} \sim \log(\Lambda/0),$$

  a logarithmic IR divergence.
- If $n - 3 < -1$, i.e. $n < 2$, then the integral diverges as a power in the IR.

Therefore

$$I_{\text{IR}}(d-p) \text{ diverges if and only if } d - p \le 2.$$

Step 5: Conclude the higher-form CMW theorem

If $d - p \leq 2$, then

$$\left\langle \left(\int_{\Gamma_p} B_p \right)^2 \right\rangle_{\mathrm{IR}} = \infty,$$

so for large Γ_p,

$$\langle W(\Gamma_p) \rangle = \exp\left\{ -\frac{1}{2} \left\langle \left(\int_{\Gamma_p} B_p \right)^2 \right\rangle \right\} = \exp\{-\infty\} = 0.$$

So a putative long-distance "order parameter" for broken p-form symmetry is destroyed by IR fluctuations whenever $d - p \leq 2$. This contradicts the assumption of spontaneous breaking.

Hence we obtain the higher-form Coleman–Mermin–Wagner theorem:

$$\boxed{\text{A continuous } p\text{-form global symmetry cannot be spontaneously broken if } d - p \leq 2.}$$

Immediate corollary used in many notes

In $d = 4$,

$$d - p \leq 2 \quad \Longleftrightarrow \quad p \geq 2,$$

so a continuous p-form symmetry with $p \geq 2$ cannot be spontaneously broken in four dimensions. This is a strong, purely kinematical constraint on possible phases.